

**CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
LOS ANGELES REGION**

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**ORDER R4-2019-0135
NPDES NO. CA0055221**

**WASTE DISCHARGE REQUIREMENTS
FOR THE CITY OF SIMI VALLEY
SIMI VALLEY WATER QUALITY CONTROL PLANT
DISCHARGE TO THE ARROYO SIMI**

The following entity is subject to waste discharge requirements (WDRs) set forth in this Order:

Table 1. Discharger Information

Discharger	City of Simi Valley (Discharger or Permittee)
Name of Facility	Simi Valley Water Quality Control Plant (Simi Valley WQCP or Facility) including its associated wastewater collection system and outfalls
Facility Address	600 West Los Angeles Avenue, Simi Valley, CA 93065 Ventura County

Table 2. Discharge Location

Discharge Point	Effluent Description	Discharge Point Latitude (North)	Discharge Point Longitude (West)	Receiving Water
001	Tertiary treated wastewater	34.28200°	-118.81290°	Arroyo Simi
002	Tertiary treated wastewater	34.28169°	-118.81350°	Arroyo Simi

Table 3. Administrative Information

This Order was adopted on:	November 14, 2019
This Order shall become effective on:	January 1, 2020
This Order shall expire on:	December 31, 2024
The Discharger shall file a Report of Waste Discharge as an application for reissuance of WDRs in accordance with title 23, California Code of Regulations, and an application for reissuance of a National Pollutant Discharge Elimination System (NPDES) permit no later than:	180 days prior to the Order expiration date
The United States Environmental Protection Agency (USEPA) and the California Regional Water Quality Control Board, Los Angeles Region have classified this discharge as follows:	Major

I, Renee Purdy, Executive Officer, do hereby certify that this Order with all attachments is a full, true, and correct copy of the Order adopted by the California Regional Water Quality Control Board, Los Angeles Region, on **the date indicated above**.

A handwritten signature in black ink, appearing to read "R Purdy", is positioned above a horizontal line.

Renee Purdy, Executive Officer

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I. FACILITY INFORMATION

Information describing the Simi Valley Water Quality Control Plant (Simi Valley WQCP or Facility) is summarized in Table 1 and in sections I and II of the Fact Sheet (Attachment F). Section I of the Fact Sheet also includes information regarding the Facility's permit application.

II. FINDINGS

The California Regional Water Quality Control Board, Los Angeles Region (Regional Water Board), finds:

- A. Legal Authorities.** This Order serves as waste discharge requirements (WDRs) pursuant to article 4, chapter 4, division 7 of the California Water Code (CWC) (commencing with section 13260). This Order is also issued pursuant to section 402 of the federal Clean Water Act (CWA) and implementing regulations adopted by the U.S. EPA and chapter 5.5, division 7 of the Water Code (commencing with section 13370). It shall serve as a National Pollutant Discharge Elimination System (NPDES) permit authorizing the Discharger to discharge into waters of the United States at the discharge location described in Table 2 subject to the WDRs in this Order.
- B. Background and Rationale for Requirements.** The Regional Water Board developed the requirements in this Order based on information submitted as part of the application, through monitoring and reporting programs, and other available information. The Fact Sheet (Attachment F), which contains background information and rationale for the requirements in this Order, is hereby incorporated into and constitutes Findings for this Order. Attachments A through E and G through I are also incorporated into this Order.
- C. Provisions and Requirements Implementing State Law.** The provisions/requirements in subsection IV.B, IV.C, and V.B are included to implement state law only.
- D. Notification of Interested Parties.** The Regional Water Board has notified the Discharger and interested agencies and persons of its intent to prescribe WDRs for the discharge and has provided them with an opportunity to submit their written comments and recommendations. Details of the notification are provided in the Fact Sheet
- E. Consideration of Public Comment.** The Regional Water Board, in a public meeting, heard and considered all comments pertaining to the discharge. Details of the Public Hearing are provided in the Fact Sheet.

THEREFORE, IT IS HEREBY ORDERED that this Order supersedes Order R4-2014-0066-A01 except for enforcement purposes, and, in order to meet the provisions contained in Division 7 of the CWC (commencing with section 13000) and regulations adopted thereunder, and the provisions of the CWA and regulations and guidelines adopted thereunder, the Permittee is authorized to discharge from the identified facility and outfalls into waters of the United States and shall comply with the requirements in this

Order. This action in no way prevents the Regional Water Board from taking enforcement action for past violations of the previous Order.

III. DISCHARGE PROHIBITIONS

- A. Discharge of treated wastewater at a location different from that described in this Order is prohibited.
- B. The bypass or overflow of untreated wastewater or wastes to surface waters or surface water drainage courses is prohibited, except as allowed in Standard Provision I.G. of Attachment D, Standard Provisions.
- C. The monthly average effluent dry weather discharge flow rate from the facility shall not exceed the 12.5 million gallons per day (MGD) design capacity.
- D. The Permittee shall not cause degradation of any water supply, except as consistent with State Water Board Resolution No. 68-16.
- E. The treatment or disposal of wastes from the facility shall not cause pollution or nuisance as defined in section 13050, subdivisions (l) and (m), of the CWC.
- F. The discharge of any substances in concentrations toxic to animal or plant is prohibited.
- G. The discharge of any radiological, chemical, or biological warfare agent or high level radiological waste is prohibited.

IV. EFFLUENT LIMITATIONS AND DISCHARGE SPECIFICATIONS

A. Effluent Limitations – Discharge Points 001 and 002

1. Final Effluent Limitations – Discharge Points 001 and 002

- a. The Permittee shall maintain compliance with the following effluent limitations at Discharge Points 001 and 002, into the Arroyo Simi, with compliance measured at Monitoring Location EFF-001 as described in the Monitoring and Reporting Program (MRP), Attachment E:

Table 4. Effluent Limitations

Parameters	Units	Average Monthly	Average Weekly	Maximum Daily
Biochemical Oxygen Demand (BOD ₅ 20°C)	mg/L	20	30	45
BOD ₅ 20°C	lbs/day	2,080	3,130	4,690
Total Suspended Solids (TSS)	mg/L	15	40	45
TSS	lbs/day	1,560	4,170	4,690
Temperature	°F	--	--	86

Parameters	Units	Average Monthly	Average Weekly	Maximum Daily
Radioactivity ¹	--	--	--	--
Combined Radium-226 and Radium 228	pCi/L	5	--	--
Gross Alpha particle activity (excluding radon and uranium)	pCi/L	15	--	--
Uranium	pCi/L	20	--	--
Gross Beta/photon emitters	millirem/year	4	--	--
Strontium-90	pCi/L	8	--	--
Tritium	pCi/L	20,000	--	--
Removal Efficiency for BOD	%	≥85	--	--
Removal Efficiency for TSS	%	≥85	--	--
Oil and Grease	mg/L	10	--	15
Oil and Grease	lbs/day	1,040	--	1,560
Settleable Solids	ml/L	0.1	--	0.3
Total Residual Chlorine	mg/L	--	--	0.1
Total Dissolved Solids (TDS) (dry-weather)	lbs/day	88,610	--	--
TDS (wet-weather)	mg/L	850	--	--
Sulfate (dry-weather)	lbs/day	26,060	--	--
Sulfate (wet-weather)	mg/L	250	--	--
Chloride (dry-weather)	lbs/day	15,640	--	--
Chloride (wet-weather)	mg/L	150	--	--
Boron (dry-weather)	lbs/day	104	--	--
Boron (wet-weather)	mg/l	1.0	--	--
MBAS	mg/L	0.5	--	--
MBAS	lbs/day	50	--	--

¹ These effluent limitations are derived from Title 22, chapter 15, article 5, sections 64442 and 64443, of the California Code of Regulations (CCR). The incorporation by reference is prospective including future changes to the incorporated provisions as the changes take effect.

Parameters	Units	Average Monthly	Average Weekly	Maximum Daily
Ammonia Nitrogen	mg/L	2.4	--	3.3
Ammonia Nitrogen	lbs/day	--	--	$2.9 \times Q^2$
Nitrate + Nitrite (as N)	mg/L	9	--	--
Nitrate (as N)	mg/L	9	--	--
Nitrite (as N)	mg/L	0.9	--	--
Copper	µg/L	30.5	--	31
Nickel	µg/L	169	--	960
Mercury	lbs/month	0.031	--	--
Selenium	µg/L	4.5	--	6.8
Selenium	lbs/day	0.47	--	0.71
Cyanide	µg/L	4.1	--	8.9
Cyanide	lbs/day	0.43	--	0.93
Bis (2-Ethylhexyl) Phthalate	µg/L	4	--	--
Bis (2-Ethylhexyl) Phthalate	lbs/day	0.42	--	--
Chlorpyrifos	µg/L	0.014	--	0.025
Diazinon	µg/L	0.1	--	0.1

² Q represents the POTW effluent flow at the time the water quality measurement is collected (not to exceed the design flow of 12.5 MGD) and a conversion factor to lbs/day based on the units of measure for the flow.

Parameters	Units	Average Monthly	Average Weekly	Maximum Daily
Chronic Toxicity ^{3, 4}	Pass or Fail, % Effect (Test of Significant Toxicity, (TST))	Pass ⁵	--	Pass or % Effect <50
Chlordane	µg/L	0.00059	--	0.0012
4,4'-DDD	µg/L	0.00084	--	0.0017
4,4'-DDE	µg/L	0.00059	--	0.0012
4,4'-DDT	µg/L	0.00059	--	0.0012
Dieldrin	µg/L	0.00014	--	0.00028
PCBs	µg/L	0.00017	--	0.00034
Toxaphene	µg/L	0.00016	--	0.00033

- b. The mass based effluent limitations are based on the plant design flow rate of 12.5 MGD and are calculated as follows: Flow (MGD) x Concentration (mg/L) x 8.34 (conversion factor) = lbs/day. During wet-weather storm events in which the flow exceeds the design capacity, the mass discharge rate limitations shall not apply, and concentration limitations shall be the only applicable effluent limitations.
- c. The effluent values for pH shall be maintained within the limits of 6.5 standard units and 8.5 standard units.

³ The Median Monthly Effluent Limitation (MMEL) shall be reported as "Pass" or "Fail." The Maximum Daily Effluent Limitation (MDEL) shall be reported as "Pass" or "Fail" and "% Effect." The MMEL for chronic toxicity shall only apply when there is a discharge on more than one day in a calendar month period. During such calendar months, up to three independent toxicity tests may be conducted when one toxicity test results in "Fail."

⁴ The *Calleguas Creek Watershed Toxicity TMDL* includes a WLA of 1.0 TUC for toxicity, which is required to be implemented in accordance with USEPA, State Water Board, and Regional Water Board resolutions, guidance and policy at the time of permit issuance or renewal. In addition, a numeric water quality-based effluent limitation (WQBEL) is established because effluent data showed that there is reasonable potential for the effluent to cause or contribute to an exceedance of the chronic toxicity water quality objective. The numeric WLA is protective of both the numeric acute toxicity and the narrative toxicity Basin Plan water quality objectives. Consistent with the *Toxicity TMDL*, these chronic toxicity WLA-based final effluent limitations shall be implemented using the *Short Term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Freshwater Organisms* (U.S. EPA 2002, EPA-821-R-02-013), and current USEPA guidance in *National Pollutant Discharge Elimination System Test of Significant Toxicity Implementation Document* (EPA 833-R-10-003, June /2010) and EPA Regions 8, 9 and 10 Toxicity Training Tool (January 2010), <http://www2.epa.gov/region8/epa-regions-8-9-and-10-toxicity-training-tool-january-2010>.

⁵ This is a Median Monthly Effluent Limitation.

- d. The wastes discharged to water courses shall at all times be adequately disinfected. For the purpose of this requirement, the wastes shall be considered adequately disinfected if: (1) the median number of total coliform bacteria at some point in the treatment process does not exceed a 7-day median of 2.2 Most Probable Number (MPN) or Colony Forming Unit (CFU) per 100 milliliters utilizing the bacteriological results of the last seven (7) days for which an analysis has been completed, (2) the number of total coliform bacteria does not exceed 23 MPN or CFU per 100 milliliters in more than one sample within any 30-day period, and (3) no sample shall exceed 240 MPN or CFU of total coliform bacteria per 100 milliliters. Samples shall be collected at a time when wastewater flow and characteristics are most demanding on treatment facilities and disinfection processes.
- e. For the protection of the water contact recreation beneficial use, the wastes discharged to water courses shall have received adequate treatment, so that the turbidity of the treated wastewater does not exceed any of the following: (a) an average of 2 Nephelometric turbidity units (NTU) within a 24-hour period, (b) 5 NTU more than 5 percent of the time (72 minutes) within a 24-hour period, and (c) 10 NTU at any time.
- f. To protect the existing beneficial use of groundwater recharge (GWR) and the underlying ground water basins, pollutants shall not be present in the wastes discharged at concentrations that pose a threat to ground water quality.

2. Interim Effluent Limitations

- a. **Metals TMDL-based Interim limits:** As of March 27, 2017, the effluent from the Simi Valley WQCP has consistently achieved the final waste load allocations for the Metals TMDL. Therefore, no interim effluent limitations are included in this permit, and only the final effluent limitations for these pollutants are applicable in this Order.
- b. **Organo Chlorine Pesticides, PCBs, and Siltation TMDL-based Interim limits:** Interim WLAs are included in the OC Pesticides, PCBs, and Siltation TMDL for chlordane, 4,4'-DDD, 4,4'-DDE, 4,4'-DDT, dieldrin, PCBs, and toxaphene applicable to Simi Valley WQCP effluent discharge. However, existing data indicate that the facility is consistently meeting the final effluent limitations for the aforementioned parameters. Therefore, the interim effluent limitations are not warranted and only the final effluent limitations for these pollutants are applicable in this permit.
- c. **Boron, Chloride, Sulfate, and TDS (Salts) TMDL-based Interim limits:** The interim effluent limitations for salts are derived from the interim WLAs as set forth in the *Calleguas Creek Watershed Salts TMDL* (Salts TMDL), established by the Regional Water Board on October 4, 2007, which became effective on December 8, 2008. The TMDL interim effluent limitation are set equal to the 95th percentile of available discharge data. There is no interim effluent limitation for boron because the 95th percentile concentration is below the Basin Plan objective. TDS, sulfate, and chloride data indicate that

the facility is complying with the Basin Plan water quality objectives (WQOs). Therefore, the interim effluent limitations for TDS, sulfate, chloride, and boron are not warranted and only the final effluent limitations for TDS, sulfate, chloride, and boron are applicable in this permit.

B. Land Discharge Specifications – Not Applicable

C. Recycling Specifications

The Permittee shall continue to investigate the feasibility of recycling, conservation, and/or alternative disposal methods for wastewater (such as groundwater injection), and/or beneficial use of storm water and dry-weather urban runoff. The Facility captures and treats storm water on the premises through its wastewater treatment plant. The Permittee shall submit an update to this feasibility study as part of the submittal of the Report of Waste Discharge (ROWD) for the next permit renewal.

V. RECEIVING WATER LIMITATIONS

A. Surface Water Limitations

Receiving water limitations are based on water quality objectives contained in the Basin Plan and are a required part of this Order. The discharge shall not cause the following in the receiving water:

1. The natural receiving water temperature of all regional waters shall not be altered unless it can be demonstrated to the satisfaction of the Regional Water Board that such alteration in temperature does not adversely affect beneficial uses. Alterations that are allowed must meet the following requirements at the receiving water point of compliance: For waters designated with a warm freshwater habitat (WARM) beneficial use, water temperature shall not be altered by more than 5°F above the natural temperature. At no time shall these WARM-designated waters be raised above 80°F as a result of waste discharge.
2. The pH of inland surface waters shall not be depressed below 6.5 or raised above 8.5 as a result of wastes discharged. Ambient pH levels shall not be changed more than 0.5 units from natural conditions as a result of wastes discharged. Natural conditions shall be determined on a case-by-case basis.
3. The dissolved oxygen in the receiving water shall not be depressed below 5 mg/L as a result of the wastes discharged.
4. The total residual chlorine shall not exceed 0.1 mg/L in the receiving waters and shall not persist in the receiving water at any concentration that causes impairment of beneficial uses as a result of the wastes discharged.
5. The *Escherichia coli* (E. coli) concentration in the receiving water shall not exceed the following, as a result of wastes discharged:
 - a. Geometric Mean (Six-week rolling) Limits
 - i. E. coli shall not exceed 100 cfu/100 mL.
 - b. Statistical Threshold Value (STV)

- i. E. coli STV of 320 cfu/100 mL shall not be exceeded by more than 10 percent of the samples collected in a calendar month, calculated in a static manner.
6. Waters shall be free of changes in turbidity that cause nuisance or adversely affect beneficial uses. Increases in natural turbidity attributable to controllable water quality factors shall not exceed the following limits, as a result of wastes discharged:
 - a. Where natural turbidity is between 0 and 50 NTU, increases shall not exceed 20%.
 - b. Where natural turbidity is greater than 50 NTU, increases shall not exceed 10%.
7. The wastes discharged shall not produce concentrations of substances in the receiving water that are toxic to or cause detrimental physiological responses in human, animal, or aquatic life.
8. The wastes discharged shall not cause concentrations of contaminants to occur at levels that are harmful to human health in waters which are existing or potential sources of drinking water.
9. The concentrations of toxic pollutants in the water column, sediments, or biota shall not adversely affect beneficial uses as a result of the wastes discharged.
10. The wastes discharged shall not contain substances that result in increases in BOD, which adversely affect the beneficial uses of the receiving waters.
11. Waters discharged shall not contain biostimulatory substances in concentrations that promote aquatic growth to the extent that such growth causes nuisance or adversely affects beneficial uses.
12. The dissolved sulfide concentration of waters in and near sediments shall not be significantly increased above that present under natural conditions as a result of waters discharged.
13. The wastes discharged shall not cause the receiving waters to contain any substance in concentrations that adversely affect any designated beneficial use.
14. The wastes discharged shall not degrade surface water communities and populations, including vertebrate, invertebrate, and plant species.
15. The wastes discharged shall not alter the natural taste, odor, or color of fish, shellfish, or other surface water resources used for human consumption.
16. The wastes discharged shall not result in problems due to breeding of mosquitoes, gnats, black flies, midges, or other pests.
17. The wastes discharged shall not result in visible floating particulates, foams, or oil and grease in the receiving waters.
18. The wastes discharged shall not cause objectionable aquatic growths or degrade indigenous biota.

19. The wastes discharged shall not alter the color of the receiving waters; create a visual contrast with the natural appearance of the water; or cause aesthetically undesirable discoloration of the receiving waters.
20. The wastes discharged shall not contain any individual pesticide or combination of pesticides in concentrations that adversely affect beneficial uses of the receiving waters. There shall be no increase in pesticide concentrations found in bottom sediments or aquatic life as a result of the wastes discharged.
21. Ammonia shall not be present at levels that, when oxidized to nitrate, pose a threat to groundwater quality.
22. Chronic Toxicity Receiving Water Quality Objective
 - a. There shall be no chronic toxicity in ambient waters as a result of wastes discharged.
 - b. Receiving water and effluent toxicity testing shall be performed on the same day as close to concurrently as possible.
 - c. If the chronic toxicity median monthly threshold at the immediate downstream receiving water location is not met and the toxicity cannot be attributed to upstream toxicity, as assessed by the Permittee, then the Permittee shall initiate accelerated monitoring according to Attachment E – MRP section V.A.7.
 - d. If the chronic toxicity median monthly threshold of the receiving water at both upstream and downstream stations is not met, but the effluent chronic toxicity median monthly effluent limitation was met, then accelerated monitoring need not be implemented.

B. Groundwater Limitations

1. The discharge shall not cause the underlying groundwater to be degraded except as consistent with State Water Board Resolution No. 68-16. The discharge to groundwater shall not exceed WQOs, unreasonably affect beneficial uses, or cause a condition of pollution or nuisance.

VI. PROVISIONS

A. Standard Provisions

1. The Permittee shall comply with all Standard Provisions included in Attachment D.
2. **Regional Water Board Standard Provisions.** The Permittee shall comply with the following provisions. In the event that there is any conflict, duplication, or overlap between provisions specified by this Order, the more stringent provision shall apply:
 - a. Neither the treatment nor the discharge of pollutants shall create a pollution, contamination, or nuisance as defined by section 13050 of the CWC.
 - b. Odors, vectors, and other nuisances of sewage or sludge origin beyond the limits of the treatment plant site or the sewage collection system due to

improper operation of facilities, as determined by the Regional Water Board, are prohibited.

- c. All facilities used for collection, transport, treatment, or disposal of wastes shall be adequately protected against damage resulting from overflow, washout, or inundation from a storm or flood having a recurrence interval of once in 100 years.
- d. Collection, treatment, and disposal systems shall be operated in a manner that precludes or impedes public contact with wastewater.
- e. Collected screenings, sludges, and other solids removed from liquid wastes shall be disposed of in a manner approved by the Executive Officer of the Regional Water Board.
- f. The provisions of this order are severable. If any provision of this Order is found invalid, the remainder of this Order shall not be affected.
- g. Nothing in this permit shall be construed to preclude the institution of any legal action or relieve the Permittee from any responsibilities, liabilities or penalties established pursuant to any applicable state law or regulation under authority preserved by section 510 of the CWA, related to oil and hazardous substances liability.
- h. The Discharger must comply with the lawful requirements of municipalities, counties, drainage districts, and other local agencies regarding discharges of storm water to storm drain systems or other water courses under their jurisdiction, including applicable requirements in municipal storm water management programs developed to comply with NPDES permits issued by the Regional Water Board to local agencies.
- i. Discharge of wastes to any point other than specifically described in this Order is prohibited and constitutes a violation thereof.
- j. The Permittee shall comply with all applicable effluent limitations, national standards of performance, toxic effluent standards, and all federal regulations established pursuant to sections 301, 302, 303(d), 304, 306, 307, 316, 403, and 405 of the federal CWA and amendments thereto.
- k. These requirements do not exempt the operator of the waste disposal facility from compliance with any other laws, regulations, or ordinances which may be applicable; they do not legalize this waste disposal facility; and they leave unaffected any further restraints on the disposal of wastes at this site which may be contained in other statutes or required by other agencies.
- l. The Facility shall be protected to reduce infrastructure vulnerability to current and future impacts resulting from climate change, including but not limited to extreme wet weather events, flooding, storm surges, and projected sea level rise when the facility is located near the ocean or discharges to the ocean.
- m. Oil or oily material, chemicals, refuse, or other polluting materials shall not be stored or deposited in areas where they may be picked up by rainfall and carried off of the property and/or discharged to surface waters. Any such spill of such materials shall be contained and removed immediately.

- n. A copy of these waste discharge specifications shall be maintained at the discharge Facility so as to be available at all times to operating personnel.
- o. If there is any storage of hazardous or toxic materials or hydrocarbons at this Facility and if the Facility is not manned at all times, a 24-hour emergency response telephone number shall be prominently posted where it can easily be read from the outside.
- p. The Permittee shall file with the Regional Water Board a report of waste discharge at least 120 days before making any proposed change in the character, location or volume of the discharge.
- q. In the event of any change in name, ownership, or control of these waste disposal facilities, the Permittee shall notify the Regional Water Board of such change and shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be forwarded to the Regional Water Board, 30 days prior to taking effect.
- r. The discharge of any waste resulting from the combustion of toxic or hazardous wastes to any waste stream that ultimately discharges to waters of the United States is prohibited, unless specifically authorized elsewhere in this Order.
- s. The Permittee shall notify the Executive Officer in writing no later than 6 months prior to planned discharge of any chemical, other than the products previously reported to the Executive Officer, which may be toxic to aquatic life. Such notification shall include:
 - i. Name and general composition of the chemical,
 - ii. Frequency of use,
 - iii. Quantities to be used,
 - iv. Proposed discharge concentrations, and
 - v. USEPA registration number, if applicable.
- t. Failure to comply with provisions or requirements of this Order, or violation of other applicable laws or regulations governing discharges from this Facility, may subject the Permittee to administrative or civil liabilities, criminal penalties, and/or other enforcement remedies to ensure compliance. Additionally, certain violations may subject the Permittee to civil or criminal enforcement from appropriate local, state, or federal law enforcement entities.
- u. CWC section 13385(h)(i) requires the Regional Water Board to assess a mandatory minimum penalty of three-thousand dollars (\$3,000) for each serious violation. Pursuant to CWC section 13385(h)(2), a "serious violation" is defined as any waste discharge that violates the effluent limitations contained in the applicable waste discharge requirements for a Group II pollutant by 20 percent or more, or for a Group I pollutant by 40 percent or more. Appendix A of 40 CFR section 123.45 specifies the Group I and II pollutants. Pursuant to CWC section 13385.1(a)(1), a "serious violation" is

also defined as “a failure to file a discharge monitoring report required pursuant to section 13383 for each complete period of 30 days following the deadline for submitting the report, if the report is designed to ensure compliance with limitations contained in waste discharge requirements that contain effluent limitations.”

- v. CWC section 13385(i) requires the Regional Water Board to assess a mandatory minimum penalty of three-thousand dollars (\$3,000) for each violation whenever a person violates a waste discharge requirement effluent limitation in any period of six consecutive months, except that the requirement to assess the mandatory minimum penalty shall not be applicable to the first three violations within that time period.
- w. Pursuant to CWC section 13385.1(d), for the purposes of section 13385.1 and subdivisions (h), (i), and (j) of section 13385, “effluent limitation” means a numeric restriction or a numerically expressed narrative restriction, on the quantity, discharge rate, concentration, or toxicity units of a pollutant or pollutants that may be discharged from an authorized location. An effluent limitation may be final or interim, and may be expressed as a prohibition. An effluent limitation, for these purposes, does not include a receiving water limitation, a compliance schedule, or a best management practice.
- x. CWC section 13387(e) provides that any person who knowingly makes any false statement, representation, or certification in any record or other document submitted or required to be maintained under this order, including monitoring reports or reports of compliance or noncompliance, or who knowingly falsifies, tampers with, or renders inaccurate any monitoring device or method required to be maintained in this order shall be punished by a fine of not more than twenty-five thousand dollars (\$25,000), imprisonment pursuant to subdivision (h) of Section 1170 of the Penal Code for 16, 20, or 24 months, or by both that fine and imprisonment. For a subsequent conviction, such a person shall be punished by a fine of not more than twenty-five thousand dollars (\$25,000) per day of violation, by imprisonment pursuant to subdivision (h) of Section 1170 of the Penal Code for two, three, or four years, or by both that fine and imprisonment.
- y. In the event the Permittee does not comply or will be unable to comply for any reason, with any prohibition, effluent limitation, or receiving water limitation of this Order, the Permittee shall notify the Chief of the Watershed Regulatory Section at the Regional Water Board by telephone (213) 620-2083 or by fax at (213) 576-6660 within 24 hours of having knowledge of such noncompliance, and shall confirm this notification in writing to the Regional Water Board within five days, unless the Regional Water Board waives confirmation. The written notification shall state the nature, time, duration, and cause of noncompliance, and shall describe the measures being taken to remedy the current noncompliance and, prevent recurrence including, where applicable, a schedule of implementation. The written notification shall also be submitted via email with reference to CI-3021 to losangeles@waterboards.ca.gov. Other noncompliance requires written notification as above at the time of the normal monitoring report.

B. Monitoring and Reporting Program (MRP) Requirements

The Permittee shall comply with the MRP, and future revisions thereto, in Attachment E.

C. Special Provisions

1. Reopener Provisions

- a. This Order may be modified, revoked and reissued, or terminated for cause, including, but not limited to:
 - i. Violation of any term or condition contained in this Order;
 - ii. Obtaining this Order by misrepresentation, or by failure to disclose fully all relevant facts; or
 - iii. A change in any condition that requires either a temporary or permanent reduction or elimination of the authorized discharge.

The filing of a request by the Permittee for an Order modification, revocation, and issuance or termination, or a notification of planned changes or anticipated noncompliance does not stay any condition of this Order.

- b. This Order may be reopened for modification, or revocation and reissuance, as a result of the detection of a reportable priority pollutant generated by special conditions included in this Order. These special conditions may be, but are not limited to, fish tissue sampling, whole effluent toxicity testing, monitoring of internal waste stream(s), and monitoring for surrogate parameters. Additional requirements may be included in this Order as a result of the special condition monitoring data.
- c. This Order may be modified, in accordance with the provisions set forth in title 40 of the Code of Federal Regulations (40 CFR) parts 122 and 124 to include requirements for the implementation of a watershed protection management approach.
- d. The Board may modify, or revoke and reissue this Order if present or future investigations demonstrate that the discharge(s) governed by this Order have or will have a reasonable potential to cause or contribute to adverse impacts on water quality or beneficial uses of the receiving waters.
- e. This Order may also be modified, revoked, and reissued or terminated in accordance with the provisions of 40 CFR sections 122.44, 122.62 to 122.64, 125.62, and 125.64. Causes for taking such actions include, but are not limited to, failure to comply with any condition of this Order, endangerment to human health or the environment resulting from the permitted activity, or acquisition of newly obtained information which would have justified the application of different conditions if known at the time of Order adoption. The filing of a request by the Permittee for an Order modification, revocation and issuance, or termination, or a notification of planned changes or anticipated noncompliance does not stay any condition of this Order.

- f. This Order may be modified, in accordance with the provisions set forth in 40 CFR parts 122 to 124, to include new minimum levels (MLs).
- g. If an applicable toxic effluent standard or prohibition (including any schedule of compliance specified in such effluent standard or prohibition) is promulgated under section 307(a) of the CWA for a toxic pollutant and that standard or prohibition is more stringent than any limitation on the pollutant in this Order, the Regional Water Board may institute proceedings under these regulations to modify or revoke and reissue the Orders to conform to the toxic effluent standard or prohibition.
- h. If more stringent applicable water quality standards are promulgated or approved pursuant to section 303 of the CWA, or amendments, thereto, the Regional Water Board will revise and modify this Order in accordance with such standards.
- i. This Order may be reopened and modified, to add or revise effluent limitations as a result of future Basin Plan Amendments, such as an update of a water quality objective, or the adoption/revision of any of the Calleguas Creek Watershed TMDLs.
- j. This Order may be reopened to modify the total dissolved solids, sulfate, chloride, and boron final effluent limitations to include an adjustment factor (AF), following approval of an AF for the Facility by the Regional Water Board.
- k. This Order may be reopened and modified, to revise effluent limitations as a result of the delisting of a pollutant from the CWA section 303(d) list.
- l. This Order may be reopened and modified to revise any and all of the chronic toxicity testing provisions and effluent limitations, to the extent necessary, to be consistent with any Toxicity Provisions that are subsequently adopted by the State Water Board promptly after USEPA approval of such Provisions.
- m. This Order will be reopened and modified to the extent necessary, to be consistent with new policies, new state-wide plans, new laws, or new regulations.

2. Special Studies, Technical Reports and Additional Monitoring Requirements

a. Calleguas Creek TMDL Monitoring Requirements

The POTWs within the Calleguas Creek Watershed (CCW) have developed a watershed monitoring program to implement the requirements for monitoring, conducting special studies, and implementing actions to reduce discharges of pollutants covered by the TMDLs. This watershed monitoring program has been approved by the Regional Water Board. The responsible parties to the CCW TMDLs have signed a Memorandum of Agreement to jointly fund and complete the implementation of the TMDL Calleguas Creek Watershed Monitoring Program (CCWTMP), which began in August 2008. The CCWTMP was created to better facilitate a coordinated monitoring effort where multiple TMDL monitoring requirements could be addressed via a single program that would carry out and manage all aspects of the

monitoring activities. This monitoring program has been developed to easily integrate new TMDL monitoring efforts as TMDLs are adopted and/or special study monitoring efforts are required.

The CCWTMP Annual Monitoring Report has been submitted since 2009. The annual monitoring reports summarize the monitoring for five of the six TMDLs currently effective in the CCW. These TMDLs include nitrogen compounds and related effects, toxicity, organochlorine pesticides and PCBs, metals and selenium, and salts. A separate annual report is submitted for the trash TMDL. These reports are submitted to the Regional Water Board TMDL staff for review.

Since 2009, all sampling has followed the Standard Operating Procedures outlined in the Executive Officer approved *Calleguas Creek Watershed Management Plan Quality Assurance Project Plan (QAPP)*, with the following exception: the methods for the salts compliance monitoring that began on September 9, 2012, are not currently contained in the QAPP but were described in detail in the final Salts Monitoring Approach submitted to the Regional Water Board on June 29, 2012. The QAPP revision is in the process of being revised to incorporate the methods, sites, and schedule for compliance salts monitoring described in the final approach document.

In addition, the majority of the TMDLs include requirements for monitoring, conducting special studies, and implementing actions to reduce discharges of pollutants covered by the TMDLs. Many of these activities overlap and provide benefits for numerous TMDLs in the watershed. The CCWTMP annual reports included an appendix that summarizes workplan and study submittal dates, dates of responses to comments received by the Regional Water Board, and actions that have been taken to reduce pollutant discharges to the waterbodies. Additionally, the report provides a mechanism for providing the Regional Water Board with required progress reports for some of the TMDLs.

b. Treatment Plant Capacity

The Permittee shall submit a written report to the Executive Officer of the Regional Water Board within 90 days after the “30-day (monthly) average” daily dry-weather flow equals or exceeds 75 percent of the design capacity of waste treatment and/or disposal facilities. The Permittee's senior administrative officer shall sign a letter, which transmits that report and certify that the Permittee's policy-making body is adequately informed of the report's contents. The report shall include the following:

- i. The average daily flow for the month, the date on which the peak flow occurred, the rate of that peak flow, and the total flow for the day;
- ii. The best estimate of when the monthly average daily dry-weather flow rate will equal or exceed the design capacity of the facilities; and,
- iii. A schedule for studies, design, and other steps needed to provide additional capacity for waste treatment and/or disposal facilities before the waste flow rate equals the capacity of present units.

This requirement is applicable in the case where the facility has not reached 75 percent of capacity as of the effective date of this Order. If the facility has reached 75 percent of capacity by that date but has not previously submitted such report, such a report shall be filed within 90 days of the issuance of this Order.

3. Best Management Practices and Pollution Prevention

- a. **Storm Water Pollution Prevention Plan (SWPPP) – (Not Applicable)** The facility captures and treats storm water collected on the premises. The captured storm water is mixed with the domestic wastewater for treatment at the facility.

- b. **Spill Clean-up Contingency Plan (SCCP)**

Within 90 days of the effective date of this Order, the Permittee is required to submit a SCCP, which describes the activities and protocols to address clean-up of spills, overflows, and bypasses of untreated or partially treated wastewater from the Permittee's collection system or treatment facilities that reach water bodies, including dry channels and beach sands. At a minimum, the plan shall include sections on spill clean-up and containment measures, public notification, and monitoring. The Permittee shall review and amend the plan as appropriate after each spill from the Facility or in the service area of the Facility. The Permittee shall include a discussion in the annual summary report of any modifications to the Plan and the application of the Plan to all spills during the year.

- c. **Pollutant Minimization Program (PMP)**

Reporting protocols in MRP section X.C.4 describe sample results that are to be reported as Detected but Not Quantified (DNQ) or Not Detected (ND). Definitions for a reported Minimum Level (ML) and Method Detection Limit (MDL) are provided in Attachment A. These reporting protocols and definitions are used in determining the need to conduct a PMP as follows:

The Permittee shall develop and conduct a PMP as further described below when there is evidence (e.g., sample results reported as DNQ when the effluent limitation is less than the MDL; sample results from analytical methods more sensitive than those methods required by this Order; presence of whole effluent toxicity; health advisories for fish consumption; or, results of benthic or aquatic organism tissue sampling) that a pollutant is present in the effluent above an effluent limitation and either of the following is true:

The concentration of the pollutant is reported as DNQ and the effluent limitation is less than the reported ML; or,

The concentration of the pollutant is reported as ND and the effluent limitation is less than the MDL, using definitions described in Attachment A and reporting protocols described in the MRP.

The goal of the PMP shall be to reduce all potential sources of a pollutant through pollutant minimization (control) strategies, including pollution prevention measures as appropriate, to maintain the effluent concentration at or below the effluent limitation. Pollution prevention measures may be particularly appropriate for persistent bioaccumulative priority pollutants where there is evidence that beneficial uses are being impacted. The Regional Water Board may consider cost-effectiveness when establishing the requirements of a PMP. The completion and implementation of a Pollution Prevention Plan (PPP), if required pursuant to CWC section 13263.3(d), shall be considered to fulfill the PMP requirements.

The PMP shall include, but not be limited to, the following actions and submittals acceptable to the Regional Water Board:

An annual review and semi-annual monitoring of potential sources of the reportable pollutant(s), which may include fish tissue monitoring and other bio-uptake sampling.

Quarterly monitoring for the reportable pollutant(s) in the influent to the wastewater treatment system.

Submittal of a control strategy designed to proceed toward the goal of maintaining concentrations of the reportable pollutant(s) in the effluent at or below the effluent limitation.

Implementation of appropriate cost-effective control measures for the reportable pollutant(s), consistent with the control strategy.

An annual status report that shall be sent to the Regional Water Board including:

- i. All PMP monitoring results for the previous year.
- ii. A list of potential sources of the reportable pollutant(s).
- iii. A summary of all actions undertaken pursuant to the control strategy.
- iv. A description of actions to be taken in the following year.

4. Construction, Operation and Maintenance Specifications

- a. **Certified Wastewater Treatment Plant Operator:** Wastewater treatment facilities subject to this Order shall be supervised and operated by persons possessing certificates of appropriate grade pursuant to California Code of Regulations (CCR), title 23, division 3, chapter 26 (CWC sections 13625 – 13633).
- b. **Climate Change Effects Vulnerability Assessment and Mitigation Plan:** The Permittee shall consider the impacts of climate change as they affect the operation of the treatment facility due to flooding, wildfire, or other climate-related changes. The Permittee shall develop a Climate Change Effects Vulnerability Assessment and Mitigation Plan (Climate Change Plan) to assess and manage climate change-related effects that may impact the wastewater treatment facility's operation, water supplies, its collection

system, and water quality, including any projected changes to the influent water temperature and pollutant concentrations, and beneficial uses. For facilities that discharge to the ocean including desalination plants, the Climate Change Plan shall also include the impacts from sea level rise. The Climate Change Plan is due 12 months after adoption of this Order.

- c. **Alternate Power Source:** The Permittee shall maintain in good working order a sufficient alternate power source for operating the wastewater treatment and disposal facilities. All equipment shall be located and secured to minimize failure due to moisture, liquid spray, flooding, wildfires, and other physical phenomena. The alternate power source shall be designed to allow inspection and maintenance and shall provide for periodic testing. If such alternate power source is not in existence, the discharger shall halt, reduce, or otherwise control all discharges upon the reduction, loss, or failure of the primary source of power. The Permittee shall provide standby or emergency power facilities and/or storage capacity or other means so that in the event of plant upset or outage due to power failure or other cause, discharge of raw or inadequately treated sewage does not occur.

5. Special Provisions for Municipal Facilities (POTWs Only)

a. Biosolids Disposal Requirements

- i. All biosolids generated at the wastewater treatment plant must be disposed of, treated, or applied to land in accordance with federal regulations contained in 40 CFR part 503. These requirements are enforced by USEPA. 40 CFR part 503 requirements identified in Attachment H are for information only and are not regulated by this Order.
- ii. The Permittee is separately required to comply with the requirements in State Water Board Order No. 2004-0012-DWQ, General WDRs for the Discharge of Biosolids to Land for Use as a Soil Amendment in Agricultural, Silvicultural, Horticultural and Land Reclamation Activities for those sites receiving the Permittee's biosolids which a Regional Water Board has placed under this general order, and with the requirements in individual WDRs issued by a Regional Water Board for sites receiving the Permittee's biosolids.

b. Pretreatment Requirements

- i. The Permittee has developed and implemented an approved Pretreatment Program that was submitted to the Regional Water Board. This Order requires implementation of the approved Pretreatment Program. Any violation of the Pretreatment Program will be considered a violation of this Order.
- ii. On March 21, 2011, the City Council of the City of Simi Valley added Chapter 13 to Title 6 of the Simi Valley Municipal Code regulating sewer use by adopting Ordinance No. 1170, providing the Simi Valley WQCP the legal authority to continue to implement and enforce its Pretreatment

Program. On March 21, 2011, based on the conditions of Chapter 13 of Title 6 of the Simi Valley Municipal Code, the City Council of the City of Simi Valley approved the local limits through Ordinance No. 1170 and incorporated them into its Pretreatment Program. Based on the legal authority provided by the Simi Valley Municipal Code, the Simi Valley WQCP prepared an Enforcement Response Plan on February 14, 2011, that is part of the City's Pretreatment Program.

- iii. Any change to the program shall be reported to the Regional Water Board in writing and shall not become effective until approved by the Executive Officer in accordance with procedures established in 40 CFR section 403.18.
- iv. Applications for renewal or modification of this Order must contain information about industrial discharges to the POTW pursuant to 40 CFR section 122.21(j)(6). Pursuant to 40 CFR section 122.42(b) and provision VII.A of Attachment D, Standard Provisions, of this Order, the Permittee shall provide adequate notice of any new introduction of pollutants or substantial change in the volume or character of pollutants from industrial discharges which were not included in the permit application. Pursuant to 40 CFR section 122.44(j)(1), the Permittee shall annually identify and report, in terms of character and volume of pollutants, any Significant Industrial Users discharging to the POTW subject to Pretreatment Standards under section 307(b) of the CWA and 40 CFR part 403.
- v. The Permittee shall comply with Attachment I – Pretreatment Reporting Requirements.

c. Collection System Requirements

The Permittee's collection system is part of the system that is subject to this Order. As such, the Permittee must properly operate and maintain its collection system (40 CFR section 122.41(e)). The Permittee must report any non-compliance (40 CFR section 122.41(l)(6) and (7)) and mitigate any discharge from the collection system in violation of this Order (40 CFR section 122.41(d)). See the Order at Attachment D, subsections I.D, V.E, V.H, and I.C., and the following section of this Order.

d. Filter Bypass

Conditions pertaining to bypass are contained in Attachment D, Section I. Standard Provisions – Permit Compliance, subsection G. The bypass or overflow of untreated or partially treated wastewater to waters of the State is prohibited, except as allowed under conditions stated in 40 CFR section 122.41(m) and (n). During periods of elevated, wet weather flows, a portion of the secondary treated wastewater is diverted around the tertiary filters. These anticipated discharges are approved under the bypass conditions when the resulting combined discharge of fully treated (tertiary) and partially treated (secondary) wastewater complies with the effluent and receiving water limitations in this Order.

6. Spill Reporting Requirements

a. Initial Notification

Although State and Regional Water Board staff do not have duties as first responders, this requirement is an appropriate mechanism to ensure that the agencies that do have first responder duties are notified in a timely manner in order to protect public health and beneficial uses. For certain spills, overflows and bypasses, the Permittee shall make notifications as required below:

- i. In accordance with the requirements of Health and Safety Code section 5411.5, the Permittee shall provide notification to the local health officer or the director of environmental health with jurisdiction over the affected water body of any unauthorized release of sewage or other waste that causes, or probably will cause, a discharge to any waters of the state as soon as possible, but no later than **two hours** after becoming aware of the release.
- ii. In accordance with the requirements of CWC section 13271, the Permittee shall provide notification to the California Office of Emergency Services(OES) of the release of reportable amounts of hazardous substances or sewage that causes, or probably will cause, a discharge to any waters of the state as soon as possible, but not later than two hours after becoming aware of the release. The CCR, Title 23, section 2250, defines a reportable amount of sewage as being 1,000 gallons. The phone number for reporting these releases to the OES is (800) 852-7550. In addition, the Permittee shall notify other interested persons of any such sewage spill by maintaining an email list of those interested persons that have requested such notification.
- iii. The Permittee shall notify the Regional Water Board of any unauthorized release of sewage from its POTW that causes, or probably will cause, a discharge to a water of the state as soon as possible, but not later than **two hours** after becoming aware of the release. This initial notification does not need to be made if the Permittee has notified OES and the local health officer or the director of environmental health with jurisdiction over the affected waterbody. The phone number for reporting these releases of sewage to the Regional Water Board is (213) 576-6657. The phone numbers for after hours and weekend reporting of releases of sewage to the Regional Water Board are (213) 305-2284 and (213) 305-2253.

At a minimum, the following information shall be provided to the Regional Water Board:

- (a) The location, date, and time of the release.
- (b) The water body that received or will receive the discharge.
- (c) An estimate of the amount of sewage or other waste released and the amount that reached a surface water at the time of notification.

- (d) If ongoing, the estimated flow rate of the release at the time of the notification.
- (e) The name, organization, phone number and email address of the reporting representative.

b. Monitoring

For spills, overflows and bypasses reported under section VI.C.6.a, the Permittee shall monitor as required below:

- i. To define the geographical extent of the spill's impact, the Permittee shall obtain grab samples for all spills, overflows or bypasses of any volume that reach any waters of the state (including surface and ground waters). If a grab sample cannot be obtained due to accessibility or safety concerns that cannot be addressed with the appropriate personal protective equipment or following proper sampling procedures, the sample shall be obtained as soon as it becomes safe to do so. For spills that reach surface freshwaters, the Permittee shall monitor for *E. coli* density. For spills that reach marine water, the Permittee shall monitor for total coliform, fecal coliform, and enterococcus density. The Permittee shall also analyze the samples for relevant pollutants of concern, upstream and downstream of the point of entry of the spill (if feasible, accessible, and safe). This monitoring shall be done on a daily basis from the time the spill is known until the results of two consecutive sets of bacteriological monitoring indicate the return to the background level or the County Department of Public Health authorizes cessation of monitoring.

c. Reporting

The initial notification required under section VI.C.6.a shall be followed by:

- i. As soon as possible, but not later than twenty-four hours after becoming aware of an unauthorized discharge of sewage or other waste from its wastewater treatment plant to a water of the state, the Permittee shall submit a statement to the Regional Water Board by email at augustine.anijelo@waterboards.ca.gov. If the discharge is 1,000 gallons or more, this statement shall certify that OES has been notified of the discharge in accordance with CWC section 13271. The statement shall also certify that the local health officer or director of environmental health with jurisdiction over the affected water bodies has been notified of the discharge in accordance with Health and Safety Code section 5411.5. The statement shall also include at a minimum the following information:
 - (a) Agency, NPDES No., Order No., and MRP CI No., if applicable.
 - (b) The location, date, and time of the discharge.
 - (c) The water body that received the discharge.
 - (d) A description of the level of treatment of the sewage or other waste discharged.

- (e) An initial estimate of the amount of sewage or other waste released and the amount that reached a surface water.
 - (f) The OES control number and the date and time that notification of the incident was provided to OES.
 - (g) The name of the local health officer or director of environmental health representative notified (if contacted directly); the date and time of notification; and the method of notification (e.g., phone, fax, email).
- ii. A written preliminary report five working days after disclosure of the incident is required. Submission to the Regional Water Board of the California Integrated Water Quality System (CIWQS) Sanitary Sewer Overflow (SSO) event number shall satisfy this requirement. Within 30 days after submitting the preliminary report, the Permittee shall submit the final written report to this Regional Water Board. (A copy of the final written report, for a given incident, already submitted pursuant to a statewide General WDRs for Wastewater Collection System Agencies (SSO WDR), may be submitted to the Regional Water Board to satisfy this requirement.) The written report shall document the information required in paragraph d below, monitoring results and any other information required in provisions of the Standard Provisions document including corrective measures implemented or proposed to be implemented to prevent/minimize future occurrences. The Executive Officer for just cause can grant an extension for submittal of the final written report.
 - iii. The Permittee shall include a certification in the annual summary report (due according to the schedule in the MRP) that states that the sewer system emergency equipment, including alarm systems, backup pumps, standby power generators, and other critical emergency pump station components were maintained and tested in accordance with the Permittee's preventive maintenance plan. Any deviations from or modifications to the plan shall be discussed.

d. **Records**

The Permittee shall develop and maintain a record of all spills, overflows or bypasses of raw or partially treated sewage from its collection system or treatment plant. This record shall be made available to the Regional Water Board upon request and a spill summary shall be included in the annual summary report. The records shall contain:

- i. The date and time of each spill, overflow, or bypass.
- ii. The location of each spill, overflow, or bypass.
- iii. The estimated volume of each spill, overflow, and bypass including gross volume, amount recovered and amount not recovered, monitoring results as required by section VI.C.6.b.
- iv. The cause of each spill, overflow, or bypass.

- v. Whether each spill, overflow, or bypass entered a receiving water and, if so, the name of the water body and whether it entered via storm drains or other man-made conveyances.
- vi. Any mitigation measures implemented.
- vii. Any corrective measures implemented or proposed to be implemented to prevent/minimize future occurrences.
- viii. The mandatory information included in SSO online reporting for finalizing and certifying the SSO report for each spill, overflow, or bypass under the SSO WDR.

e. Activities Coordination

Although not required by this Order, Regional Water Board also expects the CCW watershed group to continue to work together regarding activities related to desalters, water uses, and the use of the brine line in order to comply with the requirements of this Order, in addition to meeting the deadlines in the Salts TMDL.

f. Consistency with SSO WDRs

The CWA prohibits the discharge of pollutants from point sources to surface waters of the United States unless authorized under an NPDES permit. (33 United States Code sections 1311, 1342). The Permittee must comply with State Water Board Order No. 2006-0003-DWQ, Statewide *General Waste Discharge Requirements for Sanitary Sewer Systems*, (SSO WDR) as amended by State Water Board Order No. WQ 2013-0058-exec and any subsequent order updating these requirements. These statewide WDRs require public agencies that own or operate sanitary sewer systems with greater than one mile of sewer lines to enroll for coverage and comply with requirements, to develop and implement sewer system management plans, and report all SSOs to the State Water Board's online SSOs database. The Permittee enrolled in the SSO WDRs in 2006, and the collection systems of the Permittee are covered under the SSO WDRs. The Permittee must properly operate and maintain its collection system (40 CFR section 122.41 (e)), report any non-compliance (40 CFR section 122.41(1)(6) and (7)), and mitigate any discharge from the collection system in violation of this NPDES permit (40 CFR section 122.41(d)).

The requirements contained in this Order in sections VI.C.3.b (SCCP Plan section), VI.C.4 (Construction, Operation and Maintenance Specifications section), and VI.C.6 (Spill Reporting Requirements section) are intended to be consistent with the requirements of the SSO WDR. The Regional Water Board recognizes that there may be some overlap between these NPDES permit provisions and SSO WDR requirements, related to the collection systems. The requirements of the SSO WDR are considered the minimum thresholds (see finding 11 of State Water Board Order No. 2006-0003-DWQ). To encourage efficiency, the Regional Water Board will accept the documentation prepared by the Permittees under the SSO WDR for compliance purposes as satisfying the requirements in sections VI.C.3.b,

VI.C.4, and VI.C.6 provided the more stringent provisions contained in this NPDES permit are also addressed. Pursuant to SSO WDR, section D, provision 2(iii) and (iv), the provisions of this NPDES permit supersede the SSO WDR, for all purposes, including enforcement, to the extent the requirements may be deemed duplicative.

7. Compliance Schedules – Not Applicable

VII. COMPLIANCE DETERMINATION

Compliance with the effluent limitations contained in section IV of this Order will be determined as specified below:

A. General

Compliance with effluent limitations for priority pollutants shall be determined using sample reporting protocols defined in the MRP and Attachment A of this Order. For purposes of reporting and administrative enforcement by the Regional and State Water Boards, the Permittee shall be deemed out of compliance with effluent limitations if the concentration of the priority pollutant in the monitoring sample is greater than the effluent limitation and greater than or equal to the reporting level (RL).

B. Multiple Sample Data

When determining compliance with a measure of central tendency (arithmetic mean, geometric mean, median, etc.) of multiple sample analyses and the data set contains one or more reported determinations of DNQ or ND. In those cases, the Permittee shall compute the median in place of the arithmetic mean in accordance with the following procedure:

1. The data set shall be ranked from low to high, ranking the reported ND determinations lowest, DNQ determinations next, followed by quantified values (if any). The order of the individual ND or DNQ determinations is unimportant.
2. The median value of the data set shall be determined. If the data set has an odd number of data points, then the median is the middle value. If the data set has an even number of data points, then the median is the average of the two values around the middle unless one or both of the points are ND or DNQ, in which case the median value shall be the lower of the two data points where DNQ is lower than a value and ND is lower than DNQ.

C. Average Monthly Effluent Limitation (AMEL)

If the average (or when applicable, the median determined by subsection B above for multiple sample data) of daily discharges over a calendar month exceeds the AMEL for a given parameter, this will represent a single violation for the purpose of calculating mandatory minimum penalties, though the Permittee may be considered out of compliance for each day of that month for that parameter (e.g., resulting in 31 days of non-compliance in a 31-day month) in cases where discretionary administrative civil liabilities are appropriate. If only a single sample is taken during

the calendar month and the analytical result for that sample exceeds the AMEL, the Permittee may be considered out of compliance for that calendar month. The Permittee will only be considered out of compliance for days when the discharge occurs. For any one calendar month during which no sample (daily discharge) is taken, no compliance determination can be made for that calendar month with respect to the AMEL.

If the analytical result of a single sample, monitored monthly, quarterly, semiannually, or annually, does not exceed the AMEL for a given parameter, the Permittee will have demonstrated compliance with the AMEL for each day of that month for that parameter.

If the analytical result of any single sample, monitored monthly, quarterly, semiannually, or annually, exceeds the AMEL for any parameter, the Permittee may collect up to four additional samples within the same calendar month. All analytical results shall be reported in the monitoring report for that month. The concentration of pollutant (an arithmetic mean or a median) in these samples estimated from the "Multiple Sample Data Reduction" section above, will be used for compliance determination.

In the event of noncompliance with an AMEL, the sampling frequency for that parameter shall be increased to weekly and shall continue at this level until compliance with the AMEL has been demonstrated.

D. Average Weekly Effluent Limitation (AWEL)

If the average of daily discharges over a calendar week exceeds the AWEL for a given parameter, an alleged violation will be flagged and the Permittee will be considered out of compliance for each day of that week for that parameter, resulting in 7 days of non-compliance. The average of daily discharges over the calendar week that exceeds the AWEL for a parameter will be considered out of compliance for that week only. If only a single sample is taken during the calendar week and the analytical result for that sample exceeds the AWEL, the Permittee will be considered out of compliance for that calendar week. For any one calendar week during which no sample (daily discharge) is taken, no compliance determination can be made for that calendar week with respect to the AWEL.

A calendar week will begin on Sunday and end on Saturday. Partial calendar weeks at the end of calendar month will be carried forward to the next month in order to calculate and report a consecutive seven-day average value on Saturday.

E. Maximum Daily Effluent Limitation (MDEL)

If a daily discharge on a calendar day exceeds the MDEL for a given parameter, an alleged violation will be flagged and the Permittee will be considered out of compliance for that day for that parameter. If no sample (daily discharge) is taken over a calendar day, no compliance determination can be made for that day with respect to effluent violation determination, but compliance determination can be made for that day with respect to reporting violation determination.

F. Instantaneous Minimum Effluent Limitation

If the analytical result of a single grab sample is lower than the instantaneous minimum effluent limitation for a parameter, a potential violation will be flagged and the Permittee will be considered out of compliance for that parameter for that single sample. Non-compliance for each sample will be considered separately (e.g., the results of two grab samples taken within a calendar day that both are lower than the instantaneous minimum effluent limitation would result in two instances of non-compliance with the instantaneous minimum effluent limitation).

G. Instantaneous Maximum Effluent Limitation

If the analytical result of a single grab sample is higher than the instantaneous maximum effluent limitation for a parameter, a potential violation will be flagged and the Permittee will be considered out of compliance for that parameter for that single sample. Non-compliance for each sample will be considered separately (e.g., the results of two grab samples taken within a calendar day that both exceed the instantaneous maximum effluent limitation would result in two instances of non-compliance with the instantaneous maximum effluent limitation).

H. Six-month Median Effluent Limitation

If the median of daily discharges over any 180-day period exceeds the six-month median effluent limitation for a given parameter, a potential violation will be flagged and the Permittee will be considered out of compliance for each day of that 180-day period for that parameter. The next assessment of compliance will occur after the next sample is taken. If only a single sample is taken during a given 180-day period and the analytical result for that sample exceeds the six-month median, the Permittee will be considered out of compliance for the 180-day period. For any 180-period during which no sample is taken, no compliance determination can be made for the six-month median effluent limitation.

I. Median Monthly Effluent Limitation (MMEL)

If the median of daily discharges over a calendar month exceeds the MMEL for a given parameter, an alleged violation will be flagged and the Permittee will be considered out of compliance for each day of that month for that parameter (e.g., resulting in 31 days of non-compliance in a 31-day month). However, an alleged violation of the MMEL will be considered one violation for the purpose of assessing State mandatory minimum penalties. If no sample (daily discharge) is taken over a calendar month, no compliance determination can be made for that month with respect to effluent violation determination, but compliance determination can be made for that month with respect to reporting violation determination.

J. Chronic Toxicity

The discharge is subject to determination of "Pass" or "Fail" and "Percent Effect" from a chronic toxicity test using the Test of Significant Toxicity (TST) statistical t-test approach described in *National Pollutant Discharge Elimination System Test of Significant Toxicity Implementation Document* (EPA 833-R-10-003, 2010), Appendix A, Figure A-1, Table A-1, and Appendix B, Table B-1. The null hypothesis (Ho) for the

TST statistical approach is: Mean discharge In-stream Waste Concentration (IWC) response $\leq 0.75 \times$ Mean control response. A test result that rejects this null hypothesis is reported as "Pass." A test result that does not reject this null hypothesis is reported as "Fail." The relative "Percent Effect" at the discharge IWC is defined and reported as: $((\text{Mean control response} - \text{Mean discharge IWC response}) \div \text{Mean control response}) \times 100$. This is a t-test (formally Student's t-Test), a statistical analysis comparing two sets of replicate observations - in the case of Whole Effluent Toxicity (WET), only two test concentrations (i.e., a control and IWC). The purpose of this statistical test is to determine if the means of the two sets of observations are different (i.e., if the IWC or receiving water concentration differs from the control (the test result is "Pass" or "Fail")). The Welch's t-test employed by the TST statistical approach is an adaptation of Student's t-test and is used with two samples having unequal variances.

The Maximum Daily Effluent Limitation (MDEL) for chronic toxicity is exceeded and a violation will be flagged when a chronic toxicity test, analyzed using the TST statistical approach, results in "Fail" and the "Percent Effect" is ≥ 0.50 .

The MMEL for chronic toxicity is exceeded and a violation will be flagged when the median of no more than three independent chronic toxicity tests, conducted within the same calendar month and analyzed using the TST statistical approach, results in "Fail." The MMEL for chronic toxicity shall only apply when there is a discharge on more than one day in a calendar month period. During such calendar months, up to three independent toxicity tests may be conducted when one toxicity test results in "Fail."

The chronic toxicity MDEL and MMEL are set at the IWC for the discharge (100% effluent) and expressed in units of the TST statistical approach ("Pass" or "Fail", "Percent Effect"). All NPDES effluent compliance monitoring for the chronic toxicity MDEL and MMEL shall be reported using only the 100% effluent concentration and negative control, expressed in units of the TST. The TST hypothesis (H_0) (see above) is statistically analyzed using the IWC and a negative control. Effluent toxicity tests shall be run using a multi-concentration test design when required by *Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Freshwater Organisms* (USEPA 2002, EPA-821-R-02-013). The Regional Water Board's review of reported toxicity test results will include review of concentration-response patterns as appropriate (see Fact Sheet discussion at IV.C.5). As described in the bioassay laboratory audit correspondence from the State Water Resources Control Board dated August 7, 2014, and from the USEPA dated December 24, 2013, the Percent Minimum Significant Difference (PMSD) criteria only apply to compliance reporting for the No Observable Effect Concentration (NOEC) and the sublethal statistical endpoints of the NOEC, and therefore are not used to interpret TST results. Standard Operating Procedures used by the toxicity testing laboratory to identify and report valid, invalid, anomalous, or inconclusive effluent (and receiving water) toxicity test measurement results from the TST statistical approach, including those that incorporate a consideration of concentration-response patterns, must be submitted to the Regional Water Board (40 CFR section 122.41(h)). The Regional Water Board will make a final determination as to whether a toxicity test result is valid, and may consult with the Permittee, the USEPA, the State Water Board's Quality Assurance Officer, or the State Water Board's Environmental Laboratory Accreditation Program (ELAP) as

needed. The Board may consider the results of any TIE/TRE studies in an enforcement action.

K. Percent Removal

The average monthly percent removal is the removal efficiency expressed in percentage across a treatment plant for a given pollutant parameter, as determined from the 30-day average values of pollutant concentrations (C in mg/L) of influent and effluent samples collected at about the same time using the following equation:

$$\text{Percent Removal (\%)} = [1 - (C_{\text{Effluent}}/C_{\text{Influent}})] \times 100\%$$

When preferred, the Permittee may substitute mass loadings and mass emissions for the concentrations.

L. Mass and Concentration Limitations

Compliance with mass and concentration effluent limitations for the same parameter shall be determined separately with their respective limitations. When the concentration of a constituent in an effluent sample is determined to be ND or DNQ, the corresponding mass emission rate determined from that sample concentration shall also be reported as ND or DNQ.

M. Compliance with single constituent effluent limitations

Permittees may be considered out of compliance with the effluent limitation if the concentration of the pollutant (see section B “Multiple Sample Data Reduction” above) in the monitoring sample is greater than the effluent limitation and greater than or equal to the RL.

N. Compliance with effluent limitations expressed as a sum of several constituents

Permittees are out of compliance with an effluent limitation which applies to the sum of a group of chemicals (e.g., PCB's) if the sum of the individual pollutant concentrations is greater than the effluent limitation. Individual pollutants of the group will be considered to have a concentration of zero if the constituent is reported as ND or DNQ.

O. Compliance with Calleguas Creek Salts TMDL effluent limitations

Simi Valley WQCP discharges to Arroyo Simi, Reach 7 of the Calleguas Creek. The Calleguas Creek and its tributaries are on the CWA section 303(d) list as impaired for TDS, sulfate, chloride, and boron. For this discharge, the Calleguas Creek Salts TMDL has established seasonal WLAs for TDS, sulfate, chloride, and boron. Federal regulations require that NPDES permits incorporate water quality based effluent limitations (WQBELs) consistent with the requirements and assumptions of any available WLAs.

WLAs established for Simi Valley WQCP in the *Salts TMDL* will be implemented through final effluent limitations contained in the NPDES permit. No interim effluent

limitations are provided. Compliance will be determined through monitoring of final effluent discharge as defined in this NPDES permit. The effluent limitations will be applied as end-of pipe mass-based and concentration-based monthly average effluent limitations. A daily maximum effluent limit is not required because salts are not expected to have an immediate or acute effect on the beneficial uses. Compliance with the minimum salt export requirements for Simi Valley WQCP will be based on the salt export from the subwatershed to which they discharge. The mechanisms for meeting the minimum salt export requirements and for monitoring progress towards meeting those requirements shall be included in the monitoring program work plan submitted for the approval of the Executive Officer.

Simi Valley WQCP's mass-based WLAs are calculated as the POTW effluent flow rate multiplied by the water quality objective and include a mass-based adjustment factor (AF) that is subtracted from the product of the flow-rate and the water quality objective. AF is set equal to the difference between the minimum salts export requirement to attain a salt balance in the subject reaches and the actual salts export.

Dry-weather definition. The WLAs apply to Simi Valley WQCP during dry weather when the average flows in the receiving water are below the 86th percentile flow and there is no measurable precipitation. Dry weather conditions exist when flow in Calleguas Creek at Ventura County Watershed Protection station 805 is less than 31 cubic feet per second (cfs). During wet weather, the loading capacity of the stream is significantly increased by storm water flows with very low salt concentrations. Any discharges from the Facility during wet weather would be assimilated by these large storm flows and would not cause exceedances of water quality objectives for salts. The rainfall precipitation shall be obtained from an existing rainfall gauging station located at the Simi Valley WQCP, Station ID: 246A. The stream flow and rainfall gauging stations are operated and maintained by the Ventura County Watershed Protection District. The required stream flow and rainfall data are available online at <http://www.vcwatershed.net/fws/>. The dry-weather final effluent limitation for Salts will be calculated as follows:

The dry-weather final effluent limitation for salts will be calculated as follows:

Given: Minimum Salt Export Requirements for AF

Chloride = 460 lbs/day

TDS = 3,220 lbs/day

Sulfate = 9,120 lbs/day

Boron = 3.3 lbs/day

The formula for determining final effluent limitation (dry-weather) applied as monthly average is as follows:

Chloride, lbs/day = 150 x Q-AF

TDS, lbs/day = 850 x Q-AF

Sulfate, lbs/day = 250 x Q-AF

Boron, lbs/day = 1.0 x Q-AF

where;

Q = the Facility's flow at the time the water quality measurement is collected and a conversion factor to lbs/day based on the units of measurement for the flow.

AF = (minimum salt export requirement – actual salt export)

However, the use of AFs is subject to approval by the Regional Water Board, following the demonstration of evidence presented by the Discharger. POTWs wanting to use AFs must apply to the Regional Water Board for approval and submit the following documentation together with their request: (1) water supply salt concentrations, (2) receiving water salt concentrations, (3) the effluent mass, and (4) evidence of increased salt exports to offset the increased discharges from the POTW.

Simi Valley WQCP is currently not connected to the brine line. The City of Simi Valley has not applied to the Regional Water Board for AFs. As a result, the AF term in the formula above will be set equal to zero until the City of Simi Valley requests and the Regional Board approves an AF for the Simi Valley WQCP. As a result, the AF term will drop out of the equation, and the final effluent limitations are expressed as follows:

Chloride, lbs/day = $150 \times Q = 150 \times 12.5 \times 8.34 = 15,640$

TDS, lbs/day = $850 \times Q = 850 \times 12.5 \times 8.34 = 88,612$

Sulfate, lbs/day = $250 \times Q = 250 \times 12.5 \times 8.34 = 26,062$

Boron, lbs/day = $1.0 \times Q = 1.0 \times 12.5 \times 8.34 = 104$

where;

Q = represents the product of the Facility's design capacity and a conversion factor, to convert from MGD to lbs/day.

If AFs are approved, the permit will be reopened to adjust the final effluent limitations to reflect the approved AFs.

Wet-weather definition. Wet-weather is any day when the average flow in the receiving water is equal to or greater than the 86th percentile flow of the receiving water measured at Calleguas Creek at Ventura County Watershed Protection station 805near. The 86th percentile flow is equal to 31 cfs. During wet weather, the loading capacity of the stream is significantly increased by storm water flows with very low salt concentrations. Any discharges from the Facility during wet weather would be assimilated by these large storm flows and would not cause exceedances of water quality objectives.

The wet-weather final effluent limitation for Salts will be as followed:

Parameter	Units	Effluent Limitations (Average Monthly)
Chloride	mg/L	150
TDS	mg/L	850

Parameter	Units	Effluent Limitations (Average Monthly)
Sulfate	mg/L	250
Boron	mg/L	1

The wet-weather final effluent limitations listed above for TDS, chloride, sulfate, and boron shall apply on the effective date of this Order.

P. Compliance with Calleguas Creek Metals TMDL for Mercury in Suspended Solids

A mass-based limit is developed for mercury expressed in lbs/month. The final waste load allocation for Simi Valley WQCP for mercury is based on median monthly mercury effluent concentration, which is more stringent than the *Metals TMDL* numeric target for mercury multiplied by the design flow. The *Metals TMDL* assumes that the total load of mercury in water is equal to the suspended sediment load multiplied by the mercury concentration. In addition to the water column final effluent monitoring, sediment sampling of mercury in the effluent is required, as specified in the Monitoring and Reporting Program, if both the TSS and the mercury final effluent limitations are exceeded.

Q. Compliance with 2,3,7,8-TCDD Equivalents

TCDD equivalents shall be calculated using the following formula, where the MLs, and toxicity equivalency factors (TEFs) are as provided in the table below. The Permittee shall report all measured values of individual congeners, including data qualifiers. When calculating TCDD equivalents, the Permittee shall set congener concentrations below the minimum levels to zero. USEPA method 1613 may be used to analyze dioxin and furan congeners.

$$\text{Dioxin Concentration} = \sum_{i=1}^{17} (TEQi) = \sum_{i=1}^{17} (Ci)(TEFi)$$

where:

Ci = individual concentration of a dioxin or furan congener

TEFi = individual TEF for a congener

MLs and TEFs

Congeners	MLs (pg/L)	TEFs
2,3,7,8-TetraCDD	10	1.0
1,2,3,7,8-PentaCDD	50	1.0
1,2,3,4,7,8-HexaCDD	50	0.1
1,2,3,6,7,8-HexaCDD	50	0.1
1,2,3,7,8,9-HexaCDD	50	0.1

Congeners	MLs (pg/L)	TEFs
1,2,3,4,6,7,8-HeptaCDD	50	0.01
OctaCDD	100	0.0001
2,3,7,8-TetraCDF	10	0.1
1,2,3,7,8-PentaCDF	50	0.05
2,3,4,7,8-PentaCDF	50	0.5
1,2,3,4,7,8-HexaCDF	50	0.1
1,2,3,6,7,8-HexaCDF	50	0.1
1,2,3,7,8,9-HexaCDF	50	0.1
2,3,4,6,7,8-HexaCDF	50	0.1
1,2,3,4,6,7,8-HeptaCDFs	50	0.01
1,2,3,4,7,8,9-HeptaCDFs	50	0.01
OctaCDF	100	0.0001

R. Compliance with Gross Beta/photon Emitters

The monthly average effluent limitation for gross beta/photon is equal to 4 millirem/year with a screening level of 50 picoCuries per Liter (pCi/L). Due to naturally occurring Potassium-40, the results of the Potassium-40 gross beta activity may be subtracted from the total gross beta activity to determine if the screening level is exceeded. The Potassium-40 beta particle activity must be calculated by multiplying elemental potassium concentrations (in mg/L) by a factor of 0.82 to determine activity from Potassium-40. The Potassium-40 must be analyzed from the same or equivalent sample used for the gross beta analysis.

If the gross beta particle activity minus the naturally occurring Potassium-40 is less than or equal to 50 pCi/L, the facility is in compliance and the value shall be reported as <4 millirem/year. If the gross beta particle activity minus the naturally occurring Potassium-40 beta particle activity exceeds the screening level, the Permittee must have the samples further analyzed for the *individual* nuclides. The calculation for the sum of the fractions is presented below.

The maximum contaminant level (MCL) for gross beta/photon emitters is equal 4 millirem per year. A millirem is a dose of energy to the body or any internal organ. USEPA regulates 179 man-made nuclides, and each of them has a concentration of radiation measured in pCi/L, which produces the 4 millirem dose. These concentrations are listed on table, *Derived Concentrations of (pCi/L) of Beta and Photon Emitters in Drinking Water*, which shall be used to determine compliance.

Yielding a Dose of 4 mrem/yr to the Total Body or to any Critical Organ as defined in NBS Handbook 69

Nuclide	pCi/l	Nuclide	pCi/l	Nuclide	pCi/l	Nuclide	pCi/l	Nuclide	pCi/l	Nuclide	pCi/l
H-3	20,000	Ni-65	300	Nb-95	300	Sb-124	60	Nd-147	200	Os-191	600
Be-7	6,000	Cu-64	900	Nb-97	3,000	Sb-125	300	Nd-149	900	Os-191m	9,000
C-14	2,000	Zn-65	300	Mo-99	600	Te-125m	600	Pm-147	600	Os-193	200
F-18	2,000	Zn-69	6,000	Tc-96	300	Te-127	900	Pm-149	100	Ir-190	600
Na-22	400	Zn-69m	200	Tc-96m	30,000	Te-127m	200	Sm-151	1,000	Ir-192	100
Na-24	600	Ga-72	100	Tc-97	6,000	Te-129	2,000	Sm-153	200	Ir-194	900
Si-31	3,000	Ge-71	6,000	Tc-97m	1,000	Te-129m	90	Eu-152	200	Pt-191	300
P-32	30	As-73	1,000	Tc-99	900	Te-131m	200	Eu-154	60	Pt-193	3,000
S-35 inorg	500	As-74	100	Tc-99m	20,000	Te-132	90	Eu-155	600	Pt-193m	3,000
Cl-36	700	As-76	60	Ru-97	1,000	I-126	3	Gd-153	600	Pt-197	300
Cl-38	1,000	As-77	200	Ru-103	200	I-129	1	Gd-159	200	Pt-197m	3,000
K-42	900	Se-75	900	Ru-105	200	I-131	3	Tb-160	100	Au-196	600
Ca-45	10	Br-82	100	Ru-106	30	I-132	90	Dy-165	1,000	Au-198	100
Ca-47	80	Rb-86	600	Rh-103m	30,000	I-133	10	Dy-166	100	Au-199	600
Sc-46	100	Rb-87	300	Rh-105	300	I-134	100	Ho-166	90	Hg-197	900
Sc-47	300	Sr-85m	20,000	Pd-103	900	I-135	30	Er-169	300	Hg-197m	600
Sc-48	80	Sr-85	900	Pd-109	300	Cs-131	20,000	Er-171	300	Hg-203	600
V-48	90	Sr-89	20	Ag-105	300	Cs-134	80	Tm-170	100	Ti-200	1,000
Cr-51	6,000	Sr-90	8	Ag-110m	90	Cs-134m	20,000	Tm-171	1,000	Ti-201	900
Mn-52	90	Sr-91	200	Ag-111	100	Cs-135	900	Yb-175	300	Ti-202	300
Mn-54	300	Sr-92	200	Cd-109	600	Cs-136	800	Lu-177	300	Ti-204	300
Mn-56	300	Y-90	60	Cd-115	90	Cs-137	200	Hf-181	200	Pb-203	1,000
Fe-55	2,000	Y-91	90	Cd-115m	90	Ba-131	600	Ta-182	100	Bi-206	100
Fe-59	200	Y-91m	9,000	In-113m	3,000	Ba-140	90	W-181	1,000	Bi-207	200
Co-57	1,000	Y-92	200	In-114m	60	La-140	60	W-185	300	Pa-230	600
Co-58	300	Y-93	90	In-115	300	Ce-141	300	W-187	200	Pa-233	300
Co-58m	9000	Zr-93	2,000	In-115m	1,000	Ce-143	100	Re-186	300	Np-239	300
Co-60	300	Zr-95	200	Sn-113	300	Ce-144	30	Re-187	9,000	Pu-241	300
Ni-59	300	Zr-97	60	Sn-125	60	Pr-142	90	Re-188	200	Bk-249	2,000
Ni-63	50	Nb-93m	1,000	Sb-122	90	Pr-143	100	Os-185	200		

Each nuclide has a different concentration that produces 4 millirem dose because different radionuclides have different energy levels. Some nuclides need to be in a higher concentration to give the same 4 millirem dose.

Fraction of the maximum

$$4 \text{ millirem/year exposure limit} = \frac{\text{pCi/L found in sample (from laboratory results)}}{\text{pCi/L equivalent from 4 millirem of exposure (from conversion table)}}$$

A sample calculation is presented in the table below:

	X	Y	X/Y	4(X/Y)
Emitter	Lab Analysis (pCi/L)	Conversion from table (pCi/4millirem)	Calculate Fraction	Calculate Total (millirem)
Cs-134	5,023	20,000	0.25115	1.0
Cs-137	30	200	0.150	0.6
Sr-90	4	8	0.5	2.0
I-131	2	3	0.7	2.8
		Sum of the Fractions	1.60115	6.4

In the example above, the system would be considered in violation of the gross beta/photon effluent limitation because the “sum-of-the-fractions” is 6.4 millirem, which means that the sum of the annual dose equivalent to the total body, or to any internal organ, exceeds 4 millirem/year.

S. Mass Emission Rate

The mass emission rate shall be obtained from the following calculation for any calendar day:

$$\text{Mass emission rate (lb/day)} = \frac{8.34}{N} \sum_{i=1}^N Q_i C_i$$

$$\text{Mass emission rate (kg/day)} = \frac{3.79}{N} \sum_{i=1}^N Q_i C_i$$

in which 'N' is the number of samples analyzed in any calendar day. 'Q_i' and 'C_i' are the flow rate (MGD) and the constituent concentration (mg/L), respectively, which are associated with each of the 'N' grab samples, which may be taken in any calendar day. If a composite sample is taken, 'C_i' is the concentration measured in the composite sample and 'Q_i' is the average flow rate occurring during the period over which samples are composited.

The daily concentration of all constituents shall be determined from the flow-weighted average of the same constituents in the combined waste streams as follows:

$$\text{Daily concentration} = \frac{1}{Q_t} \sum_{i=1}^N Q_i C_i$$

in which 'N' is the number of component waste streams. 'Q_i' and 'C_i' are the flow rate (MGD) and the constituent concentration (mg/L), respectively, which are associated with each of the 'N' waste streams. 'Q_t' is the total flow rate of the combined waste streams.

T. Bacterial Standards and Analysis

1. The geometric mean used for determining compliance with bacterial standards is calculated with the following equation:

$$\text{Geometric Mean} = (C_1 \times C_2 \times \dots \times C_n)^{1/n}$$

2. where n is the number of days samples were collected during the period and C is the concentration of bacteria (MPN/100 mL or CFU/100 mL) found on each day of sampling.
3. For bacterial analyses, sample dilutions should be performed so the expected range of values is bracketed (for example, with multiple tube fermentation method or membrane filtration method, 2 to 16,000 per 100 ml for total and fecal coliform, at a minimum, and 1 to 1000 per 100 ml for enterococcus). The detection methods used for each analysis shall be reported with the results of the analyses.
4. Detection methods used for coliforms (total and fecal) shall be those presented in Table 1A of 40 CFR part 136, unless alternate methods have been approved by USEPA pursuant to 40 CFR part 136, or improved methods have been determined by the Executive Officer and/or USEPA.
5. Detection methods used for enterococcus shall be those presented in Table 1A of 40 CFR part 136 or in the USEPA publication EPA 600/4-85/076, Test Methods for Escherichia coli and Enterococci in Water By Membrane Filter Procedure or any improved method determined by the Executive Officer and/or USEPA to be appropriate.

U. Single Operational Upset (SOU)

A SOU that leads to simultaneous violations of more than one pollutant parameter shall be treated as a single violation and limits the Permittee's liability in accordance with the following conditions:

1. A SOU is broadly defined as a single unusual event that temporarily disrupts the usually satisfactory operation of a system in such a way that it results in violation of multiple pollutant parameters.
2. A Permittee may assert SOU to limit liability only for those violations which the Permittee submitted notice of the upset as required in Provision V.E.2(b) of Attachment D – Standard Provisions.
3. For purpose outside of CWC section 13385 subdivisions (h) and (i), determination of compliance and civil liability (including any more specific definition of SOU, the requirements for Permittees to assert the SOU limitation of liability, and the manner of counting violations) shall be in accordance with USEPA Memorandum "Issuance of Guidance Interpreting Single Operational Upset" (September 27, 1989).

4. For purpose of CWC section 13385 (h) and (i), determination of compliance and civil liability (including any more specific definition of SOU, the requirements for Permittees to assert the SOU limitation of liability, and the manner of counting violations) shall be in accordance with CWC section 13385 (f)(2).

ATTACHMENT A - DEFINITIONS

ARITHMETIC MEAN (μ)

Also called the average, is the sum of measured values divided by the number of samples. For ambient water concentrations, the arithmetic mean is calculated as follows:

Arithmetic mean = $\mu = \Sigma x / n$ where: Σx is the sum of the measured ambient water concentrations, and n is the number of samples.

AVERAGE MONTHLY EFFLUENT LIMITATION (AMEL)

The highest allowable average of daily discharges over a calendar month, calculated as the sum of all daily discharges measured during a calendar month divided by the number of daily discharges measured during that month.

AVERAGE WEEKLY EFFLUENT LIMITATION (AWEL)

The highest allowable average of daily discharges over a calendar week (Sunday through Saturday), calculated as the sum of all daily discharges measured during a calendar week divided by the number of daily discharges measured during that week.

BIOACCUMULATIVE

Those substances taken up by an organism from its surrounding medium through gill membranes, epithelial tissue, or from food and subsequently concentrated and retained in the body of the organism.

BIOSOLIDS

Biosolids refer to sewage sludge that has been treated and tested and shown to be capable of being beneficially and legally used pursuant to federal and state regulations as a soil amendment for agricultural, silvicultural, horticultural, and land reclamation activities as specified under 40 CFR part 503.

CARCINOGENIC

Pollutants are substances that are known to cause cancer in living organisms.

COEFFICIENT OF VARIATION (CV)

CV is a measure of the data variability and is calculated as the estimated standard deviation divided by the arithmetic mean of the observed values.

DAILY DISCHARGE

Daily Discharge is defined as either: (1) the total mass of the constituent discharged over the calendar day (12:00 am through 11:59 pm) or any 24-hour period that reasonably represents a calendar day for purposes of sampling (as specified in the permit), for a constituent with limitations expressed in units of mass or; (2) the unweighted arithmetic mean measurement of the constituent over the day for a constituent with limitations expressed in other units of measurement (e.g., concentration).

The daily discharge may be determined by the analytical results of a composite sample taken over the course of one day (a calendar day or other 24-hour period defined as a day) or by the

arithmetic mean of analytical results from one or more grab samples taken over the course of the day.

For composite sampling, if 1 day is defined as a 24-hour period other than a calendar day, the analytical result for the 24-hour period will be considered as the result for the calendar day in which the 24-hour period ends.

DETECTED, BUT NOT QUANTIFIED (DNQ)

DNQ are those sample results less than the RL, but greater than or equal to the laboratory's MDL. Sample results reported as DNQ are estimated concentrations.

DILUTION CREDIT

Dilution Credit is the amount of dilution granted to a discharge in the calculation of a water quality-based effluent limitation, based on the allowance of a specified mixing zone. It is calculated from the dilution ratio or determined through conducting a mixing zone study or modeling of the discharge and receiving water.

EFFLUENT CONCENTRATION ALLOWANCE (ECA)

ECA is a value derived from the water quality criterion/objective, dilution credit, and ambient background concentration that is used, in conjunction with the coefficient of variation for the effluent monitoring data, to calculate a long-term average (LTA) discharge concentration. The ECA has the same meaning as waste load allocation (WLA) as used in U.S. EPA guidance (Technical Support Document For Water Quality-based Toxics Control, March 1991, second printing, EPA/505/2-90-001).

ENCLOSED BAYS

Enclosed Bays means indentations along the coast that enclose an area of oceanic water within distinct headlands or harbor works. Enclosed bays include all bays where the narrowest distance between the headlands or outermost harbor works is less than 75 percent of the greatest dimension of the enclosed portion of the bay. Enclosed bays include, but are not limited to, Humboldt Bay, Bodega Harbor, Tomales Bay, Drake's Estero, San Francisco Bay, Morro Bay, Los Angeles-Long Beach Harbor, Upper and Lower Newport Bay, Mission Bay, and San Diego Bay. Enclosed bays do not include inland surface waters or ocean waters.

ESTIMATED CHEMICAL CONCENTRATION

The estimated chemical concentration that results from the confirmed detection of the substance by the analytical method below the ML value.

ESTUARIES

Estuaries means waters, including coastal lagoons, located at the mouths of streams that serve as areas of mixing for fresh and ocean waters. Coastal lagoons and mouths of streams that are temporarily separated from the ocean by sandbars shall be considered estuaries. Estuarine waters shall be considered to extend from a bay or the open ocean to a point upstream where there is no significant mixing of fresh water and seawater. Estuarine waters included, but are not limited to, the Sacramento-San Joaquin Delta, as defined in Water Code section 12220, Suisun Bay, Carquinez Strait downstream to the Carquinez Bridge, and appropriate areas of the Smith, Mad, Eel, Noyo, Russian, Klamath, San Diego, and Otay rivers. Estuaries do not include inland surface waters or ocean waters.

INLAND SURFACE WATERS

All surface waters of the state that do not include the ocean, enclosed bays, or estuaries.

INSTANTANEOUS MAXIMUM EFFLUENT LIMITATION

The highest allowable value for any single grab sample or aliquot (i.e., each grab sample or aliquot is independently compared to the instantaneous maximum limitation).

INSTANTANEOUS MINIMUM EFFLUENT LIMITATION

The lowest allowable value for any single grab sample or aliquot (i.e., each grab sample or aliquot is independently compared to the instantaneous minimum limitation).

MAXIMUM DAILY EFFLUENT LIMITATION (MDEL)

The highest allowable daily discharge of a pollutant, over a calendar day (or 24-hour period). For pollutants with limitations expressed in units of mass, the daily discharge is calculated as the total mass of the pollutant discharged over the day. For pollutants with limitations expressed in other units of measurement, the daily discharge is calculated as the arithmetic mean measurement of the pollutant over the day.

MEDIAN

The middle measurement in a set of data. The median of a set of data is found by first arranging the measurements in order of magnitude (either increasing or decreasing order). If the number of measurements (n) is odd, then the median = $X_{(n+1)/2}$. If n is even, then the median = $(X_{n/2} + X_{(n/2)+1})/2$ (i.e., the midpoint between the $n/2$ and $n/2+1$).

METHOD DETECTION LIMIT (MDL)

MDL is the minimum concentration of a substance that can be measured and reported with 99 percent confidence that the analyte concentration is greater than zero, as defined in 40 CFR part 136, Attachment B, revised as of July 3, 1999.

MINIMUM LEVEL (ML)

ML is the concentration at which the entire analytical system must give a recognizable signal and acceptable calibration point. The ML is the concentration in a sample that is equivalent to the concentration of the lowest calibration standard analyzed by a specific analytical procedure, assuming that all the method specified sample weights, volumes, and processing steps have been followed.

MIXING ZONE

Mixing Zone is a limited volume of receiving water that is allocated for mixing with a wastewater discharge where water quality criteria can be exceeded without causing adverse effects to the overall water body.

NOT DETECTED (ND)

Sample results which are less than the laboratory's MDL.

PERSISTENT POLLUTANTS

Persistent pollutants are substances for which degradation or decomposition in the environment is nonexistent or very slow.

POLLUTANT MINIMIZATION PROGRAM (PMP)

PMP means waste minimization and pollution prevention actions that include, but are not limited to, product substitution, waste stream recycling, alternative waste management methods, and education of the public and businesses. The goal of the PMP shall be to reduce all potential sources of a priority pollutant(s) through pollutant minimization (control) strategies, including pollution prevention measures as appropriate, to maintain the effluent concentration at or below the water quality-based effluent limitation. Pollution prevention measures may be particularly appropriate for persistent bioaccumulative priority pollutants where there is evidence that beneficial uses are being impacted. The Regional Water Board may consider cost effectiveness when establishing the requirements of a PMP. The completion and implementation of a Pollution Prevention Plan, if required pursuant to Water Code section 13263.3(d), shall be considered to fulfill the PMP requirements.

POLLUTION PREVENTION

Pollution Prevention means any action that causes a net reduction in the use or generation of a hazardous substance or other pollutant that is discharged into water and includes, but is not limited to, input change, operational improvement, production process change, and product reformulation (as defined in Water Code section 13263.3). Pollution prevention does not include actions that merely shift a pollutant in wastewater from one environmental medium to another environmental medium, unless clear environmental benefits of such an approach are identified to the satisfaction of the State Water Resources Control Board (State Water Board) or Regional Water Board.

REPORTING LEVEL (RL)

The RL is the ML (and its associated analytical method) chosen by the Permittee for reporting and compliance determination from the MLs included in this Order, including an additional factor if applicable as discussed herein. The MLs included in this Order correspond to approved analytical methods for reporting a sample result that are selected by the Regional Water Board either from Appendix 4 of the SIP in accordance with section 2.4.2 of the SIP or established in accordance with section 2.4.3 of the SIP. The ML is based on the proper application of method-based analytical procedures for sample preparation and the absence of any matrix interferences. Other factors may be applied to the ML depending on the specific sample preparation steps employed. For example, the treatment typically applied in cases where there are matrix-effects is to dilute the sample or sample aliquot by a factor of ten. In such cases, this additional factor must be applied to the ML in the computation of the RL.

SOURCE OF DRINKING WATER

Any water designated as municipal or domestic supply (MUN) in a Regional Water Board Basin Plan.

STANDARD DEVIATION (σ)

Standard Deviation is a measure of variability that is calculated as follows:

$$\sigma = (\sum[(x - \mu)^2]/(n - 1))^{0.5}$$

where:

- x is the observed value;
- μ is the arithmetic mean of the observed values; and
- n is the number of samples.

STATISTICAL THRESHOLD VALUE (STV)

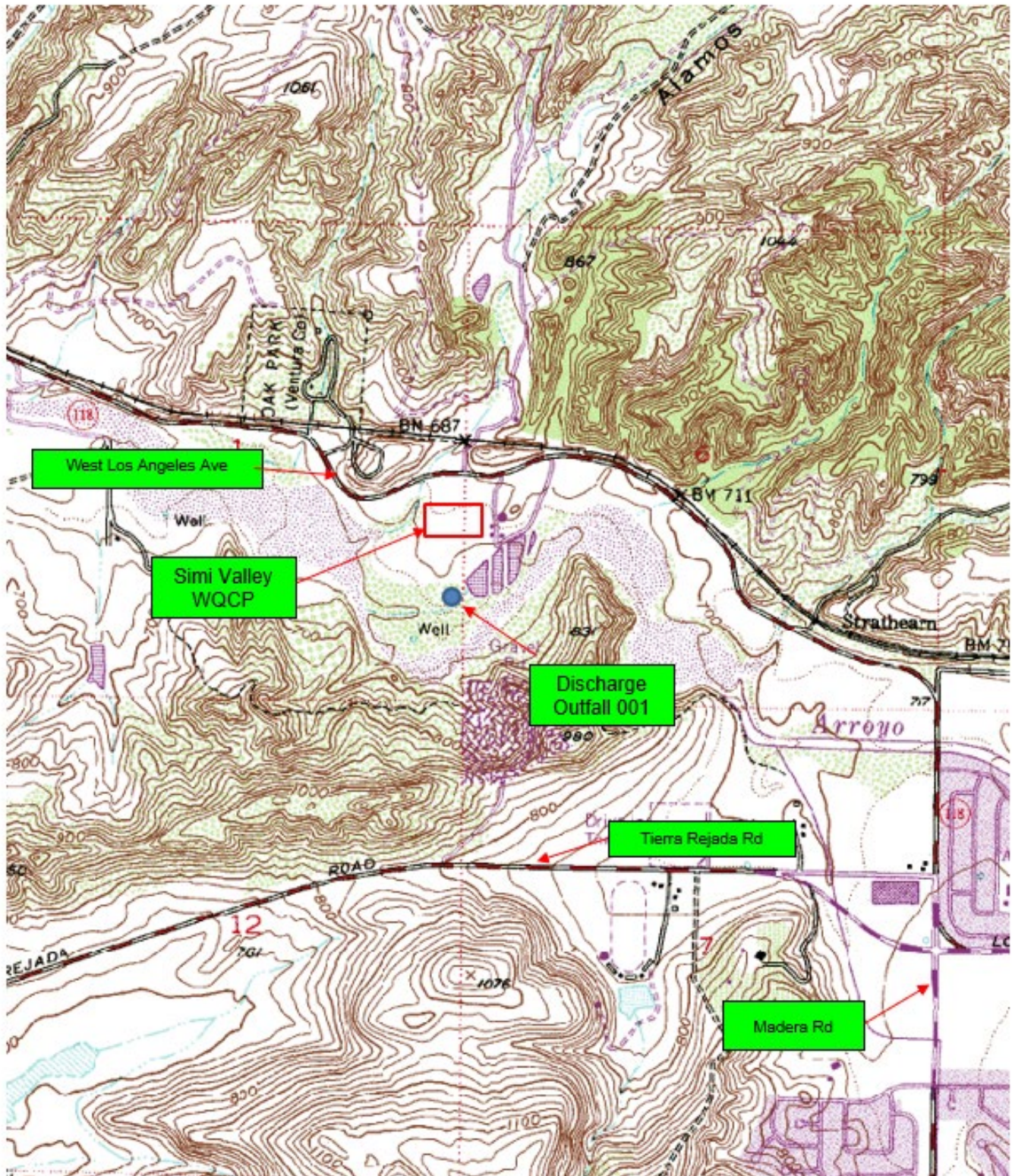
The STV for the bacteria water quality objectives is a set value that approximates the 90th percentile of the water quality distribution of a bacterial population.

TOXICITY REDUCTION EVALUATION (TRE)

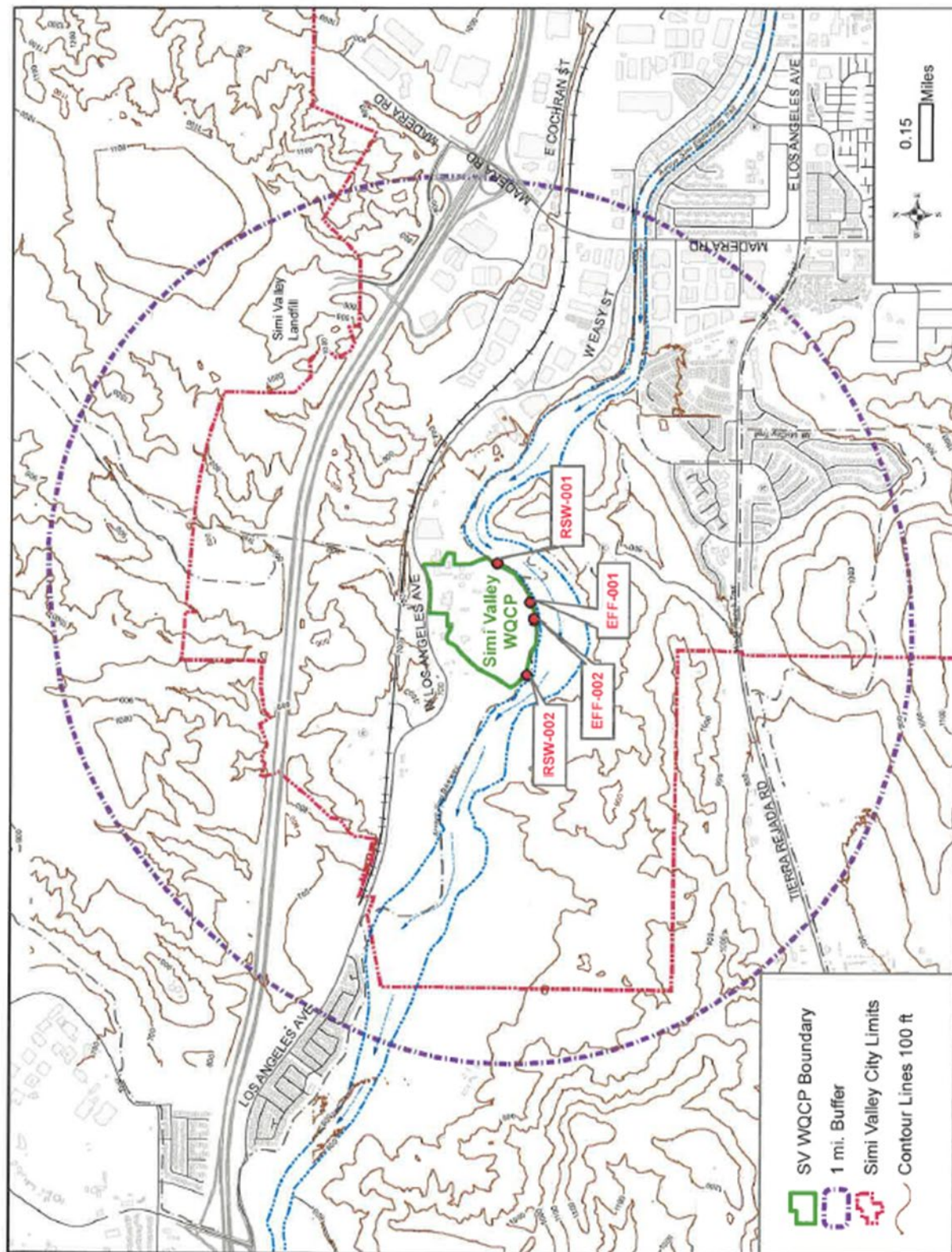
TRE is a study conducted in a step-wise process designed to identify the causative agents of effluent or ambient toxicity, isolate the sources of toxicity, evaluate the effectiveness of toxicity control options, and then confirm the reduction in toxicity. The first steps of the TRE consist of the collection of data relevant to the toxicity, including additional toxicity testing, and an evaluation of facility operations and maintenance practices, and best management practices. A Toxicity Identification Evaluation (TIE) may be required as part of the TRE, if appropriate. (A TIE is a set of procedures to identify the specific chemical(s) responsible for toxicity. These procedures are performed in three phases (characterization, identification, and confirmation) using aquatic organism toxicity tests.)

ATTACHMENT B - MAPS

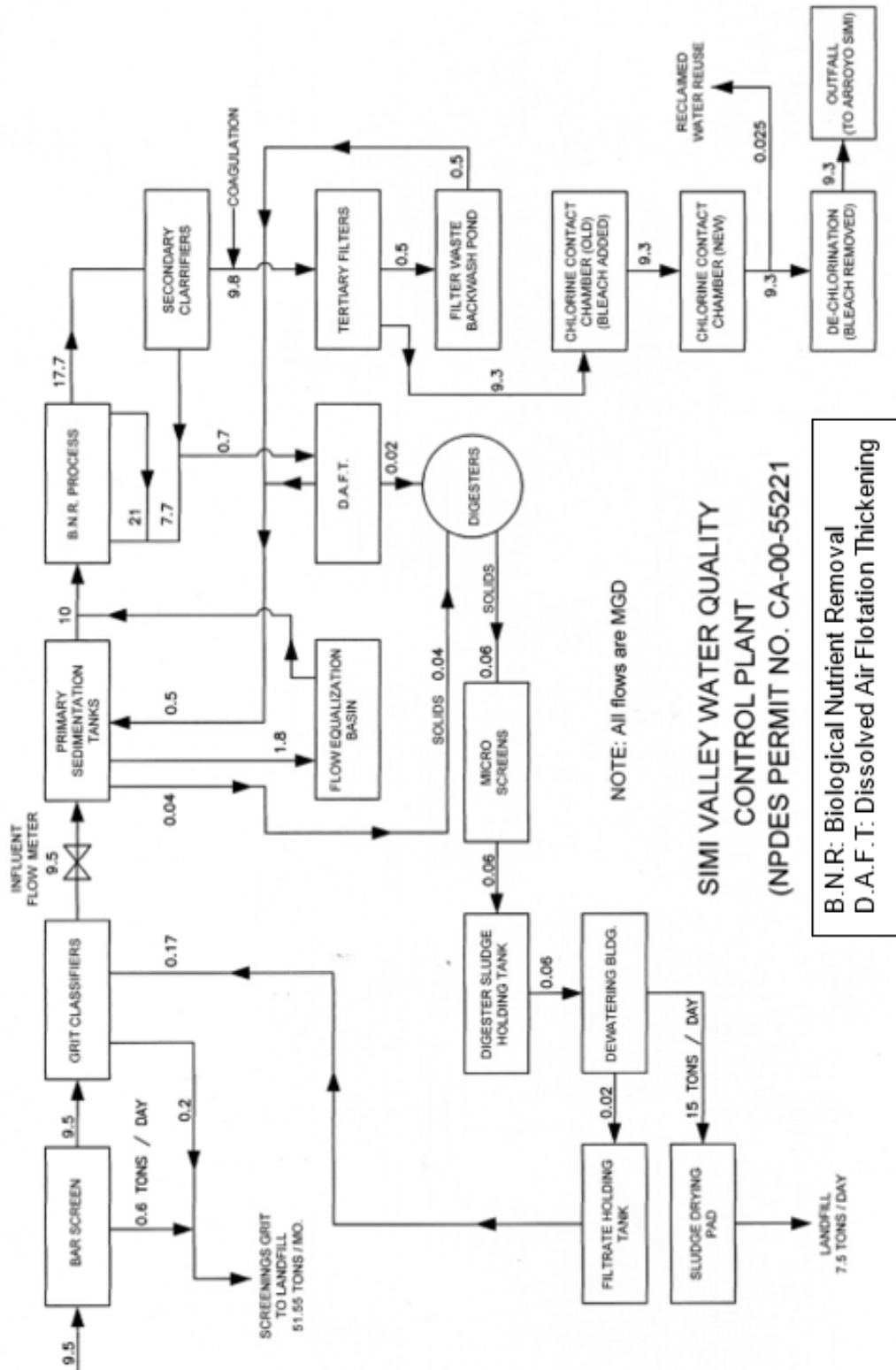
VICINITY MAP



MAP OF OUTFALLS AND SAMPLING LOCATIONS



ATTACHMENT C - FLOW SCHEMATIC



ATTACHMENT D - STANDARD PROVISIONS

I. STANDARD PROVISIONS – PERMIT COMPLIANCE

A. Duty to Comply

1. The Permittee must comply with all of the terms, requirements, and conditions of this Order. Any noncompliance constitutes a violation of the Clean Water Act (CWA), its regulations, and the California Water Code (CWC) and is grounds for enforcement action, for permit termination, revocation and reissuance, or modification; denial of a permit renewal application; or a combination thereof. (40 CFR section 122.41(a); California Water Code (CWC) sections 13261, 13263, 13264, 13265, 13268, 13000, 13001, 13304, 13350, 13385.)
2. The Permittee shall comply with effluent standards or prohibitions established under Part 307(a) of the CWA for toxic pollutants and with standards for sewage sludge use or disposal established under section 405(d) of the CWA within the time provided in the regulations that establish these standards or prohibitions, even if this Order has not yet been modified to incorporate the requirement. (Title 40 of the Code of Federal Regulations (40 CFR) section 122.41(a)(1).)

B. Need to Halt or Reduce Activity Not a Defense

It shall not be a defense for a Permittee in an enforcement action that it would have been necessary to halt or reduce the permitted activity in order to maintain compliance with the conditions of this Order. (40 CFR section 122.41(c).)

C. Duty to Mitigate

The Permittee shall take all reasonable steps to minimize or prevent any discharge or sludge use or disposal in violation of this Order that has a reasonable likelihood of adversely affecting human health or the environment. (40 CFR section 122.41(d).)

D. Proper Operation and Maintenance

The Permittee shall at all times properly operate and maintain all facilities and systems of treatment and control (and related appurtenances) which are installed or used by the Permittee to achieve compliance with the conditions of this Order. Proper operation and maintenance also include adequate laboratory controls and appropriate quality assurance procedures. This provision requires the operation of backup or auxiliary facilities or similar systems that are installed by a Permittee only when necessary to achieve compliance with the conditions of this Order. (40 CFR section 122.41(e).)

E. Property Rights

1. This Order does not convey any property rights of any sort or any exclusive privileges. (40 CFR section 122.41(g).)
2. The issuance of this Order does not authorize any injury to persons or property or invasion of other private rights, or any infringement of state or local law or regulations. (40 CFR section 122.5(c).)

F. Inspection and Entry

The Permittee shall allow the Regional Water Board, State Water Board, USEPA, and/or their authorized representatives (including an authorized contractor acting as their representative), upon the presentation of credentials and other documents, as may be required by law, to (33 U.S.C. section 1318(a)(B); 40 CFR section 122.41(i); CWC sections 13267 and 13383):

1. Enter upon the Permittee's premises where a regulated facility or activity is located or conducted, or where records are kept under the conditions of this Order (33 U.S.C. section 1318(a)(4)(B)(i); 40 CFR section 122.41(i)(1); CWC sections 13267 and 13383);
2. Have access to and copy, at reasonable times, any records that must be kept under the conditions of this Order (33 U.S.C. section 1318(a)(4)(B)(ii); 40 CFR section 122.41(i)(2); CWC sections 13267 and 13383);
3. Inspect and photograph, at reasonable times, any facilities, equipment (including monitoring and control equipment), practices, or operations regulated or required under this Order (33 U.S.C. section 1318(a)(4)(B)(ii); 40 CFR section 122.41(i)(3); CWC sections 13267 and 13383); and
4. Sample or monitor, at reasonable times, for the purposes of assuring Order compliance or as otherwise authorized by the CWA or the CWC, any substances or parameters at any location. (33 U.S.C. section 1318(a)(4)(B); 40 CFR section 122.41(i)(4); CWC sections 13267 and 13383)

G. Bypass

1. Definitions
 - a. "Bypass" means the intentional diversion of waste streams from any portion of a treatment facility. (40 CFR section 122.41(m)(1)(i).)
 - b. "Severe property damage" means substantial physical damage to property, damage to the treatment facilities, which causes them to become inoperable, or substantial and permanent loss of natural resources that can reasonably be expected to occur in the absence of a bypass. Severe property damage does not mean economic loss caused by delays in production. (40 CFR section 122.41(m)(1)(ii).)
2. Bypass not exceeding limitations. The Permittee may allow any bypass to occur which does not cause exceedances of effluent limitations, but only if it is for essential maintenance to assure efficient operation. These bypasses are not subject to the provisions listed in Standard Provisions – Permit Compliance I.G.3, I.G.4, and I.G.5 below. (40 CFR section 122.41(m)(2).)
3. Prohibition of bypass. Bypass is prohibited, and the Regional Water Board may take enforcement action against a Permittee for bypass, unless (40 CFR section 122.41(m)(4)(i)):
 - a. Bypass was unavoidable to prevent loss of life, personal injury, or severe property damage (40 CFR section 122.41(m)(4)(i)(A));

- b. There were no feasible alternatives to the bypass, such as the use of auxiliary treatment facilities, retention of untreated wastes, or maintenance during normal periods of equipment downtime. This condition is not satisfied if adequate back-up equipment should have been installed in the exercise of reasonable engineering judgment to prevent a bypass that occurred during normal periods of equipment downtime or preventive maintenance (40 CFR section 122.41(m)(4)(i)(B)); and
 - c. The Permittee submitted notice to the Regional Water Board as required under Standard Provisions – Permit Compliance I.G.5 below. (40 CFR section 122.41(m)(4)(i)(C).)
- 4. The Regional Water Board may approve an anticipated bypass, after considering its adverse effects, if the Regional Water Board determines that it will meet the three conditions listed in Standard Provisions – Permit Compliance I.G.3 above. (40 CFR section 122.41(m)(4)(ii).)
- 5. Notice
 - a. Anticipated bypass. If the Permittee knows in advance of the need for a bypass, it shall submit a notice, if possible at least 10 days before the date of the bypass. As of December 21, 2020, all notices must be submitted electronically to the initial recipient defined in Standard Provisions – Reporting V.J below. Notices shall comply with 40 CFR part 3, 40 CFR section 122.22, and 40 CFR section 127. (40 CFR section 122.41(m)(3)(i).)
 - b. Unanticipated bypass. The Permittee shall submit notice of an unanticipated bypass as required in Standard Provisions - Reporting V.E below (24-hour notice). As of December 21, 2020, all notices must be submitted electronically to the initial recipient defined in Standard Provisions – Reporting V.J below. Notices shall comply with 40 CFR part 3, 40 CFR section 122.22, and 40 CFR section 127. (40 CFR section 122.41(m)(3)(ii).)

H. Upset

Upset means an exceptional incident in which there is unintentional and temporary noncompliance with technology based permit effluent limitations because of factors beyond the reasonable control of the Permittee. An upset does not include noncompliance to the extent caused by operational error, improperly designed treatment facilities, inadequate treatment facilities, lack of preventive maintenance, or careless or improper operation. (40 CFR section 122.41(n)(1).)

- 1. Effect of an upset. An upset constitutes an affirmative defense to an action brought for noncompliance with such technology based permit effluent limitations if the requirements of Standard Provisions – Permit Compliance I.H.2 below are met. No determination made during administrative review of claims that noncompliance was caused by upset, and before an action for noncompliance, is final administrative action subject to judicial review. (40 CFR section 122.41(n)(2).)
- 2. Conditions necessary for a demonstration of upset. A Permittee who wishes to establish the affirmative defense of upset shall demonstrate, through properly

signed, contemporaneous operating logs or other relevant evidence that (40 CFR section 122.41(n)(3)):

- a. An upset occurred and that the Permittee can identify the cause(s) of the upset (40 CFR section 122.41(n)(3)(i));
 - b. The permitted facility was, at the time, being properly operated (40 CFR section 122.41(n)(3)(ii));
 - c. The Permittee submitted notice of the upset as required in Standard Provisions – Reporting V.E.2.b below (24-hour notice) (40 CFR section 122.41(n)(3)(iii)); and
 - d. The Permittee complied with any remedial measures required under Standard Provisions – Permit Compliance I.C above. (40 CFR section 122.41(n)(3)(iv).)
3. Burden of proof. In any enforcement proceeding, the Permittee seeking to establish the occurrence of an upset has the burden of proof. (40 CFR section 122.41(n)(4).)

II. STANDARD PROVISIONS – PERMIT ACTION

A. General

This Order may be modified, revoked and reissued, or terminated for cause. The filing of a request by the Permittee for modification, revocation and reissuance, or termination, or a notification of planned changes or anticipated noncompliance does not stay any Order condition. (40 CFR section 122.41(f).)

B. Duty to Reapply

If the Permittee wishes to continue an activity regulated by this Order after the expiration date of this Order, the Permittee must apply for and obtain a new permit. (40 CFR section 122.41(b).)

C. Transfers

This Order is not transferable to any person except after notice to the Regional Water Board. The Regional Water Board may require modification or revocation and reissuance of the Order to change the name of the Permittee and incorporate such other requirements as may be necessary under the CWA and the CWC. (40 CFR sections 122.41(l)(3); and 122.61.)

III. STANDARD PROVISIONS – MONITORING

- A. Samples and measurements taken for the purpose of monitoring shall be representative of the monitored activity. (40 CFR section 122.41(j)(1).)
- B. Monitoring must be conducted according to sufficiently sensitive test methods approved under 40 CFR part 136 for the analysis of pollutants or pollutant parameters or as required under 40 CFR chapter 1, subchapter N or O. For the purposes of this paragraph, a method is sufficiently sensitive when:

1. The method minimum level (ML) is at or below the level of the most stringent effluent limitation established in the permit for the measured pollutant or pollutant parameter, and either the method ML is at or below the level of the most stringent applicable water quality criterion for the measured pollutant or pollutant parameter or the method ML is above the applicable water quality criterion but the amount of the pollutant or pollutant parameter in the facility's discharge is high enough that the method detects and quantifies the level of the pollutant or pollutant parameter in the discharge; or
2. The method has the lowest ML of the analytical methods approved under 40 CFR part 136 when approved by this Regional Water Board and the State Water Board, or required under 40 CFR chapter 1, subchapter N or O for the measured pollutant or pollutant parameter.

In the case of pollutants or pollutant parameters for which there are no approved methods under 40 CFR part 136, or otherwise required under 40 CFR chapter 1, subchapters N or O, monitoring must be conducted according to a test procedure specified in this Order for such pollutants or pollutant parameters. (40 CFR section 122.21(e)(3).

IV. STANDARD PROVISIONS – RECORDS

A. Records of monitoring information shall include:

1. The date, exact place, and time of sampling or measurements (40 CFR section 122.41(j)(3)(i));
2. The individual(s) who performed the sampling or measurements (40 CFR section 122.41(j)(3)(ii));
3. The date(s) analyses were performed (40 CFR section 122.41(j)(3)(iii));
4. The individual(s) who performed the analyses (40 CFR section 122.41(j)(3)(iv));
5. The analytical techniques or methods used (40 CFR section 122.41(j)(3)(v)); and
6. The results of such analyses. (40 CFR section 122.41(j)(3)(vi).)

B. Claims of confidentiality for the following information will be denied (40 CFR section 122.7(b)):

1. The name and address of any permit applicant or Permittee (40 CFR section 122.7(b)(1)); and
2. Permit applications and attachments, permits and effluent data. (40 CFR section 122.7(b)(2).)

V. STANDARD PROVISIONS – REPORTING

A. Duty to Provide Information

The Permittee shall furnish to the Regional Water Board, State Water Board, or USEPA within a reasonable time, any information which the Regional Water Board, State Water Board, or USEPA may request to determine whether cause exists for modifying, revoking and reissuing, or terminating this Order or to determine

compliance with this Order. Upon request, the Permittee shall also furnish to the Regional Water Board, State Water Board, or USEPA copies of records required to be kept by this Order. (40 CFR section 122.41(h); CWC sections 13267 and 13383.)

B. Signatory and Certification Requirements

1. All applications, reports, or information submitted to the Regional Water Board, State Water Board, and/or USEPA shall be signed and certified in accordance with Standard Provisions – Reporting V.B.2, V.B.3, V.B.4, and V.B.5 below. (40 CFR section 122.41(k).)
2. All permit applications shall be signed by either a principal executive officer or ranking elected official. For purposes of this provision, a principal executive officer of a federal agency includes: (i) the chief executive officer of the agency, or (ii) a senior executive officer having responsibility for the overall operations of a principal geographic unit of the agency (e.g., Regional Administrators of USEPA). (40 CFR section 122.22(a)(3).)
3. All reports required by this Order and other information requested by the Regional Water Board, State Water Board, or USEPA shall be signed by a person described in Standard Provisions – Reporting V.B.2 above, or by a duly authorized representative of that person. A person is a duly authorized representative only if:
 - a. The authorization is made in writing by a person described in Standard Provisions – Reporting V.B.2 above (40 CFR section 122.22(b)(1));
 - b. The authorization specifies either an individual or a position having responsibility for the overall operation of the regulated facility or activity such as the position of plant manager, operator of a well or a well field, superintendent, position of equivalent responsibility, or an individual or position having overall responsibility for environmental matters for the company. (A duly authorized representative may thus be either a named individual or any individual occupying a named position.) (40 CFR section 122.22(b)(2)); and
 - c. The written authorization is submitted to the Regional Water Board and State Water Board. (40 CFR section 122.22(b)(3).)
4. If an authorization under Standard Provisions – Reporting V.B.3 above is no longer accurate because a different individual or position has responsibility for the overall operation of the facility, a new authorization satisfying the requirements of Standard Provisions – Reporting V.B.3 above must be submitted to the Regional Water Board and State Water Board prior to or together with any reports, information, or applications, to be signed by an authorized representative. (40 CFR section 122.22(c).)
5. Any person signing a document under Standard Provisions – Reporting V.B.2 or V.B.3 above shall make the following certification:

“I certify under penalty of law that this document and all attachments were prepared under my direction or supervision in accordance with a system designed to assure that qualified personnel properly gather and evaluate the

information submitted. Based on my inquiry of the person or persons who manage the system or those persons directly responsible for gathering the information, the information submitted is, to the best of my knowledge and belief, true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment for knowing violations.” (40 CFR section 122.22(d).)

6. Any person providing the electronic signature for documents described in Standard Provisions – V.B.1, V.B.2, or V.B.3 that are submitted electronically shall meet all relevant requirements of Standard Provisions – Reporting V.B, and shall ensure that all relevant requirements of 40 CFR section 3 (Cross-Media Electronic Reporting) and 40 CFR section 127 (NPDES Electronic Reporting Requirements) are met for that submission. (40 CFR section 122.22(e).)

C. Monitoring Reports

1. Monitoring results shall be reported at the intervals specified in the Monitoring and Reporting Program (Attachment E) in this Order. (40 CFR section 122.41(l)(4).)
2. Monitoring results must be reported on a Discharge Monitoring Report (DMR) form or forms provided or specified by the Regional Water Board or State Water Board for reporting results of monitoring of sludge use or disposal practices. All reports and forms must be submitted electronically to the initial recipient defined in Standard Provisions – Reporting V.J and comply with 40 CFR section 3, 40 CFR section 122.22, and 40 CFR section 127.
3. If the Permittee monitors any pollutant more frequently than required by this Order using test procedures approved under 40 CFR section 136, or another method required for an industry-specific waste stream under 40 CFR chapter 1, subchapters N or O, the results of such monitoring shall be included in the calculation and reporting of the data submitted in the DMR or sludge reporting form specified by the Regional Water Board.
4. Calculations for all limitations, which require averaging of measurements, shall utilize an arithmetic mean unless otherwise specified in this Order.

D. Compliance Schedules

Reports of compliance or noncompliance with, or any progress reports on, interim and final requirements contained in any compliance schedule of this Order, shall be submitted no later than 14 days following each schedule date. (40 CFR section 122.41(l)(5).)

E. Twenty-Four Hour Reporting

1. The Discharger shall report any noncompliance which may endanger health or the environment. Any information shall be provided orally within 24 hours from the time the Discharger becomes aware of the circumstances. A report shall also be provided within five (5) days of the time the Discharger becomes aware of the circumstances. The report shall contain a description of the noncompliance and its cause; the period of noncompliance, including exact dates and times, and if

the noncompliance has not been corrected, the anticipated time it is expected to continue; and steps taken or planned to reduce, eliminate, and prevent reoccurrence of the noncompliance.

For noncompliance events related to combined sewer overflows, sanitary sewer overflows, or bypass events, these reports must include the data described above (with the exception of time of discovery) as well as the type of event (i.e., combined sewer overflow, sanitary sewer overflow, or bypass event), type of overflow structure (e.g., manhole, combined sewer overflow outfall), discharge volume untreated by the treatment works treating domestic sewage, types of human health and environmental impacts of the event, and whether the noncompliance was related to wet weather.

As of December 21, 2020, all reports related to combined sewer overflows, sanitary sewer overflows, or bypass events must be submitted electronically to the initial recipient defined in Standard Provisions – Reporting V.J. The reports shall comply with 40 CFR section 3, 40 CFR section 122.22, and 40 CFR section 127. The Regional Water Board may also require the Discharger to electronically submit reports not related to combined sewer overflows, sanitary sewer overflows, or bypass events under this section. (40 CFR section 122.41(I)(6)(i).)

2. The following shall be included as information that must be reported within 24 hours:
 - a. Any unanticipated bypass that exceeds any effluent limitation in this Order. (40 CFR section 122.41(I)(6)(ii)(A).)
 - b. Any upset that exceeds any effluent limitation in this Order. (40 CFR section 122.41(I)(6)(ii)(B).)
3. The Regional Water Board may waive the above-required written report under this provision on a case-by-case basis if an oral report has been received within 24 hours. (40 CFR section 122.41(I)(6)(iii).)

F. Planned Changes

The Permittee shall give notice to the Regional Water Board as soon as possible of any planned physical alterations or additions to the permitted facility. Notice is required under this provision only when (40 CFR section 122.41(I)(1)):

1. The alteration or addition to a permitted facility may meet one of the criteria for determining whether a facility is a new source in section 122.29(b) (40 CFR section 122.41(I)(1)(i)); or
2. The alteration or addition could significantly change the nature or increase the quantity of pollutants discharged. This notification applies to pollutants that are not subject to effluent limitations in this Order. (40 CFR section 122.41(I)(1)(ii).)
3. The alteration or addition results in a significant change in the Permittee's sludge use or disposal practices, and such alteration, addition, or change may justify the application of permit conditions that are different from or absent in the existing permit, including notification of additional use or disposal sites not reported during

the permit application process or not reported pursuant to an approved land application plan. (40 CFR section 122.41(l)(1)(iii).)

G. Anticipated Noncompliance

The Permittee shall give advance notice to the Regional Water Board of any planned changes in the permitted facility or activity that may result in noncompliance with this Order's requirements. (40 CFR section 122.41(l)(2).)

H. Other Noncompliance

The Permittee shall report all instances of noncompliance not reported under Standard Provisions – Reporting V.C, V.D, and V.E above at the time monitoring reports are submitted. The reports shall contain the information listed in Standard Provision – Reporting V.E above. For noncompliance events related to combined sewer overflows, sanitary sewer overflows, or bypass events, these reports shall contain the information described in Standard Provision – Reporting V.E and the applicable required data in appendix A to 40 CFR section 127. The Regional Water Board may also require the Discharger to electronically submit reports not related to combined sewer overflows, sanitary sewer overflows, or bypass events under this section (40 CFR section 122.41(l)(7).)

I. Other Information

When the Permittee becomes aware that it failed to submit any relevant facts in a permit application, or submitted incorrect information in a permit application or in any report to the Regional Water Board, State Water Board, or USEPA, the Permittee shall promptly submit such facts or information. (40 CFR section 122.41(l)(8).)

J. Initial Recipient for Electronic Reporting Data

The owner, operator, or the duly authorized representative is required to electronically submit NPDES information specified in appendix A to 40 CFR section 127 to the initial recipient defined in 40 CFR section 127.2(b). USEPA will identify and publish the list of initial recipients on its website and in the Federal Register, by state and by NPDES data group [see 40 CFR section 127.2(c)]. U.S. EPA will update and maintain this listing. (40 CFR § 122.41(l)(9).)

VI. STANDARD PROVISIONS – ENFORCEMENT

- A.** The Regional Water Board is authorized to enforce the terms of this permit under several provisions of the CWC, including, but not limited to, sections 13268, 13385, 13386, and 13387.
- B.** The CWA provides that any person who violates section 301, 302, 306, 307, 308, 318 or 405 of the CWA, or any permit condition or limitation implementing any such sections in a permit issued under section 402, or any requirement imposed in a pretreatment program approved under sections 402(a)(3) or 402(b)(8) of the CWA, is subject to a civil penalty not to exceed \$25,000 per day for each violation. The CWA provides that any person who *negligently* violates sections 301, 302, 306, 307, 308, 318, or 405 of the CWA, or any condition or limitation implementing any of such

sections in a permit issued under section 402 of the CWA, or any requirement imposed in a pretreatment program approved under section 402(a)(3) or 402(b)(8) of the CWA, is subject to criminal penalties of \$2,500 to \$25,000 per day of violation, or imprisonment of not more than one year, or both. In the case of a second or subsequent conviction for a negligent violation, a person shall be subject to criminal penalties of not more than \$50,000 per day of violation, or by imprisonment of not more than two years, or both. Any person who *knowingly* violates such conditions or limitations is subject to criminal penalties of \$5,000 to \$50,000 per day of violation, or imprisonment for not more than three years, or both. In the case of a second or subsequent conviction for a knowing violation, a person shall be subject to criminal penalties of not more than \$100,000 per day of violation, or imprisonment of not more than 6 years, or both. Any person who *knowingly* violates section 301, 302, 303, 306, 307, 308, 318 or 405 of the CWA, or any permit condition or limitation implementing any of such sections in a permit issued under section 402 of the CWA, and who knows at that time that he thereby places another person in imminent danger of death or serious bodily injury, shall, upon conviction, be subject to a fine of not more than \$250,000 or imprisonment of not more than 15 years, or both. In the case of a second or subsequent conviction for a knowing endangerment violation, a person shall be subject to a fine of not more than \$500,000 or by imprisonment of not more than 30 years, or both. An organization, as defined in section 309(c)(3)(B)(iii) of the CWA, shall, upon conviction of violating the imminent danger provision, be subject to a fine of not more than \$1,000,000 and can be fined up to \$2,000,000 for second or subsequent convictions (40 CFR section 122.41(a)(2); CWC section 13385 and 13387).

- C. Any person may be assessed an administrative penalty by the Administrator of USEPA, the Regional Water Board, or State Water Board for violating section 301, 302, 306, 307, 308, 318 or 405 of this CWA, or any permit condition or limitation implementing any of such sections in a permit issued under section 402 of the CWA. Administrative penalties for Class I violations are not to exceed \$10,000 per violation, with the maximum amount of any Class I penalty assessed not to exceed \$25,000. Penalties for Class II violations are not to exceed \$10,000 per day for each day during which the violation continues, with the maximum amount of any Class II penalty not to exceed \$125,000. (40 CFR section 122.41(a)(3))
- D. The CWA provides that any person who falsifies, tampers with, or knowingly renders inaccurate any monitoring device or method required to be maintained under this permit shall, upon conviction, be punished by a fine of not more than \$10,000, or by imprisonment for not more than two years, or both. If a conviction of a person is for a violation committed after a first conviction of such person under this paragraph, punishment is a fine of not more than \$20,000 per day of violation, or by imprisonment of not more than four years, or both. (40 CFR section 122.41(j)(5)).
- E. The CWA provides that any person who knowingly makes any false statement, representation, or certification in any record or other document submitted or required to be maintained under this permit, including monitoring reports or reports of compliance or non-compliance shall, upon conviction, be punished by a fine of not more than \$10,000 per violation, or by imprisonment for not more than six months per violation, or by both. (40 CFR section 122.41(k)(2)).

VII. ADDITIONAL PROVISIONS – NOTIFICATION LEVELS

A. Publicly-Owned Treatment Works (POTWs)

All POTWs shall provide adequate notice to the Regional Water Board of the following (40 CFR section 122.42(b)):

1. Any new introduction of pollutants into the POTW from an indirect Permittee that would be subject to sections 301 or 306 of the CWA if it were directly discharging those pollutants (40 CFR section 122.42(b)(1)); and
2. Any substantial change in the volume or character of pollutants being introduced into that POTW by a source introducing pollutants into the POTW at the time of adoption of the Order. (40 CFR section 122.42(b)(2).)
3. Adequate notice shall include information on the quality and quantity of effluent introduced into the POTW as well as any anticipated impact of the change on the quantity or quality of effluent to be discharged from the POTW. (40 CFR section 122.42(b)(3).)

ATTACHMENT E - MONITORING AND REPORTING PROGRAM

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ATTACHMENT E – MONITORING AND REPORTING PROGRAM (MRP), (CI-3021)

Section 308(a) of the federal Clean Water Act and sections 122.41(h), (j)-(l), 122.44(i), and 122.48 of Title 40 of the Code of Federal Regulations (40 CFR) requires that all NPDES permits specify monitoring and reporting requirements. California Water Code (CWC) sections 13267 and 13383 also authorizes the Regional Water Board to establish monitoring, inspection, entry, reporting, and recordkeeping requirements. This MRP establishes monitoring, reporting, and recordkeeping requirements that implement federal and California laws and/or regulations.

I. GENERAL MONITORING PROVISIONS

- A.** All samples shall be representative of the waste discharge under conditions of peak load. Quarterly effluent analyses shall be performed during the months of February, May, August, and November. Semiannual analyses shall be performed during the months of February and August. Annual analyses shall be performed during the month of August. Should there be instances when monitoring could not be done during these specified months, the Permittee must notify the Regional Water Board, state the reason why monitoring could not be conducted, and obtain approval from the Executive Officer for an alternate schedule. Results of monthly, quarterly, semiannual, and annual analyses shall be reported as due date specified in Table E-9 of MRP.
- B.** Pollutants shall be analyzed using the analytical methods described in 40 CFR sections 136.3, 136.4, and 136.5; or where no methods are specified for a given pollutant, by methods approved by this Regional Water Board or the State Water Board.
- C.** Laboratory Certification. Laboratories analyzing monitoring samples shall be certified by the State Water Resources Control Board, Division of Drinking Water (DDW) Environmental Laboratory Accreditation Program (ELAP) in accordance with the provision of Water Code section 13176, and must include quality assurance/quality control(QA/QC) data with their reports. A copy of the laboratory certification shall be provided in the Annual Report due to the Regional Water Board each time a new certification and/or renewal of the certification is obtained.
- D.** Water/wastewater samples must be analyzed within allowable holding time limits as specified in 40 CFR section 136.3. All QA/QC analyses must be run on the same dates that samples are actually analyzed. The Permittee shall retain the QA/QC documentation in its files and make available for inspection and/or submit them when requested by the Regional Water Board. Proper chain of custody procedures must be followed and a copy of that documentation shall be submitted with the monthly report.
- E.** The Permittee shall calibrate and perform maintenance procedures on all monitoring instruments and to ensure accuracy of measurements, or shall ensure that both equipment activities will be conducted.
- F.** For any analyses performed for which no procedure is specified in the United States Environmental Protection Agency (USEPA) guidelines, or in the MRP, the constituent or parameter analyzed and the method or procedure used must be specified in the monitoring report.

- G.** Each monitoring report must affirm in writing that “all analyses were conducted at a laboratory certified for such under the ELAP, or approved by the Executive Officer and in accordance with current USEPA guideline procedures or as specified in this Monitoring and Reporting Program.”
- H.** The monitoring report shall specify the USEPA analytical method used, the Method Detection Limit (MDL), and the Reporting Level (RL) [the applicable minimum level (ML) or reported Minimum Level (RML)] for each pollutant. The MLs are those published by the State Water Resources Control Board (State Water Board) in the *Policy for the Implementation of Toxics Standards for Inland Surface Waters, Enclosed Bays, and Estuaries of California, (State Implementation Policy or SIP)*, February 9, 2005, Appendix 4. The ML represents the lowest quantifiable concentration in a sample based on the proper application of all method-based analytical procedures and the absence of any matrix interference. When all specific analytical steps are followed and after appropriate application of method specific factors, the ML also represents the lowest standard in the calibration curve for that specific analytical technique. When there is deviation from the method analytical procedures, such as dilution or concentration of samples, other factors may be applied to the ML depending on the sample preparation. The resulting value is the reported ML.
- I.** The Permittee shall select the analytical method that provides a ML lower than the permit limit established for a given parameter, unless the Permittee can demonstrate that a particular ML is not attainable, in accordance with procedures set forth in 40 CFR part 136, and obtains approval for a higher ML from the Executive Officer, as provided for in section J, below. If the effluent limitation is lower than all the MLs in Appendix 4, SIP, the Discharge must select the method with the lowest ML for compliance purposes. The Permittee shall include in the Annual Summary Report a list of the analytical methods employed for each test.
- J.** The Permittee shall instruct its laboratories to establish calibration standards so that the ML (or its equivalent if there is differential treatment of samples relative to calibration standards) is the lowest calibration standard. At no time is the Permittee to use analytical data derived from extrapolation beyond the lowest point of the calibration curve. In accordance with section J, below, the Permittee’s laboratory may employ a calibration standard lower than the ML in Appendix 4 of the SIP.
- K.** In accordance with section 2.4.3 of the SIP, the Regional Water Board Executive Officer, in consultation with the State Water Board’s Quality Assurance Program Manager, may establish an ML that is not contained in Appendix 4 of the SIP to be included in the Permittee’s permit in any of the following situations:

 - 1. When the pollutant under consideration is not included in Appendix 4, SIP;
 - 2. When the Permittee and the Regional Water Board agree to include in the permit a test method that is more sensitive than those specified in 40 CFR part 136;
 - 3. When the Permittee agrees to use an ML that is lower than those listed in Appendix 4;

4. When the Permittee demonstrates that the calibration standard matrix is sufficiently different from that used to establish the ML in Appendix 4 and proposes an appropriate ML for the matrix; or,
5. When the Permittee uses a method, which quantification practices are not consistent with the definition of the ML. Examples of such methods are USEPA-approved method 1613 for dioxins, and furans, method 1624 for volatile organic substances, and method 1625 for semi-volatile organic substances. In such cases, the Permittee, the Regional Water Board, and the State Water Board shall agree on a lowest quantifiable limit and that limit will substitute for the ML for reporting and compliance determination purposes.

If there is any conflict between foregoing provisions and the SIP, the provisions stated in the SIP (section 2.4) shall prevail.

- L. If the Permittee samples and performs analyses (other than for process/operational control, startup, research, or equipment testing) on any influent, effluent, or receiving water constituent more frequently than required by this MRP using approved analytical methods, the results of those analyses shall be included in the report. These results shall be reflected in the calculation of the average used in demonstrating compliance with limitations set forth in this Order.
- M. The Permittee shall develop and maintain a record of all spills or bypasses of raw or partially treated sewage from its collection system or treatment plant according to the requirements in the WDR section of this Order. This record shall be made available to the Regional Water Board upon request and a spill summary shall be included in the annual summary report.
- N. For all bacteriological analyses, sample dilutions should be performed so the expected range of values is bracketed (for example, with multiple tube fermentation method or membrane filtration method, 2 to 16,000 per 100 ml for total coliform, at a minimum, and 1 to 1000 per 100 ml for *enterococcus*). The detection methods used for each analysis shall be reported with the results of the analyses.
 1. Detection methods used for coliforms (total and fecal) shall be those presented in Table 1A of 40 CFR part 136, unless alternate methods have been approved in advance by the USEPA pursuant to 40 CFR part 136.
 2. Detection methods used for E.coli shall be those presented in Table 1A of 40 CFR part 136 or in the USEPA publication EPA 600/4-85/076, *Test Methods for Escherichia coli and Enterococci in Water By Membrane Filter Procedure*, or any improved method determined by the Regional Water Board to be appropriate.
- O. The Permittee shall ensure that the results of the Discharge Monitoring Report-Quality Assurance (DMR-QA) Study or the most recent Water Pollution Performance Evaluation Study are submitted annually to the State Water Board at the following address:

State Water Resources Control Board
Quality Assurance Program Officer
Office of Information Management and Analysis
1001 I Street, Sacramento, CA 95814.

- P. Since compliance monitoring focuses on the effects of a point source discharge, it is not designed to assess impacts from other sources of pollution (e.g., non-point source run-off, aerial fallout) or to evaluate the current status of important ecological resources on a regional basis.

The Permittee shall continue to participate in the implementation of and comply with the Watershed-wide Monitoring Program developed by stakeholders and initiated in 2008. The City's responsibilities under the Watershed-wide Monitoring Program are described in the Receiving Water Monitoring Requirements section. To achieve the goals of the Watershed-wide Monitoring Program, revisions to the Receiving Water Monitoring Requirements may be made under the direction of USEPA and the Regional Water Board. The Permittee shall submit annual reports providing the monitoring data collected during the calendar year, as well as an interpretation of the significance of the results with respect to the health of the watershed. Annual reports shall be submitted by July 1st of each year.

Changes to the compliance monitoring program may be required to fulfill the goals of the watershed-wide monitoring program, while retaining the compliance monitoring component required to evaluate compliance with the NPDES permit. Revisions to the Permittee's program will be made under the direction of the Regional Water Board, as necessary, to accomplish the goal, and may include a reduction or increase in the number of parameters to be monitored, the frequency of monitoring, and/or the number of samples collected.

Until such time when a watershed-wide monitoring program is developed, Simi Valley WQCP shall implement the monitoring program in section IX.B of this MRP.

II. MONITORING LOCATIONS

The Permittee shall establish the following monitoring locations to demonstrate compliance with the effluent limitations, discharge specifications, and other requirements in this Order:

Table E-1. Monitoring Station Locations

Discharge Point Name	Monitoring Location Name	Monitoring Location Description
--	INF-001	Sampling stations shall be established at each point of inflow to the sewage treatment plant and shall be located upstream of any in-plant return flows and where representative samples of the influent can be obtained.
001, 002	EFF-001	The effluent sampling station shall be located downstream of any in-plant return flows and after the final disinfection process, where representative samples of the effluent can be obtained.

Discharge Point Name	Monitoring Location Name	Monitoring Location Description
--	RSW-001	Receiving Water Monitoring Station: (Upstream of Discharge Points 001 and 002). Arroyo Simi, approximately 400 feet east of Discharge Points 001 and 002, approximate coordinates: Latitude 34.28200°, Longitude -118.81290°.
--	RSW-002	Receiving Water Monitoring Station: (Downstream of Discharge Points 001 and 002). Arroyo Simi, upstream of its confluence with Alamos Canyon, approximate coordinates: Latitude 34.28169°, Longitude -118.81350°.
--	RSW-003D	TMDL Dry- and Wet-Weather Flow Monitoring Station: <i>Salts TMDL</i> stream flow monitoring station at Calleguas Creek at Ventura County Watershed Protection station 805. For the purposes of this permit, this station is also known as RSW-003D.

The North latitude and West longitude information in Table E-1 are approximate for administrative purposes. These monitoring locations are indicated in Figure E-1 below.

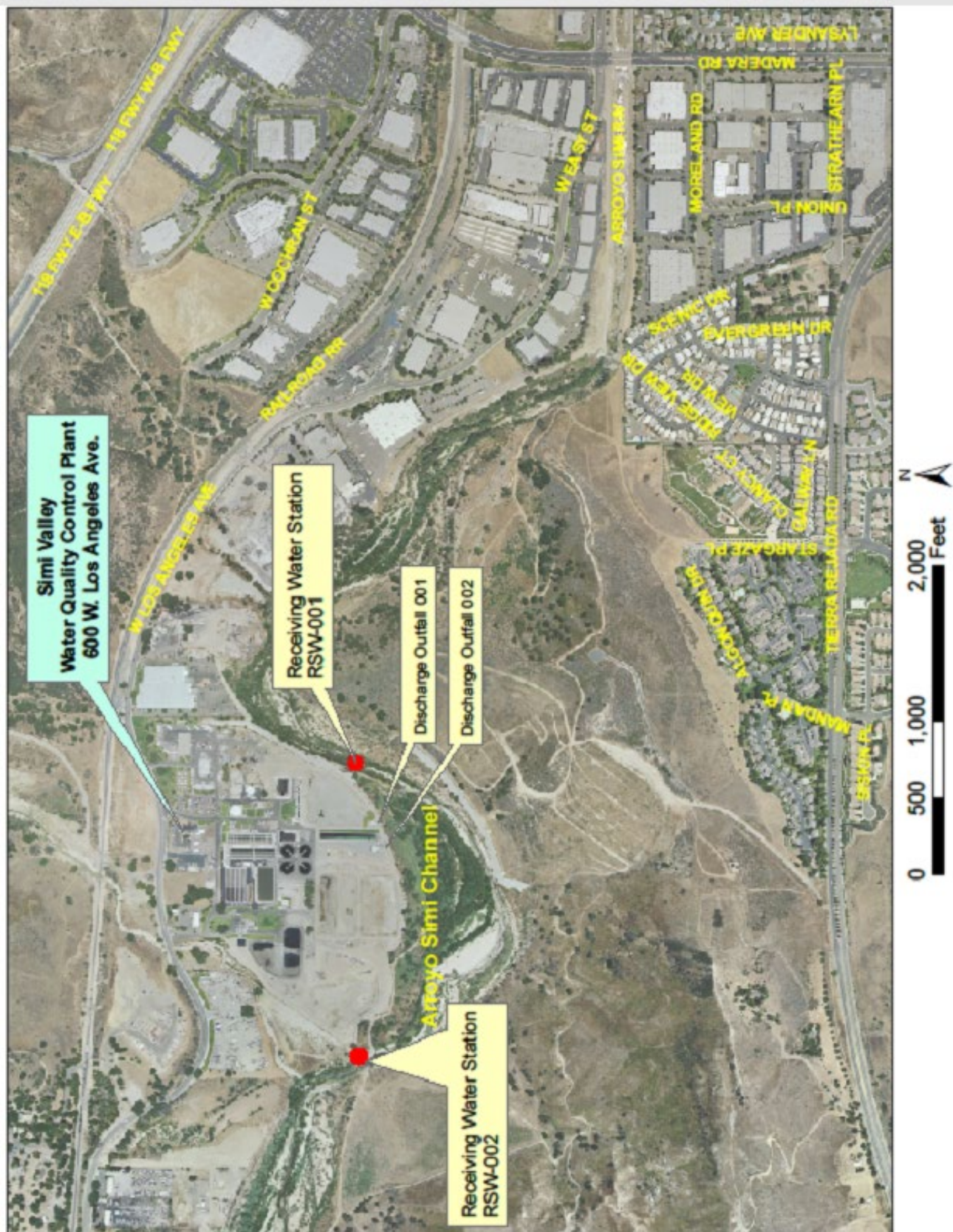


Figure E-1. Simi Valley WQCP Receiving Water Stations

III. INFLUENT MONITORING REQUIREMENTS

Influent monitoring is required to:

1. Determine compliance with NPDES permit conditions.
2. Assess treatment plant performance.
3. Assess effectiveness of the Pretreatment Program.

A. Monitoring Location INF-001

1. The Permittee shall monitor influent to the facility at INF-001 as follows:

Table E-2. Influent Monitoring

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method
Flow	MGD	recorder	continuous	See #2 below
pH	pH unit	grab	weekly	See #3 below
Total suspended solids	mg/L	24-hour composite	weekly	See #3 below
Biochemical oxygen demand (BOD ₅ 20°C)	mg/L	24-hour composite	weekly	See #3 below
PCBs as aroclors	µg/L	24-hour composite	annually	See #3, #4, #6 below
PCBs as congeners	pg/L	24-hour composite	annually	See #3, #5, #6 below
Remaining USEPA priority pollutants excluding asbestos and PCBs	µg/L	24-hour composite/grab for VOCs, Cyanide, bis(2-ethylhexyl)phthalate and Chromium VI	semiannually	See #3, #7 below

2. Total daily flow and instantaneous peak daily flow (24-hr basis) shall be reported. Actual monitored flow shall also be reported (not the maximum flow, i.e., design capacity).
3. Pollutants shall be analyzed using the analytical methods described in 40 CFR part 136; where no methods are specified for a given pollutant, those methods shall be approved by this Regional Water Board or State Water Board. For any pollutant whose effluent limitation is lower than all the minimum levels (MLs) specified in Appendix 4 of the SIP, the analytical method with the lowest ML must be selected.
4. PCBs as Aroclors is the sum of PCB 1016, PCB 1221, PCB 1232, PCB 1242, PCB 1248, PCB 1254, and PCB 1260 when monitoring using USEPA method 608.

5. PCBs as congeners shall mean the sum of 41 congeners when monitoring using USEPA proposed method 1668c. PCB-18, 28, 37, 44, 49, 52, 66, 70, 74, 77, 81, 87, 99, 101, 105, 110, 114, 118, 119, 123, 126, 128, 138, 149, 151, 153, 156, 157, 158, 167, 168, 169, 170, 177, 180, 183, 187, 189, 194, 201, and 206 shall be individually quantified, or quantified as co-elutions as appropriate. PCBs as congeners shall be analyzed using method EPA 1668c for three years and may be discontinued for the remaining life of this Order if none of the PCB congeners are detected using method EPA 1668c.
6. USEPA recommends that until USEPA proposed method 1668c for PCBs is incorporated into 40 CFR 136, Permittees should use for discharge monitoring reports/State monitoring reports: (1) USEPA method 608 for monitoring data, reported as aroclor results, that will be used for assessing compliance with WQBELs (if applicable) and (2) USEPA proposed method 1668c for monitoring data, reported as 41 congener results, that will be used for informational purposes.
7. Priority pollutants are those constituents referred to in 40 CFR section 401.15; a list of these pollutants is provided as Appendix A to 40 CFR part 423.

IV. EFFLUENT MONITORING REQUIREMENTS

Effluent monitoring is required to:

1. Determine compliance with National Pollutant Discharge Elimination System (NPDES) permit conditions and water quality standards.
2. Assess plant performance, identify operational problems and improve plant performance.
3. Provide information on wastewater characteristics and flows for use in interpreting water quality and biological data.
4. Determine reasonable potential analysis for toxic pollutants.
5. Determine TMDL effectiveness in waste load allocation compliance.

A. Minimum Level (ML) and Analytical Method Selection

1. Minimum Level (ML) and Analytical Method Selection

USEPA published regulations for the Sufficiently Sensitive Methods Rule (SSM Rule) which became effective September 18, 2015. For the purposes of the NPDES program, when more than one test procedure is approved under 40 CFR part 136 for the analysis of a pollutant or pollutant parameter, the test procedure must be sufficiently sensitive as defined at 40 CFR sections 122.21(e)(3) and 122.44(i)(1)(iv). Both 40 CFR sections 122.21(e)(3) and 122.44(i)(1)(iv) apply to the selection of a sufficiently sensitive analytical method for the purposes of monitoring and reporting under NPDES permits, including review of permit applications. A USEPA-approved analytical method is sufficiently sensitive where:

- a. The ML is at or below both the level of the applicable water quality criterion/objective and the permit limitation for the measured pollutant or pollutant parameter; or

- b. In permit applications, the ML is above the applicable water quality criterion/objective, but the amount of the pollutant or pollutant parameter in a facility's discharge is high enough that the method detects and quantifies the level of the pollutant or pollutant parameter in the discharge; or
 - c. The method has the lowest ML of the USEPA-approved analytical methods where none of the USEPA-approved analytical methods for a pollutant can achieve the MLs necessary to assess the need for effluent limitations or to monitor compliance with a permit limitation.
2. The MLs in SIP Appendix 4 remain applicable. However, there may be situations when analytical methods are published with MLs that are more sensitive than the MLs for analytical methods listed in the SIP. For instance, USEPA Method 1631E for mercury is not currently listed in SIP Appendix 4, but it is published with an ML of 0.5 ng/L that makes it a sufficiently sensitive analytical method. Similarly, USEPA Method 245.7 for mercury is published with an ML of 5 ng/L.

B. Monitoring Location EFF-001

1. The Permittee shall monitor the discharge of tertiary-treated effluent at EFF-001 as follows. If more than one analytical test method is listed for a given parameter, the Permittee must select from the listed methods and corresponding ML:

Table E-3. Effluent Monitoring

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method and (Minimum Level, units), respectively
Total flow	MGD	recorder	continuous	See #2 below
Turbidity	NTU	recorder	continuous	See #2 & #3 below
Total residual chlorine	mg/L	recorder	continuous	See #4 below
Total residual chlorine	mg/L	grab	daily	See #3 & #5 below
Total coliform	MPN/100mL or CFU/100ml	grab	daily	See #3 & #6 below
E. coli	CFU/100ml	grab	daily	See #3 & #6 below
Temperature	°F	grab	weekly	See #3 below
pH	pH units	grab	weekly	See #3 below
Settleable Solids	mL/L	grab	weekly	See #3 below
Total Suspended Solids (TSS)	mg/L	24-hour composite	weekly	See #3 & #7 below
BOD ₅ 20°C	mg/L	24-hour composite	weekly	See #3 & #8 below
Oil and grease	mg/L	grab	quarterly	See #3 below
Dissolved oxygen	mg/L	grab	quarterly	See #3 below

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method and (Minimum Level, units), respectively
Total Dissolved Solids	mg/L	24-hour composite	monthly	See #3 below
Sulfate	mg/L	24-hour composite	monthly	See #3 below
Chloride	mg/L	24-hour composite	monthly	See #3 below
Boron	mg/L	24-hour composite	monthly	See #3 below
Ammonia Nitrogen	mg/L	24-hour composite	monthly	See #3 below
Nitrite nitrogen	mg/L	24-hour composite	monthly	See #3 below
Nitrate nitrogen	mg/L	24-hour composite	monthly	See #3 below
Organic nitrogen	mg/L	24-hour composite	monthly	See #3 below
Total nitrogen	mg/L	24-hour composite	monthly	See #3 below
Total phosphorus	mg/L	24-hour composite	monthly	See #3 below
Surfactants (MBAS)	mg/L	24-hour composite	quarterly	See #3 below
Surfactants (CTAS)	mg/L	24-hour composite	quarterly	See #3 below
Total hardness (CaCO ₃)	mg/L	24-hour composite	monthly	See #3 below
Chronic toxicity	Pass or Fail, % Effect (TST)	24-hour composite	monthly	See #3 & #9 below
Copper	µg/L	24-hour composite	monthly	See #3 below
Mercury	µg/L	24-hour composite	monthly	See #3 & #7 below
Nickel	µg/L	24-hour composite	monthly	See #3 below
Selenium	µg/L	24-hour composite	monthly	See #3 below
Cyanide	µg/L	grab	monthly	See #3 below
Zinc	µg/L	24-hour composite	quarterly	See #3 below
Bromoform	µg/L	grab	quarterly	See #3 below
Dibromochloromethane	µg/L	grab	quarterly	See #3 below
Chloroform	µg/L	grab	quarterly	See #3 below
Bromodichloromethane	µg/L	grab	quarterly	See #3 below
Bis(2-ethylhexyl) Phthalate	µg/L	grab	monthly	See #3 below
Chlorpyrifos	µg/L	24-hour composite	quarterly	See #3 below
Diazinon	µg/L	24-hour composite	quarterly	See #3 below
Chlordane	µg/L	24-hour composite	quarterly	See #3 below
4,4'-DDD	µg/L	24-hour composite	quarterly	See #3 below

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method and (Minimum Level, units), respectively
4,4'-DDE	µg/L	24-hour composite	quarterly	See #3 below
4,4'-DDT	µg/L	24-hour composite	quarterly	See #3 below
Dieldrin	µg/L	24-hour composite	quarterly	See #3 below
Radioactivity (Including gross alpha, gross beta, combined radium-226 and radium-228, tritium, strontium-90 and uranium)	pCi/L	24-hour composite	semiannually	See #10 below
PCBs as aroclors	µg/L	24-hour composite	quarterly	See #3, #11 & #12 below
PCBs as congeners	µg/L	24-hour composite	semiannually	See #3, #12 & #13 below
Toxaphene	µg/L	24-hour composite	quarterly	See #3 below
Halomethanes	µg/L	grab	quarterly	See #3 & #14 below
Antimony	µg/L	24-hour composite	semiannually	See #3 below
Arsenic	µg/L	24-hour composite	semiannually	See #3 below
Beryllium	µg/L	24-hour composite	semiannually	See #3 below
Cadmium	µg/L	24-hour composite	semiannually	See #3 below
Chromium III	µg/L	calculation	semiannually	See #3 below
Chromium VI	µg/L	grab	semiannually	See #3 below
Lead	µg/L	24-hour composite	semiannually	See #3 below
Silver	µg/L	24-hour composite	semiannually	See #3 below
Thallium	µg/L	24-hour composite	semiannually	See #3 below
2,3,7,8-TCDD	µg/L	24-hour composite	semiannually	See #3 & #15 below
Iron	mg/L	24-hour composite	semiannually	See #3 below
Fluoride	mg/L	24-hour composite	semiannually	See #3 below
Barium	mg/L	24-hour composite	semiannually	See #3 below
Methoxychlor	mg/L	24-hour composite	semiannually	See #3 below
2,4-D	mg/L	24-hour composite	semiannually	See #3 below
2,4,5-TP (Silvex)	mg/L	24-hour composite	semiannually	See #3 below
Perchlorate	µg/L	grab	semiannually	See #16 below
1,4-Dioxane	µg/L	grab	semiannually	See #16 below
1,2,3-Trichloropropane	µg/L	grab	semiannually	See #16 below

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method and (Minimum Level, units), respectively
Methyl tert-butyl-ether (MTBE)	µg/L	grab	semiannually	See #16 below
Remaining USEPA priority pollutants excluding asbestos and PCBs	µg/L	24-hour composite; grab for VOCs	semiannually	See #3 & #17 below

2. Where continuous monitoring of a constituent is required, the following shall be reported:
 Total waste flow – Total daily and peak daily flow (24-hr basis).
 Turbidity – Maximum daily value, total amount of time each day the turbidity exceeded 5 NTU, flow proportioned average daily value.
3. Pollutants shall be analyzed using the analytical methods described in 40 CFR part 136; where no methods are specified for a given pollutant, those methods shall be approved by this Regional Water Board or State Water Resources Control Board. For any pollutant whose effluent limitation is lower than all the minimum levels (MLs) specified in Appendix 4 of the SIP, the analytical method with the lowest ML must be selected.
4. Total residual chlorine shall be recorded continuously. The recorded data shall be maintained by the Permittee for at least five years. The Permittee shall extract the maximum daily peak, and average daily from the recorded media and shall be made available upon request of the Regional Water Board. The continuous monitoring data are not intended to be used for compliance determination purposes.
5. Daily grab samples shall be collected at monitoring location EFF-001, Monday through Friday only, except for holidays. Analytical results of daily grab samples will be used to determine compliance with total residual chlorine effluent limitation. Furthermore, additional monitoring requirements specified in section IV.B.18 shall be followed.
6. Total coliform shall be sampled at least once daily per CCR, Title 22 section 60321(a). However, E. coli testing shall be conducted only if total coliform testing is positive. If the total coliform analysis results in no detection, a result of less than (<) the reporting limit for total coliform will be reported for E. coli.
7. During each reporting period, if effluent monitoring results show that both the TSS and the Mercury water column final effluent limitations were exceeded, then implementation of the Sediment Monitoring, as described in Attachment E of section IV.B.20, is required. Sediment monitoring of the effluent shall begin during the first discharge event following the effluent exceedances.

The mercury effluent samples shall be analyzed using EPA method 1631E, per 40 CFR part 136.

8. If the result of the weekly BOD analysis yields a value greater than the 30-day average limit, the frequency of analysis shall be increased to daily within one week of knowledge of the test result for at least 30 days and until compliance with the 7-day and 30-day average BOD limits is demonstrated; after which the frequency shall revert to weekly.
9. The Permittee shall conduct whole effluent toxicity monitoring as outlined in section V. Please refer to section V.A.7 of this MRP for the accelerated monitoring schedule. The median monthly summary result shall be reported as "Pass" or "Fail." The maximum daily single result shall be reported as "Pass" or "Fail" and "% Effect." When there is a discharge on more than one day in a calendar period, up to three independent toxicity tests may be conducted when one toxicity test results in "Fail."
10. Analyze these radiochemicals by the following USEPA methods: method 900.0 for gross alpha and gross beta, method 903.0 or 903.1 for radium-226, method 904.0 for radium-228, method 906.0 for tritium, method 905.0 for strontium-90, and method 908.0 for uranium. Analysis for combined Radium-226 & 228 shall be conducted only if gross alpha results for the same sample exceed 15 pCi/L or beta greater than 50 pCi/L. If Radium-226 & 228 exceeds the stipulated criteria, analyze for Tritium, Strontium-90 and uranium.
11. PCBs as aroclors mean the sum of Aroclor 1016, Aroclor 1221, Aroclor 1232, Aroclor 1242, Aroclor 1248, Aroclor 1254, and Aroclor 1260 when monitoring using USEPA method 608.
12. USEPA recommends that until USEPA proposed method 1668c for PCBs is incorporated into 40 CFR part 136, Permittees should use for discharge monitoring reports/State monitoring reports: (1) USEPA method 608 for monitoring data, reported as aroclor results, that will be used for assessing compliance with WQBELs established using the WLAs, and (2) USEPA proposed method 1668c for monitoring data, reported as 41 congener results, that will be used for informational purposes for the established TMDL.
13. PCBs as congeners mean the sum of 41 congeners when monitoring using USEPA proposed method 1668c. PCB-18, 28, 37, 44, 49, 52, 66, 70, 74, 77, 81, 87, 99, 101, 105, 110, 114, 118, 119, 123, 126, 128, 138, 149, 151, 153, 156, 157, 158, 167, 168, 169, 170, 177, 180, 183, 187, 189, 194, 201, and 206 shall be individually quantified.
14. Halomethanes shall mean the sum of bromoform, bromodichloromethane, chloroform, and dibromochloromethane.
15. In accordance with the SIP, the Permittee shall conduct effluent monitoring for the seventeen 2,3,7,8-tetrachlorodibenzo-p-dioxin (2,3,7,8-TCDD or dioxin) congeners in the effluent and in the receiving water Station RSW-002, located downstream of the discharge point. The Permittee shall use the appropriate Toxicity Equivalence Factor (TEF) to determine Toxic Equivalence (TEQ). Where TEQ equals the product between each of the 17 individual congeners' (i)

concentration analytical result (C_i) and their corresponding Toxicity Equivalence Factor (TEFi), (i.e., $TEQ_i = C_i \times TEF_i$). Compliance with the Dioxin limitation shall be determined by the summation of the seventeen individual TEQs, or the following equation:

$$\text{Dioxin concentration in effluent} = \sum_{i=1}^{17} (TEQ_i) = \sum_{i=1}^{17} (C_i)(TEF_i)$$

16. Emerging chemicals include 1,4-dioxane (USEPA 8270M test method), perchlorate (USEPA 314 test method, or USEPA method 331 if a detection limit of less than 6 µg/L is achieved), 1,2,3-trichloropropane (USEPA 504.1, 8260B test method, or USEPA 524.2 in SIM mode), and methyl tert-butyl ether (USEPA 8260B test method or USEPA method 624 if a detection level of less than 5 µg/L is achieved, and if the Permittee received ELAP certification to run USEPA method 624).
17. Priority pollutants are those constituents referred to in 40 CFR section 401.15; a list of these pollutants is provided as Appendix A to 40 CFR part 423.

18. Total Residual Chlorine Additional Monitoring

Continuous monitoring of total residual chlorine at EFF-001 shall serve as an internal trigger for the increased grab sampling at EFF-001 if either of the following occurs, except as noted in item c:

- a. Total residual chlorine concentration excursions of up to 0.3 mg/L lasting greater than 15 minutes; or
- b. Total residual chlorine concentration peaks in excess of 0.3 mg/L lasting greater than 1 minute.
- c. Additional grab samples need not be taken if it can be demonstrated that a stoichiometrically appropriate amount of dechlorination chemical has been added to effectively dechlorinate the effluent to 0.1 mg/L or less for peaks in excess of 0.3 mg/L lasting more than 1 minute, but not for more than five minutes.

19. Salts Dry- and Wet-Weather Monitoring and Reporting Requirements

The Discharger shall determine the applicable wet- or dry-weather flow condition at RSW-003D and the amount of rainfall at the time of effluent sampling. The Discharger shall tabulate the date of sampling, average flow at RSW-003D, amount of rainfall, wet- or dry weather, applicable effluent limitation (wet- or dry-weather), and actual effluent concentration/mass, as presented in Table E-4.

Table E-4. Salts Monitoring and Reporting Requirements

Parameter	Date of Sampling	Flow (cfs)	Rainfall Amount (inches)	Wet or Dry Weather	Applicable Effluent Limitation	Actual Effluent Concentration/Mass
TDS (wet-weather)						

Parameter	Date of Sampling	Flow (cfs)	Rainfall Amount (inches)	Wet or Dry Weather	Applicable Effluent Limitation	Actual Effluent Concentration/ Mass
TDS (dry-weather)						
Sulfate (wet-weather)						
Sulfate (dry-weather)						
Chloride (wet-weather)						
Chloride (dry-weather)						
Boron (wet-weather)						
Boron (dry-weather)						

20. Sediment Monitoring of Effluent at Monitoring Location EFF-001

The Permittee must sample the discharge at the point following final treatment, prior to entering the receiving water. The exact location of the sampling point must be stipulated in the initial self-monitoring report. All samples shall be tested in accordance with USEPA or ASTM methodologies where such methods exist. Where no USEPA or ASTM methods exist, the State Water Board or Regional Water Board (collectively Water Boards) shall approve the use of other methods. Analytical tests shall be conducted by laboratories certified by the California Department of Public Health in accordance with Water Code Section 13176.

Table E-5. Sediment Monitoring

Parameter	Units	Sample Type	Minimum Sampling Frequency
Mercury	mg/kg	grab	Once per year

Sediment Monitoring is only required during a reporting period if effluent water column monitoring results for both TSS and Mercury are exceeded. If monitoring is not triggered because both TSS and Mercury limits were not exceeded, then at a minimum, sediment monitoring must occur at least once during the five-year permit term.

V. WHOLE EFFLUENT TOXICITY TESTING REQUIREMENTS

A. Chronic Toxicity Testing

1. Discharge In-stream Waste Concentration (IWC) for Chronic Toxicity

The chronic toxicity IWC for this discharge is 100 percent effluent.

2. Sample Volume and Holding Time

The total sample volume shall be determined by the specific toxicity test method used. Sufficient sample volume shall be collected to perform the required toxicity test. For the receiving water, sufficient sample volume shall also be collected during accelerated monitoring for subsequent TIE studies, if necessary, at each sampling event. All toxicity tests shall be conducted as soon as possible following sample collection. No more than 36 hours shall elapse before the conclusion of sample collection and test initiation.

3. Chronic Freshwater Species and Test Methods

If effluent samples are collected from outfalls discharging to receiving waters with salinity <1 ppt, the Permittee shall conduct the following chronic toxicity tests on effluent samples at the in-stream waste concentration for the discharge in accordance with species and test methods in *Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Freshwater Organisms* (EPA/821/R-02/013, 2002; Table IA, 40 CFR part 136). In no case shall these species be substituted with another test species unless written authorization from the Executive Officer is received.

- a. A static renewal toxicity test with the fathead minnow, *Pimephales promelas* (Larval Survival and Growth Test Method 1000.0).
- b. A static renewal toxicity test with the daphnid, *Ceriodaphnia dubia* (Survival and Reproduction Test Method 1002.01).
- c. A static toxicity test with the green alga, *Selenastrum capricornutum* (also named *Raphidocelis subcapitata*) (Growth Test Method 1003.0).

4. Species Sensitivity Screening

Species sensitivity screening shall be conducted during this permit's first required sample collection. The Permittee shall collect a single effluent sample to initiate and concurrently conduct three toxicity tests using the fish, an invertebrate, and the alga species previously referenced. This sample shall also be analyzed for the parameters required on a monthly frequency for the discharge, during that given month. As allowed under the test method for the *Ceriodaphnia dubia* and the *Pimephales promelas*, a second and third sample may be collected for use as test solution renewal water as the seven-day toxicity test progresses. However, that same sample shall be used to renew both the *Ceriodaphnia dubia* and the *Pimephales promelas*. If the result of all three species is "Pass", then the species that exhibits the highest "Percent Effect" at the discharge IWC during species sensitivity screening shall be used for routine monitoring during the permit cycle. If only one species fails, then that species shall be used for routine monitoring during the permit cycle. If two or more species result in "Fail," then the species that exhibits the highest "Percent Effect" at the discharge IWC during the suite of species sensitivity screening shall be used for routine monitoring during the permit cycle, until such time as a rescreening is required (24 months later).

Species sensitivity rescreening is required every 24 months if there has been discharge during dry weather conditions. If the intermittent discharge is only

during wet weather, rescreening is not required. If rescreening is necessary, the Permittee shall rescreen with the fish, an invertebrate, and the alga species previously referenced and continue to monitor with the most sensitive species. If the first suite of rescreening tests demonstrates that the same species is the most sensitive then the rescreening does not need to include more than one suite of tests. If a different species is the most sensitive or if there is ambiguity, then the Permittee shall proceed with suites of screening tests for a minimum of three, but not to exceed five suites.

During the calendar month, toxicity tests used to determine the most sensitive test species shall be reported as effluent compliance monitoring results for the chronic toxicity MDEL and MMEL.

5. Quality Assurance and Additional Requirements

Quality assurance measures, instructions, and other recommendations and requirements are found in the test methods manual previously referenced. Additional requirements are specified below.

- a. The discharge is subject to determination of "Pass" or "Fail" and "Percent Effect" from a chronic toxicity test using the Test of Significant Toxicity (TST) statistical t-test approach described in *National Pollutant Discharge Elimination System Test of Significant Toxicity Implementation Document* (EPA 833-R-10-003, 2010), Appendix A, Figure A-1, Table A-1 and Appendix B, Table B-1. The null hypothesis (H_0) for the TST approach is: Mean discharge IWC response $\leq 0.75 \times$ Mean control response. A test result that rejects this null hypothesis is reported as "Pass." A test result that does not reject this null hypothesis is reported as "Fail." The relative "Percent Effect" at the discharge IWC is defined and reported as: $((\text{Mean control response} - \text{Mean discharge IWC response}) \div \text{Mean control response}) \times 100$. This is a t-test (formally Student's t-Test), a statistical analysis comparing two sets of replicate observations - in the case of WET, only two test concentrations (i.e., a control and IWC). The purpose of this statistical test is to determine if the means of the two sets of observations are different (i.e., if the IWC or receiving water concentration differs from the control (the test result is "Pass" or "Fail")). The Welch's t-test employed by the TST statistical approach is an adaptation of Student's t-test and is used with two samples having unequal variances.
- b. The Median Monthly Effluent Limit (MMEL) for chronic toxicity only applies when there is a discharge on more than one day in a calendar month period. During such calendar months, up to three independent toxicity tests may be conducted when one toxicity test results in "Fail."
- c. If the effluent toxicity test does not meet all test acceptability criteria (TAC) specified in the referenced test method *Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Freshwater Organisms* (USEPA 2002, EPA-821-R-02-013) (See Table E.6, below), then the Permittee must re-sample and re-test within 14 days.

Table E-6. USEPA Test Methods and Test Acceptability Criteria

Species & USEPA Test Method Number	Test Acceptability Criteria (TAC)
Fathead Minnow, <i>Pimephales promelas</i> , Larval Survival and Growth Test Method 1000.0 (Table 1 of the test method, above)	80% or greater survival in controls; average dry weight per surviving organism in control chambers equals or exceeds 0.25 mg. (required)
Daphnid, <i>Ceriodaphnia dubia</i> , Survival and Reproduction Test Method 1002.0. (Table 3 of the test method, above)	80% or greater survival of all control organisms and an average of 15 or more young per surviving female in the control solutions. 60% of surviving control females must produce three broods. (required)
Green Alga, <i>Selenastrum capricornutum</i> , Growth Toxicity Test Method 1003.0. (Table 3 of the test method, above)	Mean cell density of at least 1×10^6 cells/mL in the controls; and variability (CV%) among control replicates less than or equal to 20%. (required)

- d. Dilution water and control water, including brine controls, shall be laboratory water prepared and used as specified in the test methods manual. If dilution water and control water is different from test organism culture water, then a second control using culture water shall also be used.
- e. When preparing samples for toxicity testing, in addition to the required monitoring for conductivity, etc., it is recommended that total alkalinity and total hardness be measured in the undiluted effluent, receiving water, dilution water, and culture water (following the WET methods manual), as well as the major geochemical ions (see Mount et al., 2018).
- f. Monthly reference toxicant testing is sufficient. All reference toxicant test results should be reviewed and reported using EC25. EC25 is a point estimate of the toxicant concentration that would cause an observable adverse effect (e.g., death, immobilization, or serious incapacitation) in 25 percent of the test organisms.
- g. The Permittee shall perform toxicity tests on final effluent samples. Chlorine and ammonia shall not be removed from the effluent sample prior to toxicity testing, unless explicitly authorized under this section of the Monitoring and Reporting Program and the rationale is explained in the Fact Sheet (Attachment F).

6. Preparation of an Initial Investigation TRE Work Plan

The Permittee shall prepare and submit a copy of the Permittee's initial investigation TRE work plan to the Executive Officer of the Regional Water Board for approval within 90 days of the effective date of this permit. If the Executive Officer does not disapprove the work plan within 60 days, the work plan shall

become effective. The Permittee shall use USEPA manual EPA/833B-99/002 (municipal) as guidance, or most current version, or EPA manual *Generalized Methodology for Conducting Industrial Toxicity Reduction Evaluations* (EPA/600/2-88/070, April 1989). At a minimum, the TRE Work Plan must contain the provisions in Attachment G. This work plan shall describe the steps that the Permittee intends to follow if toxicity is detected. At minimum, the work plan shall include:

- a. A description of the investigation and evaluation techniques that will be used to identify potential causes and sources of toxicity, effluent variability, and treatment system efficiency.
- b. A description of the Facility's methods of maximizing in-house treatment efficiency and good housekeeping practices, and a list of all chemicals used in the operation of the Facility.
- c. If a TIE is necessary, an indication of the person who would conduct the TIEs (i.e., an in-house expert or an outside contractor).

7. Accelerated Monitoring Schedule for Median Monthly Summary Result: "Fail"; and Accelerated Monitoring Schedule for Maximum Daily Single Result: "Fail and % Effect ≥ 50 ."

When there is discharge on more than one day in a calendar month, the Median Monthly summary result shall be used to determine if accelerated testing needs to be conducted. When there is discharge on only one day in a calendar month, the Maximum Daily single result shall be used to determine if accelerated testing needs to be conducted.

Once the Permittee becomes aware of this result, the Permittee shall implement an accelerated monitoring schedule within seven calendar days for the *Ceriodaphnia dubia* test, and within 5 calendar days for both the *Pimephales promelas* and *Selenastrum capricornutum* tests. The accelerated monitoring schedule shall consist of four toxicity tests (including IWC), conducted at approximately two week intervals, over an eight week period; in preparation for the TRE process and associated reporting, these results shall also be reported using the EC25. If each of the accelerated toxicity tests results in "Pass", the Permittee shall return to routine monitoring for the next monitoring period.

8. TRE Process

If one of the accelerated toxicity tests results in "Fail", the Permittee shall immediately implement the TRE Process conditions set forth below. During the TRE Process, monthly effluent monitoring shall resume and TST results ("Pass" or "Fail", "Percent Effect") for chronic toxicity tests shall be reported as effluent compliance monitoring results for the chronic toxicity MDEL and MMEL.

- a. **Preparation and Implementation of Detailed TRE Work Plan.** The Permittee shall immediately initiate a TRE using, according to the type of treatment facility, USEPA manual *Toxicity Reduction Evaluation Guidance for Municipal Wastewater Treatment Plants* (EPA/833/B-99/002, 1999) or EPA manual *Generalized Methodology for Conducting Industrial Toxicity*

Reduction Evaluations (EPA/600/2-88/070, April 1989) and, within 30 days, submit to the Executive Officer a Detailed TRE Work Plan, which shall follow the TRE Work Plan revised as appropriate for this toxicity event. It shall include the following information, and comply with additional conditions set by the Executive Officer:

- i. Further actions by the Permittee to investigate, identify, and correct the causes of toxicity.
 - ii. Actions the Permittee will take to mitigate the effects of the discharge and prevent the recurrence of toxicity.
 - iii. A schedule for these actions, progress reports, and the final report.
- b. **TIE Implementation.** The Permittee may initiate a TIE as part of a TRE to identify the causes of toxicity using the same species and test method and, as guidance, USEPA manuals: *Methods for Aquatic Toxicity Identification Evaluations: Phase I Toxicity Characterization Procedures* (EPA/600/6-91/003, 1991); *Chronic TIE Manual: Toxicity Identification Evaluation: Characterization of Chronically Toxic Effluents, Phase I* (EPA/600/6-91/005F, 1992); *Methods for Aquatic Toxicity Identification Evaluations, Phase II Toxicity Identification Procedures for Samples Exhibiting Acute and Chronic Toxicity* (EPA/600/R-92/080, 1993); *Methods for Aquatic Toxicity Identification Evaluations, Phase III Toxicity Confirmation Procedures for Samples Exhibiting Acute and Chronic Toxicity* (EPA/600/R-92/081, 1993); and *Marine Toxicity Identification Evaluation (TIE): Phase I Guidance Document* (EPA/600/R-96-054, 1996). The TIE should be conducted on the species demonstrating the most sensitive toxicity response.
- c. Many recommended TRE elements parallel required or recommended efforts for source control, pollution prevention, and storm water control programs. TRE efforts should be coordinated with such efforts. As toxic substances are identified or characterized, the Permittee shall continue the TRE by determining the sources and evaluating alternative strategies for reducing or eliminating the substances from the discharge. All reasonable steps shall be taken to reduce toxicity to levels consistent with toxicity evaluation parameters.
- d. The Permittee shall continue to conduct routine effluent monitoring for compliance determination purposes while the TIE and/or TRE process is taking place. Additional accelerated monitoring and TRE work plans are not required once a TRE has begun.
- e. The Regional Water Board recognizes that toxicity may be episodic and identification of causes and reduction of sources of toxicity may not be successful in all cases. The TRE may be ended at any stage if monitoring finds there is no longer toxicity.
- f. The Board may consider the results of any TIE/TRE studies in an enforcement action.

9. Reporting

The Self-Monitoring Report (SMR) shall include a full laboratory report for each toxicity test. This report shall be prepared using the format and content of the test methods manual chapter called Report Preparation, including:

- a. The valid toxicity test results for the TST statistical approach, reported as “Pass” or “Fail” and “Percent Effect” at the chronic toxicity IWC for the discharge. All toxicity test results (whether identified as valid or otherwise) conducted during the calendar month shall be reported on the SMR due date specified in Table E-10.
- b. A summary of water quality measurements for each toxicity test (e.g., pH, dissolved oxygen, temperature, conductivity, hardness, salinity, chlorine, ammonia).
- c. The statistical analysis used in National Pollutant Discharge Elimination System Test of Significant Toxicity Implementation Document (EPA 833-R-10-003, 2010) Appendix A, Figure A-1, Table A-1, and Appendix B, Table B-1.
- d. TRE/TIE results. The Executive Officer shall be notified no later than 30 days from completion of each aspect of TRE/TIE analyses. Prior to the completion of the final TIE/TRE report, the Permittee shall provide status updates in the monthly monitoring reports, indicating which TIE/TRE steps are underway and which steps have been completed.
- e. Statistical program (e.g., TST calculator, CETIS, etc.) output results, including graphical plots, for each toxicity test.
- f. Tabular data and graphical plots clearly showing the laboratory’s performance for the reference toxicant, for each solution, for the previous 20 tests and the laboratory’s performance for the control mean, control standard deviation, and control coefficient of variation, for each solution, for the previous 12-month period.
- g. Any additional QA/QC documentation or any additional chronic toxicity-related information, upon request from the Regional Water Board Chief Deputy Executive Officer or the Executive Officer.

B. Ammonia Removal

1. Except with prior approval from the Executive Officer of the Regional Water Board, ammonia shall not be removed from bioassay samples. The Permittee must demonstrate the effluent toxicity is caused by ammonia because of increasing test pH when conducting the toxicity test. It is important to distinguish the potential toxic effects of ammonia from other pH sensitive chemicals, such as certain heavy metals, sulfide, and cyanide. The following may be steps to demonstrate that the toxicity is caused by ammonia and not other toxicants before the Executive Officer would allow for control of pH in the test.
 - a. There is consistent toxicity in the effluent and the maximum pH in the toxicity test is in the range to cause toxicity due to increased pH.

- b. Chronic ammonia concentrations in the effluent are greater than 4 mg/L total ammonia.
 - c. Conduct graduated pH tests as specified in the toxicity identification evaluation methods. For example, mortality should be higher at pH 8 and lower at pH 6.
 - d. Treat the effluent with a zeolite column to remove ammonia. Mortality in the zeolite treated effluent should be lower than the non-zeolite treated effluent. Then add ammonia back to the zeolite-treated samples to confirm toxicity due to ammonia.
2. When it has been demonstrated that toxicity is due to ammonia because of increasing test pH, pH may be controlled using appropriate procedures which do not significantly alter the nature of the effluent, after submitting a written request to the Regional Water Board, and receiving written permission expressing approval from the Executive Officer of the Regional Water Board.

C. Chlorine Removal

1. Except with prior approval from the Executive Office of the Regional Water Board, chlorine shall not be removed from bioassay samples.

VI. LAND DISCHARGE MONITORING REQUIREMENTS (NOT APPLICABLE)

VII. RECYCLING MONITORING REQUIREMENTS (NOT APPLICABLE)

VIII. RECEIVING WATER MONITORING REQUIREMENTS

A. Monitoring Location RSW-001 and RSW-002

The Permittee shall monitor Arroyo Simi at RSW-001 and RSW-002 as follows:

Table E-7. Receiving Water Monitoring Requirements

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method
Total flow	cfs	calculation	monthly	-
Turbidity	NTU	grab	monthly	See #2 below
Total residual chlorine	mg/L	grab	monthly	See #2 below
E. coli	CFU/100ml	grab	monthly	See #2 below
Temperature	°F	grab	monthly	See #2 below
pH	pH units	grab	monthly	See #2 below
Settleable Solids	mL/L	grab	monthly	See #2 below
Total Suspended Solids	mg/L	grab	monthly	See #2 below
BOD ₅ 20°C	mg/L	grab	monthly	See #2 below
Oil and grease	mg/L	grab	quarterly	See #2 below
Dissolved oxygen	mg/L	grab	monthly	See #2 below

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method
Conductivity	µmho/cm	grab	monthly	See #2 below
Total Dissolved Solids	mg/L	grab	monthly	See #2 below
Sulfate	mg/L	grab	monthly	See #2 below
Chloride	mg/L	grab	monthly	See #2 below
Boron	mg/L	grab	monthly	See #2 below
Nitrate nitrogen	mg/L	grab	quarterly	See #2 below
Nitrite nitrogen	mg/L	grab	quarterly	See #2 below
Ammonia nitrogen	mg/L	grab	quarterly	See #2 below
Organic nitrogen	mg/L	grab	quarterly	See #2 below
Total kjeldahl nitrogen (TKN)	mg/L	grab	quarterly	See #2 below
Total nitrogen	mg/L	grab	quarterly	See #2 below
Total phosphorus	mg/L	grab	quarterly	See #2 below
Orthophosphate-P	mg/L	grab	quarterly	See #2 below
Algal biomass	mg/cm ²	grab	annually	See #2 & #3 below
Surfactants (MBAS)	mg/L	grab	quarterly	See #2 below
Surfactants (CTAS)	mg/L	grab	quarterly	See #2 below
Total hardness (CaCO ₃)	mg/L	grab	monthly	See #2 below
Chronic toxicity	Pass or Fail, % Effect (TST)	grab	quarterly	See #2 & #4 below
Copper	µg/L	grab	quarterly	See #2 below
Mercury	µg/L	grab	quarterly	See #2 below
Nickel	µg/L	grab	quarterly	See #2 below
Selenium	µg/L	grab	quarterly	See #2 below
Cyanide	µg/L	grab	quarterly	See #2 below
Bromoform	µg/L	grab	quarterly	See #2 below
Dibromochloromethane	µg/L	grab	quarterly	See #2 below
Chloroform	µg/L	grab	quarterly	See #2 below
Bromodichloromethane	µg/L	grab	quarterly	See #2 below
Bis(2-ethyhexyl)Phthalate	µg/L	grab	quarterly	See #2 below
Chlorpyrifos	µg/L	grab	quarterly	See #2 below
Diazinon	µg/L	grab	quarterly	See #2 below
Chlordane	µg/L	grab	quarterly	See #2 below
4,4'-DDD	µg/L	grab	quarterly	See #2 below
4,4'-DDE	µg/L	grab	quarterly	See #2 below
4,4'-DDT	µg/L	grab	quarterly	See #2 below

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method
Dieldrin	µg/L	grab	quarterly	See #2 below
PCBs as aroclors	µg/L	grab	quarterly	See #2, #5 & #6 below
PCBs as congeners	µg/L	grab	semiannually	See #2, #6 & #7 below
Toxaphene	µg/L	grab	quarterly	See #2 below
Antimony	µg/L	grab	semiannually	See #2 below
Arsenic	µg/L	grab	semiannually	See #2 below
Beryllium	µg/L	grab	semiannually	See #2 below
Cadmium	µg/L	grab	semiannually	See #2 below
Chromium III	µg/L	calculation	semiannually	See #2 below
Chromium VI	µg/L	grab	semiannually	See #2 below
Lead	µg/L	grab	semiannually	See #2 below
Silver	µg/L	grab	semiannually	See #2 below
Thallium	µg/L	grab	semiannually	See #2 below
Zinc	µg/L	grab	semiannually	See #2 below
1,4-Dioxane	µg/L	grab	semiannually	See #8 below
Perchlorate	µg/L	grab	semiannually	See #8 below
1,2,3-Trichloropropane	µg/L	grab	semiannually	See #8 below
Methyl tert-butyl-ether (MTBE)	µg/L	grab	semiannually	See #8 below
2,3,7,8-TCDD	µg/L	grab	semiannually	See #2 & #9 below
Iron	µg/L	grab	semiannually	See #2 below
Fluoride	mg/L	grab	semiannually	See #2 below
Barium	µg/L	grab	semiannually	See #2 below
Methoxychlor	µg/L	grab	semiannually	See #2 below
2,4-D	µg/L	grab	semiannually	See #2 below
2,4,5-TP (Silvex)	µg/L	grab	semiannually	See #2 below
Remaining USEPA priority pollutants excluding asbestos and PCBs	µg/L	grab	semiannually	See #2 & #10 below

2. Pollutants shall be analyzed using the analytical methods described in 40 CFR part 136; where no methods are specified for a given pollutant, those methods shall be approved by this Regional Water Board or State Water Board. For any pollutant whose effluent limitation is lower than all the minimum levels (MLs) specified in Attachment 4 of the SIP, the analytical method with the lowest ML must be selected.

3. Algal biomass or Chlorophyll A samples shall be collected by obtaining scrapings from the substrate. This will be a measure of benthic algae, rather than algae in the water column. Percent cover shall also be reported. Algal biomass monitoring shall be conducted concurrently with bioassessment monitoring.
4. The Permittee shall conduct whole effluent toxicity monitoring as outlined in section V. Please refer to section V.A.7 of this MRP for the accelerated monitoring schedule. The median monthly summary result is a threshold value for a determination of meeting the narrative receiving water objective and shall be reported as "Pass" or "Fail." The maximum daily single result is a threshold value for a determination of meeting the narrative receiving water objective and shall be reported as "Pass or Fail" and "% Effect." Up to three independent toxicity tests may be conducted when one toxicity test results in "Fail." If the chronic toxicity median monthly threshold at the immediate downstream receiving water location is not met and the toxicity cannot be attributed to upstream toxicity, as assessed by the Permittee, then the Permittee shall initiate accelerated monitoring. For example, if the chronic toxicity median monthly threshold of the receiving water at both upstream and downstream stations is not met, but the effluent chronic toxicity median monthly effluent limitation was met, then accelerated monitoring need not be implemented.
5. PCBs mean the sum of Aroclor 1016, Aroclor 1221, Aroclor 1232, Aroclor 1242, Aroclor 1248, Aroclor 1254, and Aroclor 1260 when monitoring using USEPA method 608.
6. USEPA recommends that until USEPA proposed method 1668c for PCBs is incorporated into 40 CFR part 136, Permittees should use for discharge monitoring reports/State monitoring reports: (1) USEPA method 608 for monitoring data, reported as aroclor results, that will be used for assessing compliance with WQBELs established using the WLAs, and (2) USEPA proposed method 1668c for monitoring data, reported as 41 congener results, that will be used for informational purposes for the established TMDL.
7. PCBs mean the sum of 41 congeners when monitoring using USEPA proposed method 1668c. PCB-18, 28, 37, 44, 49, 52, 66, 70, 74, 77, 81, 87, 99, 101, 105, 110, 114, 118, 119, 123, 126, 128, 138, 149, 151, 153, 156, 157, 158, 167, 168, 169, 170, 177, 180, 183, 187, 189, 194, 201, and 206 shall be individually quantified.
8. Emerging chemicals include 1,4-dioxane (USEPA 8270M test method), perchlorate (USEPA 314 test method, or USEPA method 331 if a detection limit of less than 6 µg/L is achieved), 1,2,3-trichloropropane (USEPA 504.1, 8260B test method, or USEPA 524.2 in SIM mode), and methyl tert-butyl ether (USEPA 8260B test method or USEPA method 624 if a detection level of less than 5 µg/L is achieved, and if the Permittee received ELAP certification to run USEPA method 624).
9. In accordance with the SIP, the Permittee shall conduct effluent monitoring for the seventeen 2,3,7,8-tetrachlorodibenzo-p-dioxin (2,3,7,8-TCDD or dioxin) congeners in the effluent and in the receiving water Station RSW-002, located downstream of discharge point 001. The Permittee shall use the appropriate

Toxicity Equivalence Factor (TEF) to determine Toxic Equivalence (TEQ). Where TEQ equals the product between each of the 17 individual congeners' (i) concentration analytical result (C_i) and their corresponding Toxicity Equivalence Factor (TEF_i), (i.e., $TEQ_i = C_i \times TEF_i$). Compliance with the Dioxin limitation shall be determined by the summation of the seventeen individual TEQs, or the following equation:

$$\text{Dioxin concentration in effluent} = \sum_{i=1}^{17} (TEQ_i) = \sum_{i=1}^{17} (C_i)(TEF_i)$$

10. Priority pollutants are those constituents referred to in 40 CFR section 401.15; a list of these pollutants is provided as Appendix A to 40 CFR part 423.
11. Receiving water samples shall not be taken during or within 48-hours following the flow of rainwater runoff into the Arroyo Simi. Sampling may be rescheduled at receiving water stations if weather and/or flow conditions would endanger personnel collecting receiving water samples. The monthly monitoring report shall note such occasions.

B. TMDL Stream Flow and Rainfall Monitoring

1. In order to determine the dry- and wet-weather flow conditions in the receiving water, the Permittee shall report the average daily flow at Arroyo Simi collected from an existing stream flow gauging station located at Calleguas Creek, Ventura County Watershed Protection station 805. For the purposes of this permit, this station is also known as RSW-003D. The Permittee shall also report the total daily rainfall from an existing rainfall gauging station located at the Simi Valley WQCP, Station ID: 246A. The stream flow and rainfall gauging stations are operated and maintained by the Ventura County Watershed Protection District. The required stream flow and rainfall data are available online at <http://www.vcwatershed.net/fws/>.

Calleguas Creek Salts TMDL has defined dry-weather as the condition in the receiving water when the flows are below the 86th percentile of the flow and there is no measurable precipitation. The 86th percentile of the flow was given in the TMDL staff report. The rainfall precipitation shall be obtained from an existing rainfall gauging station located at the Simi Valley WQCP, Station ID: 246A. If the gauging stations are not operational, an estimated average daily flow and rainfall may be submitted.

Table E-8. TMDL Stream Flow and Rainfall Monitoring Requirements

Parameter	Units	Sample Type	Minimum Sampling Frequency	Required Analytical Test Method
Average Daily Flow	cubic feet per second (cfs)	On-line data	daily	Not applicable
Total Daily Rainfall	inches	On-line data	daily	Not applicable

IX. OTHER MONITORING REQUIREMENTS

A. Calleguas Creek TMDLs Monitoring Requirements

1. The TMDL monitoring program is discussed in section VI.C.2.a. of the Order.

B. Watershed Monitoring

1. The goals of the Watershed-wide Monitoring Program for the Calleguas Creek Watershed are to:
 - a. Determine compliance with receiving water limits.
 - b. Monitor trends in surface water quality.
 - c. Ensure protection of beneficial uses.
 - d. Provide data for modeling contaminants of concern.
 - e. Characterize water quality including seasonal variation of surface waters within the watershed.
 - f. Assess the health of the biological community.
 - g. Determine mixing dynamics of effluent and receiving waters in the estuary.
2. The Permittee shall participate in the implementation of the Watershed-wide Monitoring Program developed by stakeholders and initiated in 2008. The City's responsibilities under the Watershed-wide Monitoring Program are described in the Receiving Water Monitoring Requirements section. To achieve the goals of the Watershed-wide Monitoring Program, revisions to the Receiving Water Monitoring Requirements may be made under the direction of USEPA and the Regional Water Board. The Permittee shall submit annual reports providing the monitoring data collected during the calendar year, as well as an interpretation of the significance of the results with respect to the health of the watershed. Annual reports shall be submitted by July 1st of each year.
3. In coordination with interested stakeholders in the Calleguas Creek Watershed, the Permittee shall conduct bioassessment program annually in the spring/summer period and include an analysis of the community structure of the instream macroinvertebrate assemblages, the community structure of the instream algal assemblages (benthic diatoms and soft-bodied algae), chlorophyll a and biomass for instream algae, and physical habitat assessment at the random monitoring stations designated by the Calleguas Creek Watershed Monitoring Program.
 - a. The bioassessment program shall include an analysis of the community structure of the instream macroinvertebrate assemblages and physical habitat assessment at monitoring stations RSW-001 and RSW-002.

This program shall be implemented by appropriately trained staff. Alternatively, a professional subcontractor qualified to conduct bioassessments may be selected to perform the bioassessment work for the Permittee. Analyses of the results of the bioassessment monitoring program, along with photographs of the monitoring site locations taken during sample

collection, shall be submitted in the corresponding annual report. If another stakeholder, or interested party in the watershed subcontracts a qualified professional to conduct bioassessment monitoring during the same season and at the same location as specified in the MRP, then the Permittee may, in lieu of duplicative sampling, submit the data, a report interpreting the data, photographs of the site, and related QA/QC documentation in the corresponding annual report.

- b. The Permittee must provide a copy of their Standard Operation Procedures (SOPs) for the Bioassessment Monitoring Program to the Regional Water Board upon request. The document must contain step-by-step field, laboratory and data entry procedures, as well as, related QA/QC procedures. The SOP must also include specific information about each bioassessment program including: assessment program description, its organization and the responsibilities of all its personnel; assessment project description and objectives; qualifications of all personnel; and the type of training each member has received.
 - c. Field sampling must conform to the SOP established for the California Stream Bioassessment Procedure (CSBP) or more recently established sampling protocols, such as used by the Surface Water Ambient Monitoring Program (SWAMP). Field crews shall be trained on aspects of the protocol and appropriate safety issues. All field data and sample Chain of Custody (COC) forms must be examined for completion and gross errors. Field inspections shall be planned with random visits and shall be performed by the Permittee or an independent auditor. These visits shall report on all aspects of the field procedure with corrective action occurring immediately.
 - d. A taxonomic identification laboratory shall process the biological samples that usually consist of subsampling organisms, enumerating and identifying taxonomic groups and entering the information into an electronic format. The Regional Water Board may require QA/QC documents from the taxonomic laboratories and examine their records regularly. Intra-laboratory QA/QC for subsampling, taxonomic validation and corrective actions shall be conducted and documented. Biological laboratories shall also maintain reference collections, vouchered specimens (the Permittee may request the return of their sample voucher collections) and remnant collections. The laboratory should participate in an (external) laboratory taxonomic validation program at a recommended level of 10% or 20%. External QA/QC may be arranged through the California Department of Fish and Wildlife's Aquatic Bioassessment Laboratory located in Rancho Cordova, California.
4. The Executive Officer of the Regional Water Board may modify Monitoring and Reporting Program to accommodate the watershed-wide monitoring.

C. Tertiary Filter Treatment Bypasses

1. During any day that filters are bypassed, the Permittee shall monitor the effluent for BOD, suspended solids, settleable solids, and oil and grease, on daily basis,

until it is demonstrated that the filter “bypass” has not caused an adverse impact on the receiving water.

2. The Permittee shall maintain chronological log of tertiary filter treatment process bypasses, to include the following:
 - a. Date and time of bypass start and end;
 - b. Total duration time; and,
 - c. Estimated total volume bypassed
3. The Permittee shall submit a written report to the Regional Water Board, according to the corresponding monthly self-monitoring report schedule. The report shall include, at a minimum, the information from the chronological log. Results from the daily effluent monitoring, required by C.1. above, shall be verbally reported to the Regional Water Board as the results become available and submitted as part of the monthly SMR.

X. REPORTING REQUIREMENTS

A. General Monitoring and Reporting Requirements

1. The Permittee shall comply with all Standard Provisions (Attachment D) related to monitoring, reporting, and recordkeeping.
2. If there is no discharge during any reporting period, the report shall so state.
3. Each monitoring report shall contain a separate section titled “Summary of Non-Compliance” which discusses the compliance record and the corrective actions taken or planned that may be needed to bring the discharge into full compliance with waste discharge requirements. This section shall clearly list all non-compliance with discharge requirements, as well as all excursions of effluent limitations.
4. The Permittee shall inform the Regional Water Board well in advance of any proposed construction activity that could potentially affect compliance with applicable requirements.

B. Calleguas Creek TMDL Monitoring and Reporting Requirements

The Calleguas Creek Watershed TMDL Monitoring Plan (CCWTMP) is designed to monitor and evaluate the implementation of multiple TMDLs within the Calleguas Creek watershed. The CCWTMP was created to better facilitate a coordinated monitoring effort where multiple TMDL monitoring requirements could be addressed via a single program that would carry out and manage all aspects of the monitoring activities. The TMDLs in this watershed include Calleguas Creek Watershed Nutrients TMDL, Toxicity TMDL, and OC Pesticide, PCBs, and Sediment TMDL. This monitoring program has been developed to easily integrate new TMDL monitoring efforts as TMDLs are adopted and/or special study monitoring efforts are required.

The Permittee shall submit reports to the Regional Water Board as required by the approved CCWTMP.

(See also section VI.C.2.a. of the Order for Monitoring and Reporting Requirements).

C. Self-Monitoring Reports (SMRs)

1. The Permittee shall electronically submit SMRs using the State Water Board's California Integrated Water Quality System (CIWQS) Program Web site (<http://www.waterboards.ca.gov/ciwqs/index.html>). The CIWQS Web site will provide additional information for SMR submittal in the event there will be a planned service interruption for electronic submittal.
2. The Permittee shall report in the SMR the results for all monitoring specified in this MRP under sections III through IX. The Permittee shall submit monthly, quarterly, semiannual, and annual SMRs including the results of all required monitoring using USEPA-approved test methods or other test methods specified in this Order. SMRs are to include all new monitoring results obtained since the last SMR was submitted. If the Permittee samples and performs analyses (other than for process/operational control, startup, research, or equipment testing) on any influent, effluent, or receiving water constituent more frequently than required by this MRP using approved analytical methods, the results of those analyses shall be included in the report. These results shall be reflected in the calculation of the average used in demonstrating compliance with limitations set forth in this Order.
3. Monitoring periods and reporting for all required monitoring shall be completed according to the following schedule:

Table E-9. Monitoring Periods and Reporting Schedule

Sampling Frequency	Monitoring Period Begins On...	Monitoring Period	SMR Due Date
Continuous	Permit effective date	All	Submit with monthly SMR
Daily	Permit effective date	(Midnight through 11:59 PM) or any 24-hour period that reasonably represents a calendar day for purposes of sampling.	Submit with monthly SMR
Weekly	Sunday following permit effective date or on permit effective date if on a Sunday	Sunday through Saturday	Submit with monthly SMR
Monthly	First day of calendar month following permit effective date or on permit effective date if that date is first day of the month	1 st day of calendar month through last day of calendar month	By the 15 th day of the third month after the month of sampling

Sampling Frequency	Monitoring Period Begins On...	Monitoring Period	SMR Due Date
Quarterly	Closest of January 1, April 1, July 1, or October 1 following (or on) permit effective date	January 1 through March 31 April 1 through June 30 July 1 through September 30 October 1 through December 31	June 15 September 15 December 15 March 15
Semiannually	Closest of January 1 or July 1 following (or on) permit effective date	January 1 through June 30 July 1 through December 31	September 15 March 15
Annually	January 1 following (or on) permit effective date	January 1 through December 31	April 15

4. Reporting Protocols. The Permittee shall report with each sample result the applicable Reporting Level (RL) and the current Method Detection Limit (MDL), as determined by the procedure in 40 CFR part 136.

The Permittee shall report the results of analytical determinations for the presence of chemical constituents in a sample using the following reporting protocols:

- Sample results greater than or equal to the RL shall be reported as measured by the laboratory (i.e., the measured chemical concentration in the sample).
- Sample results less than the RL, but greater than or equal to the laboratory's MDL, shall be reported as "Detected, but Not Quantified," or DNQ. The estimated chemical concentration of the sample shall also be reported.

For the purposes of data collection, the laboratory shall write the estimated chemical concentration next to DNQ. The laboratory may, if such information is available, include numerical estimates of the data quality for the reported result. Numerical estimates of data quality may be percent accuracy ($\pm$ a percentage of the reported value), numerical ranges (low to high), or any other means considered appropriate by the laboratory.

- Sample results less than the laboratory's MDL shall be reported as "Not Detected," or "ND".
 - Permittees are to instruct laboratories to establish calibration standards so that the ML value (or its equivalent if there is differential treatment of samples relative to calibration standards) is the lowest calibration standard. At no time is the Permittee to use analytical data derived from extrapolation beyond the lowest point of the calibration curve.
5. Compliance Determination. Compliance with effluent limitations for priority pollutants shall be determined using sample reporting protocols defined above and section VII of this Order. For purposes of reporting and administrative enforcement by the Regional Water Board and State Water Board, the Permittee shall be deemed out of compliance with effluent limitations if the concentration of

the priority pollutant in the monitoring sample is greater than the effluent limitation and greater than or equal to the RL.

6. Multiple Sample Data. When determining compliance with an AMEL, AWEL, or MDEL for priority pollutants and more than one sample result is available, the Permittee shall compute the arithmetic mean unless the data set contains one or more reported determinations of DNQ or ND. In those cases, the Permittee shall compute the median in place of the arithmetic mean in accordance with the following procedure:
 - a. The data set shall be ranked from low to high, ranking the reported ND determinations lowest, DNQ determinations next, followed by quantified values (if any). The order of the individual ND or DNQ determinations is unimportant.
 - b. The median value of the data set shall be determined. If the data set has an odd number of data points, then the median is the middle value. If the data set has an even number of data points, then the median is the average of the two values around the middle unless one or both of the points are ND or DNQ, in which case the median value shall be the lower of the two data points where DNQ is lower than a value and ND is lower than DNQ.
7. The Permittee shall submit SMRs in accordance with the following requirements:
 - a. The Permittee shall arrange all reported data in a tabular format. The data shall be summarized to clearly illustrate whether the facility is operating in compliance with interim and/or final effluent limitations. The Permittee is not required to duplicate the submittal of data that is entered in a tabular format within CIWQS. When electronic submittal of data is required and CIWQS does not provide for entry into a tabular format within the system, the Permittee shall electronically submit the data in a tabular format as an attachment.
 - b. The Permittee shall attach a cover letter to the SMR. The information contained in the cover letter shall clearly identify violations of the WDRs; discuss corrective actions taken or planned; and the proposed time schedule for corrective actions. Identified violations must include a description of the requirement that was violated and a description of the violation.

D. Discharge Monitoring Reports (DMRs)

1. DMRs are U.S. EPA reporting requirements. The Discharger shall electronically certify and submit DMRs together with SMRs using Electronic Self-Monitoring Reports module eSMR 2.5 or any upgraded version. Electronic DMR submittal shall be in addition to electronic SMR submittal. Information about electronic DMR submittal is available at the DMR website at:
http://www.waterboards.ca.gov/water_issues/programs/discharge_monitoring.

E. Other Reports

1. The Permittee shall report the results of any special studies, chronic toxicity testing, TRE/TIE, PMP, and Pollution Prevention Plan required by Special

Provisions – section VI.C. The Permittee shall submit reports in compliance with SMR reporting requirements described in subsection X.C. above.

2. Annual Summary Report

By April 15 of each year, the Permittee shall submit an annual report containing a discussion of the previous year's influent/effluent analytical results and receiving water monitoring data. The annual report shall contain an overview of any plans for upgrades to the treatment plant's collection system, the treatment processes, or the outfall system. The Permittee shall submit annual report to the Regional Water Board in accordance with the requirements described in subsection X.C.7 above.

3. Each annual monitoring report shall contain a separate section titled "Reasonable Potential Analysis" which discusses whether or not reasonable potential was triggered for pollutants which do not have a final effluent limitation in the NPDES permit. This section shall contain the following statement: "The analytical results for this sampling period did/ did not trigger reasonable potential." If reasonable potential was triggered, then the following information should also be provided:
 - a. A list of the pollutant(s) that triggered reasonable potential.
 - b. The Basin Plan or CTR criteria that was exceeded for each given pollutant.
 - c. The concentration of the pollutant(s).
 - d. The test method used to analyze the sample.
 - e. The date and time of sample collection.
4. The Permittee shall submit to the Regional Water Board, together with the monitoring report required by this permit, a list of all chemicals and proprietary additives which could affect this waste discharge, including quantities of each. Any subsequent changes in types and/or quantities shall be reported promptly.
5. The Regional Water Board requires the Permittee to file with the Regional Water Board, within 90 days after the effective date of this Order, a technical report on preventive (failsafe) and contingency (cleanup) plans for controlling accidental discharges, and for minimizing the effect of such events. The technical report should:
 - a. Identify the possible sources of accidental loss, untreated waste bypass, and contaminated drainage. Loading and storage areas, power outage, waste treatment unit outage, and failure of process equipment, tanks, and pipes should be considered.
 - b. Evaluate the effectiveness of present facilities and procedures and state when they become operational.
 - c. Describe facilities and procedures needed for effective preventive and contingency plans.
 - d. Predict the effectiveness of the proposed facilities and procedures and provide an implementation schedule contingent interim and final dates when they will be constructed, implemented, or operational.

ATTACHMENT F - FACT SHEET

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ATTACHMENT F – FACT SHEET

As described in section II.B of this Order, the Regional Water Board incorporates this Fact Sheet as findings of the Regional Water Board supporting the issuance of this Order. This Fact Sheet includes the legal requirements and technical rationale that serve as the basis for the requirements of this Order.

This Order has been prepared under a standardized format to accommodate a broad range of discharge requirements for Permittees in California. Only those sections or subsections of this Order that are specifically identified as “not applicable” have been determined not to apply to this Permittee. Sections or subsections of this Order not specifically identified as “not applicable” are fully applicable to this Permittee.

I. PERMIT INFORMATION

The following table summarizes administrative information related to the facility.

Table F-1. Facility Information

WDID	4A560110001
Discharger	City of Simi Valley
Name of Facility	Simi Valley Water Quality Control Plant including its associated wastewater collection system and outfalls (POTW)
Facility Address	600 West Los Angeles Avenue, Simi Valley, CA 93065 Ventura County
Facility Contact, Title and Phone	Alan Krieger, Deputy Director/Sanitation Services, (805) 583-6447
Authorized Person to Sign and Submit Reports	Alan Krieger, Deputy Director/Sanitation Services, (805) 583-6447
Mailing Address	2929 Tapo Canyon Road, Simi Valley, CA 93063
Billing Address	Same as Mailing Address
Type of Facility	POTW
Major or Minor Facility	Major
Threat to Water Quality	1
Complexity	A
Pretreatment Program	Yes
Recycling Requirements	Producer/User
Facility Permitted Flow	12.5 MGD
Facility Design Flow	12.5 MGD
Watershed	Calleguas Creek
Receiving Water	Arroyo Simi
Receiving Water Type	Inland surface water

- A. The City of Simi Valley (The City or Permittee) owns and operates a publicly-owned treatment works (POTW) comprised of Simi Valley Water Quality Control Plant (Simi Valley WQCP or Facility) including its associated wastewater collection system and outfalls.
- B. For the purposes of this Order, references to the “Permittee” or “permittee” in applicable federal and state laws, regulations, plans, or policy are held to be equivalent to references to the Permittee herein.
- C. The Facility discharges wastewater to Arroyo Simi, a water of the United States, within Calleguas Creek Watershed. The Permittee was previously regulated by Order R4-2014-0066 and NPDES Permit No. CA0055221 adopted on May 8, 2014. This Order was amended on July 19, 2015 by Order R4-2014-0066-A01 and expires on June 30, 2019. Attachments B-1 and B-2 provide maps of the area around the Facility and discharge locations and sampling points, respectively. Attachment C provides a flow schematic of the Facility.
- D. The Permittee filed a report of waste discharge and submitted an application for reissuance of its WDRs and NPDES permit on December 3, 2018. Supplemental information was requested via email on December 26, 2018, and the response received on January 16, 2019. The application was deemed complete on January 22, 2019. A site visit was conducted on July 2, 2019, to observe operations and collect additional data to develop permit limitations and requirements for waste discharge.
- E. Should there be an anticipated change in place of use and purpose of use of treated wastewater, state law requires dischargers to file a petition with the State Water Board, Division of Water Rights and receive approval for any change in the point of discharge, place of use, or purpose of use of treated wastewater that decreases the flow in any portion of a watercourse. The State Water Board retains separate jurisdictional authority to enforce any applicable requirements under Water Code section 1211. This is not an NPDES permit requirement.
- F. Regulations at 40 CFR section 122.46 limit the duration of NPDES permits to a fixed term not to exceed five years. Accordingly, Table 3 of this Order limits the duration of the discharge authorization. However, pursuant to California Code of Regulations, title 23, section 2235.4, the terms and conditions of an expired permit are automatically continued pending reissuance of the permit if the Discharger complies with all federal NPDES requirements for continuation of expired permits.

II. FACILITY DESCRIPTION

A. Description of Wastewater and Biosolids Treatment and Controls

1. The Simi Valley WQCP is a tertiary wastewater treatment plant with a dry weather design capacity of 12.5 MGD and serves an estimated population of 126,878 people. The Simi Valley WQCP receives wastewater from the City of Simi Valley and unincorporated sections of Ventura County.

Pretreatment Program: The wastewater is a mixture of domestic wastewater and industrial wastewater that is pre-treated pursuant to 40 CFR part 403. The City of Simi Valley modified the City’s Pretreatment Program that includes revision of Sewer Use Ordinance with local limits and an Enforcement Response

Plan. On June 2, 2011, the Regional Water Board adopted Order No. R4-2011-0098 amending the NPDES Permit pretreatment program to include the above-mentioned ordinance, local limits, and enforcement response plan. The pretreatment program is still current because there are no substantial changes or modifications to the program.

2. Treatment at the Simi Valley WQCP consists of primary sedimentation, activated sludge biological treatment with nitrification and denitrification, secondary sedimentation, dual media filtration, chlorination, and dechlorination. Treated wastewater discharged to Arroyo Simi is dechlorinated but the effluent delivered for reuse is not dechlorinated.
3. Sodium hypochlorite is used as a disinfectant in the Simi Valley WQCP. The disinfectant is dosed prior to the serpentine chlorine contact chamber and occasionally added prior to the filters to minimize algae growth. Additional disinfectant may be dosed prior to the serpentine chlorine contact chamber. Prior to discharge, sodium bisulfite is added to the treated effluent to remove residual chlorine.
4. Primary sludge is anaerobically digested. Sewage solids (sludge) separated from the wastewater are dewatered with a press and transported to Simi Valley Landfill, Simi Valley, California.
5. Simi Valley WQCP has constructed a biological nutrient removal (BNR) system with nitrification de-nitrification process (NDN) in order to achieve compliance with the ammonia Basin Plan objectives. The system was completed and has been in operation since September 2004. However, the NDN did not achieve the design objectives for nitrate + nitrite as nitrogen. The Permittee modified the NDN process and completed the modification on April 30, 2009.

B. Discharge Points and Receiving Waters

The Simi Valley WQCP discharges tertiary-treated municipal wastewater to Arroyo Simi. Treated effluent is discharged from the plant to surface waters at the following discharge points:

Discharge Point 001: Discharge to Arroyo Simi (approximate coordinates: Latitude 34.28200°, Longitude -118.81290°).

Discharge Point 002: Discharge to Arroyo Simi (approximate coordinates: Latitude 34.28169°, Longitude -118.81350°). This is an alternate downstream outfall that has never been used. It was provided as a backup discharge point in the event that discharge through Discharge Point 001 is restricted because of flooding in the Arroyo Simi. Discharge Point 002 is 225 feet downstream of Discharge Point 001.

Receiving Waters: Arroyo Simi is a tributary of Calleguas Creek. Arroyo Simi flows into Arroyo Las Posas, which then flows into Calleguas Creek Reach 3, Reach 2, and Reach 1 (Mugu Lagoon). During dry weather (May 1 – October 31), the primary sources of water flow in the receiving waters, downstream of the discharge point, is the Simi Valley WQCP effluent and other NPDES-permitted discharges, including urban runoff conveyed through the municipal separate storm sewer systems (MS4). During dry weather, surface water flow in Arroyo Simi usually percolates to

groundwater before the surface flow reaches Hitch Boulevard, which marks the beginning of Arroyo Las Posas, in the City of Moorpark, past the Moorpark Wastewater Treatment Plant. The Moorpark WWTP is another POTW in the Calleguas Creek Watershed, located downstream of the Simi Valley WQCP, that intermittently discharges to Arroyo Las Posas. Some effluent from the Simi Valley WQCP and the Moorpark WWTP may reach Seminary Road, located further downstream in Arroyo Las Posas, during parts of the year.

Calleguas Creek, including Arroyo Simi, is a water of the United States that conveys floodwater and urban runoff, along with treated waste water. Arroyo Simi is unlined near the point of discharge. Groundwater recharge may occur incidentally in unlined areas of Arroyo Simi, Arroyo Las Posas, and Calleguas Creek, where the underlying sediments maybe transmissive to water as well as pollutants.

The watershed supports a diversity of wildlife. Threatened and endangered species such as the peregrine falcon, least tern, light-footed clapper rail, and the brown pelican are found in Calleguas Creek and Mugu Lagoon.

C. Summary of Existing Requirements and Self-Monitoring Report (SMR) Data

Effluent limitations contained in the existing Order for discharges from Discharge Point 001 (Monitoring Location EFF-001) and representative monitoring data from the term of the previous Order as reported in the ROWD, are as follows:

Table F-2. Historic Effluent Limitations and Monitoring Data

Parameter	Units	Average Monthly Limit	Average Weekly Limit	Maximum Daily Limit	Highest Average Monthly Discharge	Highest Average Weekly Discharge	Highest Daily Discharge
BOD ₅ 20°C	mg/L	20	30	45	9.7	2.5	3.2
Total Suspended Solids (TSS)	mg/L	15	40	45	3.1	2.8	10.6
Oil and Grease	mg/L	10	--	15	<5	--	<5
Settleable Solids	ml/L	0.1	--	0.3	<0.1	--	< 0.1
Residual Chlorine	mg/L	--	--	0.1	<0.1	--	<0.1
Total Dissolved Solids	mg/L	850	--	--	741	--	649
MBAS	mg/L	0.5	--	--	0.21	--	0.21
Chloride	mg/L	150	--	--	168	--	168
Sulfate	mg/L	250	--	--	340	--	340
Boron	mg/L	1	--	--	0.9	--	0.9
Nitrite-N (as N)	mg/L	0.9	--	--	0.07		0.07
Nitrate + Nitrite as N	mg/L	9	--	--	9.4		9.4

Parameter	Units	Average Monthly Limit	Average Weekly Limit	Maximum Daily Limit	Highest Average Monthly Discharge	Highest Average Weekly Discharge	Highest Daily Discharge
Total Ammonia	mg/L	2.4	--	3.3	2.6		2.6
Antimony	µg/L	--	--	--	<1	--	<1
Arsenic	µg/L	--	--	--	1.5	--	1.5
Beryllium	µg/L	--	--	--	<1	--	<1
Cadmium	µg/L	--	--	--	<1	--	<1
Chromium III	µg/L	--	--	--	1.3	--	1.3
Chromium VI	µg/L	--	--	--	5	--	5
Copper	µg/L	30.5	--	31	9.5	--	9.5
Lead	µg/L	--	--	--	<1	--	<1
Mercury	lbs/day	0.031	--	--	0.0048	--	0.0048
Nickel	µg/L	169	--	960	4.5	--	4.5
Selenium	µg/L	4.4	--	7.4	2.4	--	2.4
Silver	µg/L	--	--	--	<1	--	<1
Thallium	µg/L	--	--	--	<1	--	<1
Zinc	µg/L	--	--	--	65	--	65
Cyanide	µg/L	4.3	--	8.5	1.5	--	8.1
Asbestos	µg/L	--	--	--	--	--	--
2,3,7,8-TCDD (Dioxin)	µg/L	--	--	--	< 0.000006	--	< 0.000006
Acrolein	µg/L	--	--	--	< 5	--	<5
Acrylonitrile	µg/L	--	--	--	< 2	--	< 2
Benzene	µg/L	--	--	--	< 0.5	--	<0.5
Bromoform	µg/L	--	--	--	< 1	--	<1
Carbon Tetrachloride	µg/L	--	--	--	< 0.5	--	< 0.5
Chlorobenzene	µg/L	--	--	--	< 0.5	--	<0.5
Dibromochloromethane	µg/L	--	--	--	0.8	--	0.8
Chloroethane	µg/L	--	--	--	<1	--	<1
2-chloroethyl vinyl ether	µg/L	--	--	--	<2	--	<2
Chloroform	µg/L	--	--	--	6.6	--	6.6
Dichlorobromomethane	µg/L	--	--	--	3.7	--	3.7
1,1-dichloroethane	µg/L	--	--	--	< 0.5	--	< 0.5

Parameter	Units	Average Monthly Limit	Average Weekly Limit	Maximum Daily Limit	Highest Average Monthly Discharge	Highest Average Weekly Discharge	Highest Daily Discharge
1,2-dichloroethane	µg/L	--	--	--	< 0.5	--	< 0.5
1,1-dichloroethylene	µg/L	--	--	--	< 0.5	--	< 0.5
1,2-dichloropropane	µg/L	--	--	--	< 0.5	--	< 0.5
1,3-dichloropropylene	µg/L	--	--	--	< 0.5	--	< 0.5
Ethylbenzene	µg/L	--	--	--	< 0.5	--	< 0.5
Methyl bromide	µg/L	--	--	--	< 0.5	--	< 0.5
Methyl chloride	µg/L	--	--	--	< 0.5	--	< 0.5
Methylene chloride	µg/L	--	--	--	< 2	--	< 2
1,1,2,2-tetrachloroethane	µg/L	--	--	--	< 0.5	--	< 0.5
Tetrachloroethylene	µg/L	--	--	--	< 0.5	--	< 0.5
Toluene	µg/L	--	--	--	< 0.5	--	< 0.5
Trans 1,2-Dichloroethylene	µg/L	--	--	--	< 0.25	--	< 0.25
1,1,1-Trichloroethane	µg/L	--	--	--	< 0.5	--	< 0.5
1,1,2-Trichloroethane	µg/L	--	--	--	< 0.5	--	< 0.5
Trichloroethylene	µg/L	--	--	--	< 0.5	--	< 0.5
Vinyl Chloride	µg/L	--	--	--	< 0.5	--	< 0.5
2-chlorophenol	µg/L	--	--	--	< 1	--	< 1
2,4-dichlorophenol	µg/L	--	--	--	< 2	--	< 2
2,4-dimethylphenol	µg/L	--	--	--	< 2	--	< 2
4,6-dinitro-o-resol(aka 2-methyl-4,6-Dinitrophenol)	µg/L	--	--	--	< 4.9	--	< 4.9

Parameter	Units	Average Monthly Limit	Average Weekly Limit	Maximum Daily Limit	Highest Average Monthly Discharge	Highest Average Weekly Discharge	Highest Daily Discharge
2,4-dinitrophenol	µg/L	--	--	--	< 4.9	--	< 4.9
2-nitrophenol	µg/L	--	--	--	< 0.95	--	< 0.95
4-nitrophenol	µg/L	--	--	--	< 1.9	--	< 1.9
3-Methyl-4-Chlorophenol (aka 4-chloro-m-cresol)	µg/L	--	--	--	< 2	--	< 2
Pentachlorophenol	µg/L	--	--	--	< 4.9	--	< 4.9
Phenol	µg/L	--	--	--	5.3	--	5.3
2,4,6-trichlorophenol	µg/L	--	--	--	< 1	--	< 1
Acenaphthene	µg/L	--	--	--	< 0.5	--	< 0.5
Acenaphthylene	µg/L	--	--	--	< 0.5	--	< 0.5
Anthracene	µg/L	--	--	--	< 0.5	--	< 0.5
Benzidine	µg/L	--	--	--	< 9.9	--	< 9.9
Benzo(a)-Anthracene	µg/L	--	--	--	< 4.9	--	< 4.9
Benzo(a)Pyrene	µg/L	--	--	--	< 2	--	< 2
Benzo(b)-Fluoranthene	µg/L	--	--	--	< 2	--	< 2
Benzo(ghi)Perylene	µg/L	--	--	--	< 4.9	--	< 4.9
Benzo(k)-Fluoranthene	µg/L	--	--	--	< 0.5	--	< 0.5
Bis(2-Chloroethoxy)methane	µg/L	--	--	--	< 0.5	--	< 0.5
Bis(2-Chloroethyl)Ether	µg/L	--	--	--	< 0.5	--	< 0.5
Bis(2-Chloroisopropyl)Ether	µg/L	--	--	--	< 0.5	--	< 0.5
Bis(2-Ethylhexyl)-Phthalate	µg/L	--	--	--	5.3	--	5.3

Parameter	Units	Average Monthly Limit	Average Weekly Limit	Maximum Daily Limit	Highest Average Monthly Discharge	Highest Average Weekly Discharge	Highest Daily Discharge
4-Bromophenyl Phenyl Ether	µg/L	--	--	--	< 1	--	< 1
Butylbenzyl Phthalate	µg/L	--	--	--	< 4.9	--	< 4.9
2-Chloro-naphthalene	µg/L	--	--	--	< 0.5	--	< 0.5
4-Chlorophenyl Phenyl Ether	µg/L	--	--	--	< 0.5	--	< 0.5
Chrysene	µg/L	--	--	--	< 0.5	--	< 0.5
Dibenzo(a,h) Anthracene	µg/L	--	--	--	< 4.9	--	< 4.9
1,2-Dichlorobenzene	µg/L	--	--	--	< 0.5	--	< 0.5
1,3-Dichlorobenzene	µg/L	--	--	--	< 0.5	--	< 0.5
1,4-Dichlorobenzene	µg/L	--	--	--	< 0.5	--	< 0.5
3-3'-Dichlorobenzidine	µg/L	--	--	--	< 4.9	--	< 4.9
Diethyl Phthalate	µg/L	--	--	--	< 1	--	< 1
Dimethyl Phthalate	µg/L	--	--	--	< 0.5	--	< 0.5
Di-n-Butyl Phthalate	µg/L	--	--	--	< 2	--	< 2
2-4-Dinitrotoluene	µg/L	--	--	--	< 4.9	--	< 4.9
2-6-Dinitrotoluene	µg/L	--	--	--	< 4.9	--	< 4.9
Di-n-Octyl Phthalate	µg/L	--	--	--	< 4.9	--	< 4.9
1,2-Diphenylhydrazine	µg/L	--	--	--	< 1	--	< 1
Fluoranthene	µg/L	--	--	--	< 0.5	--	< 0.5

Parameter	Units	Average Monthly Limit	Average Weekly Limit	Maximum Daily Limit	Highest Average Monthly Discharge	Highest Average Weekly Discharge	Highest Daily Discharge
Fluorene	µg/L	--	--	--	< 0.5	--	< 0.5
Hexachloro-benzene	µg/L	--	--	--	< 1	--	< 1
Hexachloro-butadiene	µg/L	--	--	--	< 2	--	< 2
Hexachloro-cyclopentadiene	µg/L	--	--	--	< 4.9	--	< 4.9
Hexachloroethane	µg/L	--	--	--	< 3	--	< 3
Indeno(1,2,3-cd)Pyrene	µg/L	--	--	--	< 2	--	< 2
Isophorone	µg/L	--	--	--	< 1	--	< 1
Naphthalene	µg/L	--	--	--	< 1	--	< 1
Nitrobenzene	µg/L	--	--	--	< 1	--	< 1
N-Nitrosodimethylamine	µg/L	--	--	--	< 2	--	< 2
N-Nitrosodimethylpropylamine	µg/L	--	--	--	< 2	--	< 2
N-Nitrosodiphenylamine	µg/L	--	--	--	< 1	--	< 1
Phenanthrene	µg/L	--	--	--	< 0.5	--	< 0.5
Pyrene	µg/L	--	--	--	< 0.5	--	< 0.5
1,2,4-Trichlorobenzene	µg/L	--	--	--	< 1	--	< 1
Aldrin	µg/L	--	--	--	0.0028	--	0.0028
Alpha-BHC	µg/L	--	--	--	<0.0024	--	<0.0024
Beta-BHC	µg/L	--	--	--	<0.0029	--	<0.0029
Gamma-BHC (aka Lindane)	µg/L	--	--	--	< 0.0028	--	< 0.0028
delta-BHC	µg/L	--	--	--	< 0.0033	--	< 0.0033
Chlordane	µg/L	0.00059	--	0.00059	< 0.075	--	< 0.075
4,4'-DDT	µg/L	0.00059	--	0.0012	< 0.0038	--	< 0.0038
4,4'-DDE	µg/L	0.00059	--	0.0012	< 0.0028	--	< 0.0028
4,4'-DDD	µg/L	0.00084	--	0.0017	< 0.0038	--	< 0.0038
Dieldrin	µg/L	0.00014	--	0.00028	< 0.0019	--	< 0.0019
Alpha-Endosulfan	µg/L	--	--	--	0.04	--	0.04

Parameter	Units	Average Monthly Limit	Average Weekly Limit	Maximum Daily Limit	Highest Average Monthly Discharge	Highest Average Weekly Discharge	Highest Daily Discharge
Beta-Endosulfan	µg/L	--	--	--	< 0.0019	--	< 0.0019
Endosulfan Sulfate	µg/L	--	--	--	< 0.0029	--	< 0.0029
Endrin	µg/L	--	--	--	< 0.0019	--	< 0.0019
Endrin Aldehyde	µg/L	--	--	--	< 0.0019	--	< 0.0019
Heptachlor	µg/L	--	--	--	< 0.0028	--	< 0.0028
Heptachlor Epoxide	µg/L	--	--	--	< 0.0024	--	< 0.0024
PCB 1016	µg/L	0.00017	--	0.00034	< 0.24	--	< 0.24
PCB 1221	µg/L	--	--	--	--	--	--
PCB 1232	µg/L	--	--	--	--	--	--
PCB 1242	µg/L	--	--	--	--	--	--
PCB 1248	µg/L	--	--	--	--	--	--
PCB 1254	µg/L	--	--	--	--	--	--
PCB 1260	µg/L	--	--	--	--	--	--
Toxaphene	µg/L	0.00016	--	0.00033	<0.24	--	<0.24
Chlorpyrifos	µg/L	0.014	--	0.025	0.012	--	0.012
Diazinon	µg/L	0.1	--	0.1	<0.0052	--	<0.0052
Iron	µg/L	--	--	--	51	--	51
Methoxychlor	µg/L	--	--	--	<0.01	--	<0.01
2,4-D	µg/L	--	--	--	< 0.71	--	< 0.71
2,4,5-TP (Sylvex)	µg/L	--	--	--	< 1	--	< 1
Halomethanes	µg/L	--	--	--	11	--	11

D. Compliance Summary

The following table lists the Facility's preliminary list of exceedances that occurred during period from July 2014 to August 2018.

Table F-3. Preliminary List of Exceedances

Date of Exceedance	Description of Exceedance	Status
02/07/2018, 02/14/2018	Chronic Toxicity monthly median limitation was exceeded with two failed toxicity tests at EFF-001.	Under Enforcement Unit review.

Date of Exceedance	Description of Exceedance	Status
12/13/2017, 12/20/2017	Chronic Toxicity monthly median limitation was exceeded with two failed toxicity tests at EFF-001.	Under Enforcement Unit review.
10/04/2017, 10/12/2017	Chronic Toxicity monthly median limitation was exceeded with two failed toxicity tests at EFF-001.	Under Enforcement Unit review.
03/31/2017	Total Dissolved Solids monthly average limitation of 850 mg/L was exceeded with a reported value of 864 mg/L at EFF-001.	Settlement Offer No. R4-2017-0193
03/31/2017	Total Sulfate monthly average limitation of 250 mg/L was exceeded with a reported value of 276 mg/L at EFF-001.	Settlement Offer No. R4-2017-0193
02/28/2017	Total Dissolved Solids monthly average limitation of 850 mg/L was exceeded with a reported value of 972 mg/L at EFF-001.	Settlement Offer No. R4-2017-0193
02/28/2017	Total Sulfate monthly average limitation of 250 mg/L was exceeded with a reported value 340 mg/L at EFF-001.	Settlement Offer No. R4-2017-0193
05/03/2016	Total Mercury average limitation of 0.031 lbs/month was exceeded with a reported value of 0.0443 lbs/month at EFF-001.	Settlement Offer No. R4-2016-0374
07/21/2015	Total Coliform monthly average limitation of 23 MPN/100 mL was exceeded with a reported value of 110 MPN/100 mL at EFF-001.	Settlement Offer No. R4-2016-0136
07/20/2015	Total Coliform monthly average limitation of 23 MPN/100 mL was exceeded with a reported value of 130 MPN/100 mL at EFF-001.	Settlement Offer No. R4-2016-0136

Date of Exceedance	Description of Exceedance	Status
06/21/2015	Total Coliform monthly average limitation of 23 MPN/100 mL was exceeded with a reported value of 30 MPN/100 mL at EFF-001.	Settlement Offer No. R4-2016-0136
03/29/2015	Total Coliform monthly average limitation of 23 MPN/100 mL was exceeded with a reported value of 900 MPN/100 mL at EFF-001.	Settlement Offer No. R4-2015-0149
10/15/2014	Ammonia, Total (as N) daily maximum limitation of 3.3 mg/L was exceeded with a reported value of 3.6 mg/L at EFF-001.	Settlement Offer No. R4-2015-0149
10/15/2014	Ammonia, Total (as N) daily maximum limitation is 174 lb/day was exceeded with a reported value of 216 lb/day at EFF-001.	Settlement Offer No. R4-2015-0149
10/14/2014	Ammonia, Total (as N) daily maximum limitation of 186 lb/day was exceeded with a reported value of 218 lb/day at EFF-001.	Settlement Offer No. R4-2015-0149
10/14/2014	Ammonia, Total (as N) daily maximum limitation of 3.3 mg/L was exceeded with a reported value of 3.4 mg/L at EFF-001.	Settlement Offer No. R4-2015-0149
09/24/2014	Total Residual Chlorine daily maximum limitation of 0.1 mg/L was exceeded with a reported value of 2.37 mg/L at EFF-001.	Settlement Offer No. R4-2015-0149

E. Planned Changes

There are no planned upgrades proposed by the Permittee.

III. APPLICABLE PLANS, POLICIES, AND REGULATIONS

The requirements contained in this Order are based on the requirements and authorities described in this section.

A. Legal Authorities

This Order serves as WDRs pursuant to article 4, chapter 4, division 7 of the California Water Code (commencing with section 13260). This Order is also issued pursuant to section 402 of the federal Clean Water Act (CWA) and implementing regulations adopted by the U.S. EPA) and chapter 5.5, division 7 of the CWC (commencing with section 13370). It shall serve as an NPDES permit authorizing the Discharger to discharge into waters of the United States at the discharge locations described in Table 2 subject to the WDRs in this Order.

B. California Environmental Quality Act (CEQA)

Under CWC section 13389, this action to adopt an NPDES permit is exempt from the provisions of Chapter 3 of CEQA, (commencing with section 21100) of Division 13 of the Public Resources Code.

C. State and Federal Laws, Regulations, Policies, and Plans

1. **Water Quality Control Plan.** The Water Quality Control Plan for the Los Angeles Region (Basin Plan) designates beneficial uses, establishes water quality objectives (WQOs), and contains implementation programs and policies to achieve those objectives for all waters addressed through the plan. Requirements in this Order implement the Basin Plan. In addition, the Basin Plan implements State Water Board Resolution 88-63, which established state policy that all waters, with certain exceptions, should be considered suitable or potentially suitable for municipal or domestic supply. On May 26, 2000, the USEPA approved the revised Basin Plan except for the implementation plan for potential MUN-designated water bodies. On August 22, 2000, the City of Los Angeles, City of Burbank, City of Simi Valley, and the County Sanitation Districts of Los Angeles County challenged USEPA's water quality standards action in the U.S. District Court. On December 18, 2011, the court issued an order remanding the matter to USEPA to take further action on the 1994 Basin Plan consistent with the court's decision. On February 15, 2002, USEPA revised its decision and approved the 1994 Basin Plan in whole. In its February 15, 2002 letter, USEPA stated:

EPA bases its approval on the court's finding that the Regional board's identification of waters with an asterisk ("*") in conjunction with the implementation language at page 2-4 of the 1994 Basin Plan, was intended "to only conditionally designate and not finally designate as MUN those water bodies identified by an (*) for the MUN use in Table 2-1 of the Basin Plan, without further action." Court Order at p. 4. Thus, the waters identified with an (*) in Table 2-1 do not have MUN as a designated use until such time as the State undertakes additional study and modifies its Basin Plan. Because this conditional use designation has no legal effect, it does not constitute a new water quality standard subject to EPA review under section 303(c)(3) of the Clean Water Act ("CWA"). 33 U.S.C. § 1313(c)(3).

USEPA's decision has no effect on the MUN designations of groundwater, which is outside of its jurisdiction. Beneficial uses applicable to Arroyo Simi, Arroyo Las Posas, and Calleguas Creek are as follows:

Table F-4. Basin Plan Beneficial Uses – Receiving Waters

Discharge Point	Receiving Water Name	Beneficial Use(s)
001, 002	Arroyo Simi (Hydro. Unit No. 403.62)	Existing: Wildlife habitat (WILD); and rare, threatened, or endangered species (RARE). Intermittent: Industrial service supply (IND); ground water recharge (GWR); freshwater replenishment (FRSH); water contact recreation (REC-1); non-contact water recreation (REC-2); and warm freshwater habitat (WARM). Potential: Municipal and domestic water supply (MUN ¹).
001, 002	Arroyo Las Posas (Hydro. Unit No. 403.62)	Existing:(GWR), (FRSH), (REC-1), (REC-2), (WARM), and (WILD). Intermittent: none Potential: (MUN), (IND), industrial process supply (PROC), agricultural supply (AGR), and cold freshwater habitat (COLD).

¹ As described above, the receiving water was designated as Potential MUN* consistent with State Water Board Resolution No. 88-63 and Regional Water Board Resolution No. 89-003. However, when designating the receiving water as Potential MUN, the Regional Water Board only conditionally designated rather than finally designated the water body as Potential MUN as indicated by the “*”. The Basin Plan states that until the Board undertakes a detailed review of the criteria in State Water Board Resolution No. 88-63, no new effluent limitations will be placed in Waste Discharge Requirements as a result of these designations.

Discharge Point	Receiving Water Name	Beneficial Use(s)
001, 002	Arroyo Las Posas (Hydro. Unit No. 403.12)	Existing: (GWR), (REC-1), (REC-2), (WARM), and (WILD). Intermittent: none Potential: (MUN), (IND), (PROC), (AGR), and (COLD).
001, 002	Calleguas Creek (Hydro. Unit No. 403.12)	Existing: (IND), (PROC), (AGR), (GWR), (REC-1 ²), (REC-2), (WARM), and (WILD). Intermittent: none Potential: (MUN)
001, 002	Calleguas Creek (Hydro. Unit No. 403.11)	Existing: (AGR), (GWR), (FRSH), (REC-1), (REC-2), (WARM), (COLD), (WILD), (RARE ³), and wetland habitat (WET). Intermittent: none Potential: (MUN)
001, 002	Calleguas Creek Estuary (Hydro. Unit No. 403.11)	Existing: (REC-2), commercial and sport fishing (COMM), estuarine habitat (EST), (WILD), (RARE ⁴), migration of aquatic organisms (MIGR ⁵), spawning, reproduction, and/or early development (SPWN), and (WET). Intermittent: none Potential: Navigation (NAV), and (REC-1 ⁶).

² Whenever flow conditions are suitable.

³ Habitat of the Clapper Rail.

⁴ One or more rare species utilize all ocean, bays, estuaries, and coastal wetlands for foraging and/or nesting.

⁵ Aquatic organisms utilize all bays, estuaries, lagoons and coastal wetlands, to a certain extent, for spawning and early development. This may include migration into areas which are heavily influenced by freshwater inputs.

⁶ Area is currently under the control of the Navy: swimming is prohibited.

Discharge Point	Receiving Water Name	Beneficial Use(s)
001, 002	Mugu Lagoon (Hydro. Unit No. 403.11)	Existing: (NAV), (REC-2), (COMM ⁷), estuarine habitat (EST), marine habitat (MAR), wildlife habitat (WILD ⁸), preservation of biological habitats (BIOL), (RARE), (MIGR), (SPWN), shellfish harvesting (SHELL), (WET). Intermittent: None Potential: (REC-1)

Beneficial uses of the receiving ground waters are as follows:

Table F-5. Basin Plan Beneficial Uses – Ground Waters

Discharge Point	Basin Name	Beneficial Use(s)
001, 002	Simi Valley Basin DWR Basin No. 4-9 <u>Confined Aquifers</u>	<u>Existing:</u> Municipal and domestic water supply (MUN), industrial service supply (IND), industrial process supply (PROC), and agricultural supply (AGR)
001, 002	Simi Valley Basin DWR Basin No. 4-9 <u>Unconfined Aquifers</u>	<u>Existing:</u> Municipal and domestic water supply (MUN), industrial service supply (IND), industrial process supply (PROC), and agricultural supply (AGR)
001, 002	Las Posas Valley DWR Basin No. 4-8 <u>South Basin</u>	<u>Existing:</u> Municipal and domestic water supply (MUN), industrial service supply (IND), industrial process supply (PROC), and agricultural supply (AGR)
001, 002	Las Posas Valley DWR Basin No. 4-8 <u>North Basin</u>	<u>Existing:</u> Municipal and domestic water supply (MUN), industrial service supply (IND), industrial process supply (PROC), and agricultural supply (AGR)
001, 002	Pleasant Valley (Ventura Central Basin) DWR Basin No. 4-6 <u>Confined Aquifers</u>	<u>Existing:</u> Municipal and domestic water supply (MUN), industrial service supply (IND), industrial process supply (PROC), and agricultural supply (AGR)

⁷ Limited public access precludes full utilization.

⁸ Marine habitats of the Channel Islands and Mugu Lagoon serve as pinniped haul-out areas for one or more species (i.e., sea lions).

Discharge Point	Basin Name	Beneficial Use(s)
001, 002	Pleasant Valley (Ventura Central Basin) DWR Basin No. 4-6 <u>Unconfined Aquifers</u>	<u>Existing:</u> Industrial service supply (IND), industrial process supply (PROC), and agricultural supply (AGR) <u>Potential:</u> Municipal and domestic water supply (MUN)
001, 002	Oxnard Plain (Ventura Central Basin) DWR Basin No. 4-4 <u>Oxnard Forebay</u>	<u>Existing:</u> Municipal and domestic water supply (MUN), industrial service supply (IND), industrial process supply (PROC), and agricultural supply (AGR)
001, 002	Oxnard Plain (Ventura Central Basin) DWR Basin No. 4-4 <u>Confined Aquifers</u>	<u>Existing:</u> Municipal and domestic water supply (MUN), industrial process supply (PROC), and agricultural supply (AGR)
001, 002	Oxnard Plain (Ventura Central Basin) DWR Basin No. 4-4 <u>Unconfined Aquifers</u>	<u>Existing:</u> Municipal and domestic water supply (MUN) and agricultural supply (AGR) <u>Potential:</u> Industrial service supply (IND)

2. **National Toxics Rule (NTR) and California Toxics Rule (CTR).** USEPA adopted the NTR on December 22, 1992, and later amended it on May 4, 1995 and November 9, 1999. About forty criteria in the NTR applied in California. On May 18, 2000, USEPA adopted the CTR. The CTR promulgated new toxics criteria for California and, in addition, incorporated the previously adopted NTR criteria that were applicable in the state. The CTR was amended on February 13, 2001. These rules contain federal water quality criteria for priority pollutants.
3. **State Implementation Policy.** On March 2, 2000, the State Water Board adopted the Policy for Implementation of Toxics Standards for Inland Surface Waters, Enclosed Bays, and Estuaries of California (State Implementation Policy or SIP). The SIP became effective on April 28, 2000, with respect to the priority pollutant criteria promulgated for California by the USEPA through the NTR and to the priority pollutant objectives established by the Regional Water Board in the Basin Plan. The SIP became effective on May 18, 2000, with respect to the priority pollutant criteria promulgated by the USEPA through the CTR. The State Water Board adopted amendments to the SIP on February 24, 2005, that became effective on July 13, 2005. The SIP establishes implementation provisions for priority pollutant criteria and objectives and provisions for chronic toxicity control. Requirements of this Order implement the SIP.
4. **Alaska Rule.** On March 30, 2000, USEPA revised its regulation that specifies when new and revised state and tribal water quality standards become effective for CWA purposes (40 CFR section 131.21, 65 Federal Register 24641 (April 27,

2000)). Under the revised regulation (also known as the Alaska Rule), new and revised standards submitted to USEPA after May 30, 2000, must be approved by USEPA before being used for CWA purposes. The final rule also provides that standards already in effect and submitted to USEPA by May 30, 2000, may be used for CWA purposes, whether or not approved by USEPA.

5. **Stringency of Requirements for Individual Pollutants.** This Order contains both technology-based effluent limitations (TBELs) and water quality-based effluent limitations (WQBELs) for individual pollutants. The TBELs consist of restrictions on BOD, TSS, oil and grease, settleable solids, turbidity, pH, and percent removal of BOD and TSS. Restrictions on BOD, TSS, oil and grease, settleable solids, turbidity, and pH are discussed in section IV.B.2 of the Fact Sheet. This Order's technology-based pollutant restrictions implement the minimum applicable federal technology-based requirements. In addition, this Order contains effluent limitations more stringent than the minimum, federal technology-based requirements that are carried over from the previous permit.

WQBELs have been scientifically derived to implement WQOs that protect beneficial uses. Both the beneficial uses and the WQOs have been approved pursuant to federal law and are the applicable federal water quality standards. All beneficial uses and WQOs contained in the Basin Plan and the Ocean Plan were approved under state law and submitted to and approved by USEPA prior to May 30, 2000. Any WQOs and beneficial uses submitted to USEPA prior to May 30, 2000, but not approved by USEPA before that date, are nonetheless "applicable water quality standards for purposes of the CWA" pursuant to 40 CFR section 131.21(c)(1). Collectively, this Order's restrictions on individual pollutants are no more stringent than required to implement the requirements of the CWA.

6. **Antidegradation Policy.** Federal regulation 40 CFR section 131.12 requires that state water quality standards include an antidegradation policy consistent with the federal antidegradation policy. The State Water Board established California's antidegradation policy in State Water Board Resolution 68-16 ("Statement of Policy with Respect to Maintaining the Quality of the Waters of the State"). Resolution 68-16 is deemed to incorporate the federal antidegradation policy where the federal policy applies under federal law. Resolution 68-16 requires that existing water quality be maintained unless degradation is justified based on specific findings. The Regional Water Board's Basin Plan implements, and incorporates by reference, both the state and federal antidegradation policies. The permitted discharges must be consistent with the antidegradation provisions of 40 CFR section 131.12 and State Water Board Resolution 68-16.
7. **Anti-Backsliding Requirements.** Sections 402(o) and 303(d)(4) of the CWA and federal regulations at 40 CFR section 122.44(l) restrict backsliding in NPDES permits. These anti-backsliding provisions require that effluent limitations in a reissued permit must be as stringent as those in the previous permit, with some exceptions in which limitations may be relaxed. The effluent limitations in this Order are at least as stringent as those in the previous permit.
8. **Endangered Species Act Requirements.** This Order prohibits any act that results in the taking of a threatened or endangered species or any act that is now

prohibited, or becomes prohibited in the future, under either the California ESA (Fish and Game Code, sections 2050 to 2097) or the Federal ESA (16 USC sections 1531 to 1544). This Order requires compliance with effluent limits, receiving water limits, and other requirements to protect the beneficial uses of waters of the state. The Permittee is responsible for meeting all requirements of the applicable ESA.

9. **Water Rights.** Prior to making any change in the point of discharge, place of use, or purpose of use of treated wastewater that results in a decrease of flow in any portion of a surface or subterranean stream, the Permittee must file a petition with the State Water Board (State Water Board), Division of Water Rights, and receive approval for such a change. The State Water Board retains the jurisdictional authority to enforce such requirements under CWC section 1211.
10. **Domestic Water Quality.** In compliance with CWC section 106.3, it is the policy of the State of California that every human being has the right to safe, clean, affordable, and accessible water adequate for human consumption, cooking, and sanitary purposes. This Order promotes that policy by requiring discharges to meet maximum contaminant levels for radionuclides, MBAS, and bis (2-ethylhexyl) phthalate developed to protect human health and ensure that water is safe for domestic use.
11. **Water Recycling** - In accordance with statewide policies concerning water reclamation (See, e.g., CWC sections 13000 and 13550-13557, State Water Board Resolution No. 77-1 (Policy with Respect to Water Reclamation in California), and State Water Board Resolution No. 2009-0011 (Recycled Water Policy)), this Regional Water Board strongly encourages, wherever practical, water recycling, water conservation, and beneficial use of storm water and dry-weather urban runoff. The facility captures and treats the storm water, after being mixed with the domestic wastewater through its treatment facility. The Permittee shall investigate the feasibility of recycling, conservation, and/or alternative disposal methods for wastewater (such as groundwater injection), and/or use of storm water and dry-weather urban runoff. The Permittee has concluded that the use of recycled water is limited to the existing users because there is no proposed expansion of water recycling pipelines to accommodate new users. However, the Permittee shall submit an update to this feasibility study as part of the submittal of the Report of Waste Discharge (ROWD) for the next permit renewal.
12. **Monitoring and Reporting.** 40 CFR section 122.48 requires that all NPDES permits specify requirements for recording and reporting monitoring results. CWC sections 13267 and 13383 authorize the Regional Water Board to require technical and monitoring reports. The Monitoring and Reporting Program (MRP) establishes monitoring and reporting requirements to implement federal and state requirements. This MRP is provided in Attachment E.
13. **Sewage Sludge/Biosolids Requirements.** Section 405 of the CWA and implementing regulations at 40 CFR part 503 require that producers of sewage sludge/biosolids meet certain reporting, handling, and use or disposal requirements. The state has not been delegated the authority to implement this

program; therefore, USEPA is the implementing agency. This Order does not contain sewage sludge/biosolids requirements (40 CFR part 503), but provides sewage sludge/biosolids requirements that are applicable to the Permittee for informational purpose only in Attachment H .

14. **Mercury Provisions.** The State Water Board adopted Part 2 of the Water Quality Control Plan for Inland Surface Waters, Enclosed Bays, and Estuaries of California- Tribal and Subsistence Fishing Beneficial Uses and Mercury Provisions (Mercury Provisions) through Resolution No. 2017-0027, which was approved by OAL on June 28, 2017 and became effective upon U.S. EPA approval on July 14, 2017. The Mercury Provisions established one narrative and four numeric water quality objectives for mercury and three new beneficial use definitions, implemented through NPDES permits issued pursuant to CWA section 402, waste discharge requirements, or waivers of waste discharge requirements. The Provisions included implementation provisions for individual non-storm water NPDES permits for municipal and industrial dischargers; storm water discharges including MS4 discharges and discharges regulated by the Industrial General Permit (NPDES No. CAS000001); mine site remediation; nonpoint source discharges; dredging activities; and wetland projects.

The Mercury Provisions contain provisions that apply to POTWs and individual industrial discharges. The Mercury Provisions converted the fish tissue-based water quality objectives to water column values, denoted as “C”. The implementation section of the Mercury Provisions requires the application of section 1.3 of the SIP with modifications to determine whether a discharge has reasonable potential to cause or contribute to an exceedance of the water column concentration for mercury and the development of effluent limitations for mercury based on the water quality objective applicable to the receiving water in accordance with Chapter IV.D.2.b. (See section IV.C.3. for SIP procedures).

However, because there is a WLA for mercury in the Calleguas Creek Watershed Metals TMDL, the Mercury Provisions were not applicable in this permit.

15. **Bacteria Provisions.** The State Water Board adopted Part 3 of the Water Quality Control Plan for Inland Surface Waters, Enclosed Bays, and Estuaries of California- Bacteria Provisions and Water Quality Standards Variance Policy through Resolution No. 2018-0038, which was approved by OAL on February 4, 2019 and became effective upon USEPA approval on March 22, 2019. The Bacteria Provisions establish *Escherichia coli* (E. coli) as the sole indicator of pathogens in freshwater. These E. coli water quality objectives supersede any numeric water quality objectives for bacteria for the protection of the REC-1 beneficial use in Regional Water Board Basin Plans prior to the effective date of the Bacteria Provisions, except in certain circumstances, such as where there are site-specific numeric water quality objectives for bacteria.

This permit includes permit effluent limitations based on Title 22 disinfected tertiary treated wastewater for the protection of human health. These Title 22 disinfection requirements are more stringent than the Bacteria Provision effluent limitations. Therefore, the limitations based on the Bacteria Provisions are not

used in this permit. However, the Bacteria Provisions limitations are used as receiving water limitations.

D. Impaired Water Bodies on CWA 303(d) List

The State Water Board proposed the California 2014 - 2016 Integrated Report from a compilation of the adopted Regional Water Boards' Integrated Reports containing CWA section 303(d) List of Impaired Waters and section 305(b) Reports following recommendations from the Regional Water Boards and information solicited from the public and other interested persons. On October 3, 2017, the State Water Board adopted the California 2014 - 2016 Integrated Report. On April 06, 2018, USEPA approved California's 2014 - 2016 Integrated Report Section 303(d) List of Impaired Waters requiring Total Maximum Daily Loads (TMDLs) for the Los Angeles Region. The CWA section 303(d) list can be found at the following link:

http://www.waterboards.ca.gov/water_issues/programs/tmdl/integrated2014_2016.shtml

Calleguas Creek and its tributaries are in the California 2014 -2016 Integrated Report. The following are the identified pollutants impacting the receiving water:

Calleguas Creek Reach 7 (was Arroyo Simi Reaches 1 and 2 on 1998 303(d) List) – Calwater Watershed 40367000

Pollutants: Ammonia, boron, chloride, chlorpyrifos, diazinon, indicator bacteria, organophosphorus pesticides, sedimentation/siltation, sulfates, total dissolved solids, toxicity, and trash.

Calleguas Creek Reach 6 (was Arroyo Las Posas Reaches 1 and 2 on 1998 303(d) List) - Calwater Watershed 40362000

Pollutants: Ammonia, chlordane, chloride, chlorpyrifos, dichlorodiphenyltrichloroethane (DDT) (sediment), diazinon, dieldrin, fecal coliform, nitrate and nitrite, nitrate as nitrate (NO₃), sedimentation/siltation, sulfates, total dissolved solids, and toxicity.

Calleguas Creek Reach 3 (Potrero Road upstream to confluence with Conejo Creek on 1998 303(d) List) - Calwater Watershed 40312000

Pollutants: Ammonia, chlordane, chloride, DDT, dieldrin, nitrate and nitrite, polychlorinated biphenyls (PCBs), sedimentation/siltation, total dissolved solids, toxaphene, and trash.

Calleguas Creek Reach 2 (Estuary to Potrero Road - was Calleguas Creek Reaches 1 and 2 on 1998 303(d) List) - Calwater Watershed 40312000

Pollutants: Ammonia, chemA (tissue), chlordane (tissue), dissolved copper, DDT, dieldrin, endosulfan (tissue), fecal coliform, nitrogen, PCBs (tissue), sediment toxicity, sedimentation/siltation, toxaphene (tissue and sediment), and trash.

E. Other Plans, Policies and Regulations

- 1. Climate Change Adaptation and Mitigation.** On March 7, 2017, the State Water Board adopted a resolution in recognition of the challenges posed by climate change that requires a proactive approach to climate change in all State Water Board actions, including drinking water regulation, water quality protection, and financial assistance (Resolution No. 2017-0012). The resolution lays the foundation for a response to climate change that is integrated into all State Water Board actions, by giving direction to the State Water Board divisions and encouraging coordination with the Regional Water Boards. In response to the State Water Board's Resolution (No. 2017-0012), the Los Angeles Water Board adopted "A Resolution to Prioritize Actions to Adapt to and Mitigate the Impacts of Climate Change on the Los Angeles Region's Water Resources and Associated Beneficial Uses" (Resolution No. R18-004) on May 10, 2018. The resolution summarizes the steps taken so far to address the impacts of climate change within the Los Angeles Water Board's programs and lists a series of steps to move forward. These include the identification of potential regulatory adaptation and mitigation measures that could be implemented on a short-term and long-term basis by each of the Los Angeles Water Board's programs to take into account, and assist in mitigating where possible, the effects of climate change on water resources and associated beneficial uses. This Order contains provisions to require planning and actions to address climate change impacts in accordance with both the State and Regional Water Boards' resolutions.

The Permittee shall develop a Climate Change Effects Vulnerability Assessment and Management Plan (Climate Change Plan) and submit the Climate Change Plan to the Regional Water Board for the Executive Officer's approval no later than 12 months after adoption of this Order. The Climate Change Plan shall include an assessment of short and long term vulnerabilities of the facility and operations as well as plans to address vulnerabilities of collection systems, facilities, treatment systems, and outfalls for predicted impacts in order to ensure that facility operations are not disrupted, compliance with permit conditions is achieved, and receiving waters are not adversely impacted by discharges. Control measures shall include, but are not limited to, emergency procedures, contingency plans, alarm/notification systems, training, backup power and equipment, and the need for planned mitigations to ameliorate climate-induced impacts including, but not limited to, changing influent and receiving water quality and conditions, as well as the impact of rising sea level (where applicable), wildfires, storm surges and back-to-back severe storms that are expected to become more frequent.

- 2. Sources of Drinking Water Policy.** On May 19, 1988, the State Water Board adopted Resolution No. 88-63, *Sources of Drinking Water (SODW) Policy*, which established a policy that all surface and ground waters, with limited exemptions, are suitable or potentially suitable for municipal and domestic supply. To be consistent with State Water Board's SODW Policy, on March 27, 1989, the Regional Water Board adopted Resolution No. 89-03, *Incorporation of Sources of Drinking Water Policy into the Water Quality Control Plans (Basin Plans) – Santa Clara River Basin (4A)/ Los Angeles River Basin (4B)*.

Consistent with Regional Water Board Resolution No. 89-03 and State Water Board Resolution No. 88-63, in 1994 the Regional Water Board conditionally designated all inland surface waters in Table 2-1 of the 1994 Basin Plan as existing, intermittent, or potential for Municipal and Domestic Supply (MUN). However, the conditional designation in the 1994 Basin Plan included the following implementation provision: “no new effluent limitations will be placed in Waste Discharge Requirements as a result of these [potential MUN designations made pursuant to the SODW policy and the Regional Water Board’s enabling resolution] until the Regional Water Board adopts [a special Basin Plan Amendment that incorporates a detailed review of the waters in the Region that should be exempted from the potential MUN designations arising from SODW policy and the Regional Water Board’s enabling resolution].” On February 15, 2002, the USEPA clarified its partial approval (May 26, 2000) of the 1994 Basin Plan amendments and acknowledged that the conditional designations do not currently have a legal effect, do not reflect new water quality standards subject to USEPA review, and do not support new effluent limitations based on the conditional designations stemming from the SODW Policy until a subsequent review by the Regional Water Board finalizes the designations for these waters. This permit is designed to be consistent with the existing Basin Plan.

3. **Title 22 of the California Code of Regulations (CCR Title 22).** The California Department of Public Health (CDPH) established primary and secondary maximum contaminant levels (MCLs) for inorganic, organic, and radioactive contaminants in drinking water. These MCLs are codified in Title 22. The Basin Plan (Chapter 3) incorporates Title 22 primary MCLs by reference. This incorporation by reference is prospective, including future changes to the incorporated provisions as the changes take effect. Title 22 primary MCLs have been used as bases for effluent limitations in WDRs and NPDES permits to protect groundwater recharge beneficial use when that receiving groundwater is designated as MUN. Also, the Basin Plan specifies that “Ground waters shall not contain taste or odor-producing substances in concentrations that cause nuisance or adversely affect beneficial uses.”
4. **Secondary Treatment Regulations.** 40 CFR part 133 establishes the minimum levels of effluent quality to be achieved by secondary treatment. These limitations, established by USEPA, are incorporated into this Order, except where more stringent limitations are required by other applicable plans, policies, or regulations or to prevent backsliding.
5. **Storm Water.** General NPDES permit No. CAS000001 is not applicable to the Simi Valley WQCP because the facility captures and treats storm water collected on the premises. In addition, when the storm water collected at the facility is mixed with the domestic wastewater for treatment, the treated wastewater will be in compliance with the NPDES permit limitations that is more stringent than the storm water effluent limitations.
6. **Sanitary Sewer Overflows (SSOs).** The CWA prohibits the discharge of pollutants from point sources to surface waters of the United States unless authorized under an NPDES permit. (33 United States Code sections 1311, 1342). The Permittee must comply with State Water Board Order No. 2006-0003-

DWQ, *Statewide General Waste Discharge Requirements for Sanitary Sewer Systems*, (SSO WDR) as amended by State Water Board Order No. WQ 2013-0058-exec and any subsequent order updating these requirements. These statewide WDRs require public agencies that own or operate sanitary sewer systems with greater than one mile of sewer lines to enroll for coverage and comply with requirements, comply with requirements to develop and implement sewer system management plans, and report all SSO to the State Water Board's online SSOs database. The Permittee enrolled in the SSO WDRs in 2006, and the collection systems of the Permittee are covered under the SSO WDRs. This NPDES permit also contains requirements pertaining to the Permittee's collection system. The Permittee must properly operate and maintain its collection system (40 CFR section 122.41 (e)), report any non-compliance (40 CFR section 122.41(1)(6) and (7)), and mitigate any discharge from the collection system in violation of this NPDES permit (40 CFR section 122.41(d)).

The requirements contained in this Order sections VI.C.3.b (Spill Cleanup Contingency Plan section), VI.C.4 (Construction, Operation and Maintenance Specifications section), and VI.C.6 (Spill Reporting Requirements section) are consistent with the requirements of the SSO WDR. The Regional Water Board recognizes that there may be some overlap between these NPDES permit provisions and SSO WDR requirements, related to the collection systems. The requirements of the SSO WDR are considered the minimum thresholds (see Finding 11 of State Water Board Order No. 2006-0003-DWQ). To encourage efficiency, the Regional Water Board will accept the documentation prepared by the Permittees under the SSO WDR for compliance purposes as satisfying the requirements in sections VI.C.3.b, VI.C.4, and VI.C.6, provided the more stringent provisions contained in this NPDES permit are also addressed. Pursuant to SSO WDR, section D, provision 2(iii) and (iv), the provisions of this NPDES permit supersede the SSO WDR, for all purposes, including enforcement, to the extent the requirements may be deemed duplicative.

The requirements of this permit are more stringent than the SSO WDR because in addition to the SSO WDR requirements, this NPDES permit requires water quality monitoring of the receiving water when the spill reaches the surface water.

7. **Watershed Management** - This Regional Water Board has been implementing a Watershed Management Approach (WMA) to address water quality protection in the Los Angeles Region. Information about watersheds in the region can be obtained at the Regional Water Board's website at http://www.waterboards.ca.gov/losangeles/water_issues/programs/regional_program/watershed/index.shtml. The WMA emphasizes cooperative relationships between regulatory agencies, the regulated community, environmental groups, and other stakeholders in the watershed to achieve the greatest environmental improvements with the resources available.

The POTWs within the Calleguas Creek Watershed (CCW) have developed a watershed monitoring program to implement the requirements for monitoring, conducting special studies, and implementing actions to reduce discharges of pollutants covered by the TMDLs. This watershed monitoring program has been approved by the Regional Water Board. The responsible parties to the CCW

TMDLs have signed a Memorandum of Agreement to jointly fund and complete the implementation of the TMDL Calleguas Creek Watershed Monitoring Program (CCWTMP), which began in August 2008. The CCWTMP was created to better facilitate a coordinated monitoring effort where multiple TMDL monitoring requirements could be addressed via a single program that would carry out and manage all aspects of the monitoring activities. This monitoring program has been developed to easily integrate new TMDL monitoring efforts as TMDLs are adopted and/or special study monitoring efforts are required.

8. **Relevant TMDLs** – Section 303(d) of the CWA requires states to identify water bodies that do not meet water quality standards and then to establish TMDLs for each waterbody for each pollutant of concern. TMDLs identify the maximum amount of pollutants that can be discharged to waterbodies without causing violations of water quality standards.
 - a. **Calleguas Creek Watershed Salts TMDL** - On October 4, 2007, the Regional Water Board adopted Resolution No. R4-2007-016, Amendment to the Water Quality Control Plan – Los Angeles Region to Incorporate the Total Maximum Daily Load for Boron, Chloride, Sulfate, and TDS (Salts) in the Calleguas Creek Watershed. This Resolution was approved by the State Water Board, Office of Administrative Law, and USEPA on May 20, 2008, November 6, 2008, and December 2, 2008, respectively. This TMDL became effective on December 2, 2008.
 - b. **Calleguas Creek Watershed Nitrogen Compounds and Related Effects TMDL** - On October 24, 2002, the Regional Water Board adopted Resolution No. 02-017, *Amendment to the Water Quality Control Plan for the Los Angeles Region to Include a TMDL for Nitrogen Compounds and Related Effects in Calleguas Creek (Nitrogen Compounds TMDL)*. This Resolution was approved by the State Water Board, Office of Administrative Law, and USEPA on March 19, 2003, June 5, 2003, and June 20, 2003, respectively.

On September 11, 2008, the Regional Water Board adopted Resolution No. R4-2008-009, Amendment to the Water Quality Control Plan for the Los Angeles Region through revision of the Waste Load Allocation for the Calleguas Creek Watershed Nitrogen Compounds and Related Effects Total Maximum Daily Load (revised Nitrogen Compounds TMDL). This Basin Plan amendment corrects the mass based daily WLAs for ammonia to be used based upon MDEL, and updates the WLAs to be consistent with the current practice of recognizing that the flow is variable. The mass based WLAs for ammonia are corrected to be based on the maximum daily effluent limit, MDEL and the actual POTW effluent flow rate at the time the monitoring is conducted. This Order includes effluent limitations for nitrogen compounds established by the revised Nitrogen Compounds TMDL which became effective on October 15, 2009.
 - c. **Calleguas Creek Toxicity, Chlorpyrifos, and Diazinon TMDL** - On July 7, 2005, the Regional Water Board adopted Resolution No. R4-2005-009, *Amendment to the Water Quality Control Plan for the Los Angeles Region to Incorporate a Total Maximum Daily Load for Toxicity, Chlorpyrifos, and*

Diazinon in Calleguas Creek, its Tributaries, and Mugu Lagoon (Toxicity TMDL). This Resolution was approved by the State Water Board, Office of Administrative Law, and USEPA on September 22, 2005, November 27, 2005, and March 14, 2006, respectively. This Order includes effluent limitations for chlorpyrifos and diazinon established by the Toxicity TMDL which became effective on March 24, 2006.

The *Toxicity TMDL* includes a WLA of 1.0 TUC for toxicity, which is required to be implemented in accordance with USEPA, State Water Board, and Regional Water Board resolutions, guidance and policy at the time of permit issuance or renewal. Consistent with the Toxicity TMDL Implementation Plan, this toxicity WLA will be implemented using the recent USEPA guidance, National Pollutant Discharge Elimination System Test of Significant Toxicity Implementation Document (EPA 833-R-10-003, June 2010).

- d. **Calleguas Creek OC Pesticides and PCBs TMDL** - On July 7, 2005, the Regional Water Board adopted Resolution No. R4-2005-010, Amendment to the Water Quality Control Plan for the Los Angeles Region to Incorporate a Total Maximum Daily Load for Organochlorine Pesticides, Polychlorinated Biphenyls, and Siltation in Calleguas Creek, its Tributaries, and Mugu Lagoon (OC Pesticides and PCBs TMDL). This Resolution was approved by the State Water Board, Office of Administrative Law, and USEPA on September 22, 2005, January 20, 2006, and March 14, 2006, respectively. This Order includes effluent limitations for OC pesticides and PCBs established by the OC Pesticides and PCBs TMDL which became effective on March 24, 2006.
- e. **Calleguas Creek Watershed Metals TMDL** – On June 8, 2006, the Regional Water Board adopted Resolution No. R4-2006-012, Amendment to the Water Quality Control Plan for the Los Angeles Region to Incorporate a Total Maximum Daily Load for Metals for the Calleguas Creek, its Tributaries, and Mugu Lagoon (Metals TMDL). This Resolution was approved by the State Water Board, Office of Administrative Law, and USEPA on October 25, 2006, February 6, 2007, and March 26, 2007, respectively. This Order includes effluent limitations for metals established by the Metals TMDL which became effective on March 26, 2007.
- i. **Calleguas Creek Copper WER** – On November 9, 2006, the Regional Water Board adopted Resolution No. R4-2006-022, Amendment to the Water Quality Control Plan for the Los Angeles Region Water Effects Ratios (WERs) for Copper in Lower Calleguas Creek and Mugu Lagoon Located in the Calleguas Creek Watershed, Ventura County (Copper WER). This Resolution was approved by the State Water Board, Office of Administrative Law, and USEPA on June 19, 2007, August 16, 2007, and August 23, 2007, respectively. However, a copper WER is not relevant to the discharge of Simi Valley WQCP because the discharged effluent does not reach lower Calleguas Creek and Mugu Lagoon. Therefore, a copper WER will not be applied in this permit.

On October 13, 2016, the Regional Water Board adopted Resolution No. R16-007, *Amendment to the Water Quality Control Plan for the Los Angeles Region to the Revise the Total Maximum Daily Load for Metals and Selenium for the Calleguas Creek, its Tributaries, and Mugu Lagoon*. This Resolution was approved by the State Water Board, Office of Administrative Law, and USEPA on February 22, 2017, May 18, 2017, and June 8, 2017, respectively. However, none of these revisions shall affect discharge requirements of Simi Valley WQCP because the discharged effluent does not reach lower Calleguas Creek and Mugu Lagoon and there are no specific revisions related to Simi Valley WQCP.

IV. RATIONALE FOR EFFLUENT LIMITATIONS AND DISCHARGE SPECIFICATIONS

The CWA requires point source Permittees to control the amount of conventional, non-conventional, and toxic pollutants that are discharged into the waters of the United States. The control of pollutants discharged is established through effluent limitations and other requirements in NPDES permits. There are two principal bases for effluent limitations in 40 CFR section 122.44(a) requires that permits include applicable TBELs and standards; and 40 CFR section 122.44(d) requires that permits include WQBELs to attain and maintain applicable numeric and narrative water quality criteria to protect the beneficial uses of the receiving water.

The variety of potential pollutants found in the Facility discharges presents a potential for aggregate toxic effects to occur. Whole effluent toxicity (WET) is an indicator of the combined effect of pollutants contained in the discharge. Chronic toxicity is a more stringent requirement than acute toxicity. Therefore, chronic toxicity is considered pollutant of concern for protection and evaluation of narrative Basin Plan Objectives.

A. Discharge Prohibitions

Effluent and receiving water limitations in this Board Order are based on the CWA, Basin Plan, State Water Board's plans and policies, USEPA guidance and regulations, and best practicable waste treatment technology. This order authorizes the discharge of tertiary-treated wastewater from Discharge Points 001 and 002 only. It does not authorize any other types of discharges.

B. Technology-Based Effluent Limitations (TBELs)

1. Scope and Authority

Technology-based effluent limits require a minimum level of treatment for industrial/municipal point sources based on currently available treatment technologies while allowing the Permittee to use any available control techniques to meet the effluent limits. The 1972 CWA required POTWs to meet performance requirements based on available wastewater treatment technology. Section 301 of the CWA established a required performance level--referred to as "secondary treatment"--that all POTWs were required to meet by July 1, 1977. More specifically, section 301(b)(1)(B) of the CWA required that USEPA develop secondary treatment standards for POTWs as defined in section 304(d)(1). Based on this statutory requirement, USEPA developed national secondary

treatment regulations which are specified in 40 CFR part 133. These technology-based regulations apply to all POTWs and identify the minimum level of effluent quality to be attained by secondary treatment in terms of BOD₅20°C, TSS, and pH.

2. Applicable TBELs

This Facility is subject to the technology-based regulations for the minimum level of effluent quality attainable by secondary treatment in terms of BOD₅20°C, TSS, and pH. However, all TBELs from the previous Order No. R4-2014-0066-A01 are based on tertiary-treated wastewater treatment standards. These effluent limitations have been carried over from the previous Order to avoid backsliding. Further, mass-based effluent limitations are based on a design flow rate of 12.5 MGD. The removal efficiency for BOD and TSS is set at the minimum level attainable by secondary treatment technology. The following Table summarizes the TBELs applicable to the Facility:

Table F-6. Summary of TBELs

Parameter	Units	Average Monthly	Average Weekly	Maximum Daily
BOD ₅ 20°C	mg/L	20	30	45
BOD ₅ 20°C	lbs/day	2,080	3,130	4,690
TSS	mg/L	15	40	45
TSS	lbs/day	1,560	4,170	4,690
Removal Efficiency for BOD and TSS	%	≥85	--	--

Note that the mass based limitations, above, are based on the plant design flow rate of 12.5 MGD, and are calculated as follows: Flow (MGD) x Concentration (mg/L) x 8.34 (conversion factor) = lbs/day. During wet-weather storm events in which the flow exceeds the design capacity, the mass discharge rate limitations shall not apply, and concentration limitations will provide the only applicable effluent limitations.

This Facility is also subject to TBELs contained in similar NPDES permits, for similar facilities, based on the treatment level achievable by tertiary-treated wastewater treatment systems. These effluent limitations are consistent with the State Water Board precedential decision, State Water Board Order No. WQ 2004-0010 (City of Woodland).

C. Water Quality-Based Effluent Limitations (WQBELs)

1. Scope and Authority

CWA section 301(b) and 40 CFR section 122.44(d) require that permits include limitations more stringent than applicable federal technology-based requirements where necessary to achieve applicable water quality standards. This Order contains more stringent requirement than secondary treatment requirements that

are necessary to meet applicable water quality standards because the facility has tertiary treatment. The Regional Water Board has considered the factors listed in CWC section 13241 in establishing these requirements. The rationale for these requirements, which consist of tertiary treatment or equivalent requirements or other provisions, is discussed beginning in section IV.C.2.

40 CFR section 122.44(d)(1)(i) requires that permits include effluent limitations for all pollutants that are or may be discharged at levels that have the reasonable potential to cause or contribute to an exceedance of a water quality standard, including numeric and narrative objectives within a standard. Where reasonable potential has been established for a pollutant, but there is no numeric criterion or objective for the pollutant, WQBELs must be established using (1) USEPA criteria guidance under CWA section 304(a), supplemented where necessary by other relevant information; (2) an indicator parameter for the pollutant of concern; or (3) a calculated numeric water quality criterion, such as a proposed state criterion or policy interpreting the state's narrative criterion, supplemented with other relevant information, as provided in section 122.44(d)(1)(vi).

The process for determining reasonable potential and calculating WQBELs when necessary is intended to protect the designated uses of the receiving water as specified in the Basin Plan, and achieve applicable WQOs and criteria that are contained in other state plans and policies, or any applicable water quality criteria contained in the CTR and NTR.

2. Applicable Beneficial Uses and Water Quality Criteria and Objectives

- a. The Basin Plan establishes the beneficial uses for surface water bodies in the Los Angeles region. The beneficial uses of the Arroyo Simi affected by the discharge have been described previously in this Fact Sheet.
- b. The Basin Plan also specifies narrative and numeric WQOs applicable to surface water as shown in the following discussions.

i. BOD₅20°C and TSS

BOD₅20°C is a measure of the quantity of the organic matter in the water and, therefore, the water's potential for becoming depleted in dissolved oxygen. As organic degradation takes place, bacteria and other decomposers use the oxygen in the water for respiration. Unless there is a steady resupply of oxygen to the system, the water will quickly become depleted of oxygen. Adequate dissolved oxygen levels are required to support aquatic life. Depressions of dissolved oxygen can lead to anaerobic conditions resulting in odors, or, in extreme cases, fish kills.

40 CFR part 133 describes the minimum level of effluent quality attainable by secondary treatment, for BOD and TSS, as:

- The 30-day average shall not exceed 30 mg/L, and
- The 7-day average shall not exceed 45 mg/L.

Simi Valley WQCP provides tertiary treatment. The Facility achieves solids removals that are better than secondary-treated wastewater by filtering the effluent.

The monthly average, the 7-day average, and the daily maximum limits cannot be removed because none of the anti-backsliding exceptions apply. Those limits were all included in the previous permit and the Simi Valley WQCP has been able to meet both limits (monthly average and the daily maximum), for both BOD and TSS.

In addition to having mass-based and concentration-based effluent limitations for BOD and TSS, the Simi Valley WQCP also has a percent removal requirement for these two constituents. In accordance with 40 CFR sections 133.102(a)(3) and 133.102(b)(3), the 30-day average percent removal shall not be less than 85 percent. Percent removal is defined as a percentage expression of the removal efficiency across a treatment plant for a given pollutant parameter, as determined from the 30-day average values of the raw wastewater influent pollutant concentrations to the Facility and the 30-day average values of the effluent pollutant concentrations for a given time period.

ii. **pH**

The hydrogen ion activity of water (pH) is measured on a logarithmic scale, ranging from 0 to 14. While the pH of “pure” water at 25°C is 7.0, the pH of natural waters is usually slightly basic due to the solubility of carbon dioxide from the atmosphere. Minor changes from natural conditions can harm aquatic life. The effluent limitation for pH in this permit requiring that the wastes discharged shall at all times be within the range of 6.5 to 8.5 is taken from the Basin Plan which reads “the pH of inland surface waters shall not be depressed below 6.5 or raised above 8.5 as a result of waste discharge.”

iii. **Settleable Solids**

Excessive deposition of sediments can destroy spawning habitat, blanket benthic (bottom dwelling) organisms, and abrade the gills of larval fish. The limits for settleable solids are based on the Basin Plan narrative, “Waters shall not contain suspended or settleable material in concentrations that cause nuisance or adversely affect beneficial uses.” The numeric limits are empirically based on results obtained from the settleable solids 1-hour test, using an Imhoff cone.

It is impracticable to use a 7-day average limitation, because short-term spikes of settleable solid levels that would be permissible under a 7-day average scheme would not be adequately protective of all beneficial uses. The monthly average and the daily maximum limits cannot be removed because none of the anti-backsliding exceptions apply. The monthly average and daily maximum limits were both included in the previous permit and the Simi Valley WQCP has been able to meet both limits.

iv. **Oil and Grease**

Oil and grease are not readily soluble in water and form a film on the water surface. Oily films can coat birds and aquatic organisms, impacting respiration and thermal regulation, and causing death. Oil and grease can also cause nuisance conditions (odors and taste), are aesthetically unpleasant, and can restrict a wide variety of beneficial uses. The limits for oil and grease are based on the Basin Plan narrative, "Waters shall not contain oils, greases, waxes, or other materials in concentrations that result in a visible film or coating on the surface of the water or on objects in the water, that cause nuisance, or that otherwise adversely affect beneficial uses."

The numeric limits are empirically based on concentrations at which an oily sheen becomes visible in water. It is impracticable to use a 7-day average limitation, because spikes that occur under a 7-day average scheme could cause a visible oil sheen. A 7-day average scheme would not be sufficiently protective of beneficial uses. The monthly average and the daily maximum limits cannot be removed because none of the anti-backsliding exceptions apply. Both limits were included in the previous permit and the Simi Valley WQCP has been able to meet both limits.

v. **Residual Chlorine**

Disinfection of wastewaters with chlorine produces a chlorine residual. Chlorine and its reaction products are toxic to aquatic life. The limit for residual chlorine is based on the Basin Plan narrative, "Chlorine residual shall not be present in surface water discharges at concentrations that exceed 0.1 mg/L and shall not persist in receiving waters at any concentration that causes impairment of beneficial uses."

It is impracticable to use a 7-day average or a 30-day average limitation, because it is not as protective as of beneficial uses as a daily maximum limitation is. Chlorine is very toxic to aquatic life and short term exposures of chlorine may cause fish kills.

vi. **TDS, Sulfate, and Boron**

During wet-weather, the limitations for TDS, sulfate, and boron are based on the Basin Plan water quality objectives for the Calleguas Creek watershed (above Potrero Road) is 850 mg/L, 250 mg/L, and 1.0 mg/L, respectively.

During dry-weather, the limitations for TDS, sulfate, and boron are based on the WLAs contained in the *Calleguas Creek Salts TMDL*. Resolution No. R4-2007- 16, *Amendment to the Water Quality Control Plant – Los Angeles Region to Incorporate the Total Maximum Daily Load for Boron, Chloride, Sulfate, and TDS (Salts) in the Calleguas Creek Watershed*, adopted by the Regional Water Board on October 4, 2007. This Resolution was approved by the State Water Board, Office of Administrative Law, and USEPA on May 20, 2008, November 6, 2008, and December 2, 2008,

respectively. This TMDL became effective on December 2, 2008.

Calleguas Creek Salts TMDL contains WLAs for TDS, sulfate, chloride and boron. However, the Facility is consistently complying with the Basin Plan WQO for TDS, sulfate, and boron. Therefore, this permit includes final effluent limitations for TDS, sulfate, and boron based on Basin Plan WQOs.

vii. **Chloride**

The WQO for chloride in the Basin Plan for Calleguas Creek Watershed (above Potrero Road) is 150 mg/L.

On January 27, 1997, the Regional Water Board adopted Resolution No. 97-02, Amendment to the Basin Plan to incorporate a Policy for Addressing Levels of Chloride in Discharges of Wastewaters. It was approved by the State Water Board (SWRCB Resolution 97-94); approved by the Office of Administrative Law (OAL) on January 8, 1998. Resolution 97-02 served to revise the chloride water quality objective to 190 mg/L in Calleguas Creek (between Potrero Road and Arroyo Las Posas) and other surface waters.

On April 13, 1998, the Regional Water Board adopted Order No. 98-027, which temporarily amended NPDES Order No. 96-043 for Simi Valley WQCP to include an interim chloride daily maximum effluent limit to 190 mg/L based on Resolution 97-02. This interim limit expired on January 9, 2001.

On December 7, 2000, the Regional Water Board adopted Resolution No. 2000-22, to extend the Interim Chloride Limits for Discharges to Calleguas Creek until March 31, 2001.

On March 22, 2002, USEPA Region 9 established the Calleguas Creek Total Maximum Daily Load for chloride which used the 150 mg/L objective in the Basin Plan to establish a waste load allocation of 10,100 lbs/day for the Simi Valley WQCP during normal conditions, and a waste load allocation of 9,200 lbs/day for the Simi Valley WQCP during drought conditions.

On August 14, 2002 the City of Simi Valley (Simi Valley WQCP), Thousand Oaks (Hill Canyon WWTP), Camarillo Sanitary District (Camarillo WRP), Camrosa Sanitation District (Camrosa WRP), Ventura County Water Works District No. 1 (Moorpark facility) and the Regional Water Board entered into a "Stipulation for Order Issuing Stay, with Conditions," which stayed the chloride final effluent limitation in NPDES Order No. 96-043. The State Water Board adopted WQO 2002-0017, which approved the August 14, 2002 stipulation.

On June 5, 2003, the NPDES permits for the City of Simi Valley (Simi Valley WQCP), Thousand Oaks (Hill Canyon WWTP), Camarillo Sanitary District (Camarillo WRP) were renewed, thereby rescinding the 1996

NPDES Orders, except for enforcement purposes. The Permittees petitioned the revised NPDES Orders to the State Water Board, requested an extension of the chloride stay, and asked that the petitions be held in abeyance.

On October 10, 2003, the City of Simi Valley (Simi Valley WQCP), Thousand Oaks (Hill Canyon WWTP), Camarillo Sanitary District (Camarillo WRP), and the Regional Water Board entered into a "Stipulation for Further Order Issuing Stay, with Conditions," which stayed the chloride final effluent limitation in NPDES Order No. R4-2003-0081. The State Water Board adopted WQO 2003-0019, which approved the October 10, 2003, stipulation and held the petitions in abeyance for three years (until November 19, 2006).

On September 28, 2006, the State Water Board granted an extension of the abeyance until July 15, 2008. The State Water Board has continued granting extensions to the abeyance.

On October 4, 2007, the Regional Water Board adopted the *Calleguas Creek Salts TMDL*, Resolution No. R4-2007-016, *Amendment to the Water Quality Control Plant – Los Angeles Region to Incorporate the Total Maximum Daily Load for Boron, Chloride, Sulfate, and TDS (Salts) in the Calleguas Creek Watershed*. This Resolution was approved by the State Water Board, Office of Administrative Law, and USEPA on May 20, 2008, November 6, 2008, and December 2, 2008, respectively. This TMDL became effective on December 2, 2008. The Salts TMDL established WLAs for chloride. During wet weather, the chloride effluent limit is based on the water quality objective found in Basin Plan for the Calleguas Creek watershed (above Potrero Road), which is 150 mg/L. During dry weather, the final effluent limitation for chloride is based on the WLAs contained in the Salts TMDL.

viii. **Methylene Blue Active Substances (MBAS) & Cobalt Thiocyanate Active Substances (CTAS)**

The existing permit effluent limitation of 0.5 mg/L for MBAS was developed based on the Basin Plan incorporation of Title 22, Drinking Water Standards. Given the nature of the Facility which accepts domestic wastewater into the sewer system and treatment plant, and the characteristics of the pollutant discharged, the discharge has reasonable potential to exceed both the numeric MBAS WQO and the narrative WQO for the prohibition of floating material such as foams and scums. Therefore, an effluent limitation is required.

Cobalt thiocyanate active substances (CTAS) are monitored in the same way as MBAS. The presence or absence of CTAS during sampling assists permit writers and the Permittee in diagnosing the source of floating materials, such as foam or scum, which are prohibited by the Basin Plan when they cause nuisance or adversely affect beneficial uses. There is no limitation or compliance requirement for CTAS because it has no established water quality objective.

ix. **Total Inorganic Nitrogen ($\text{NO}_2 + \text{NO}_3$ as N)**

Total inorganic nitrogen is the sum of Nitrate-nitrogen and Nitrite-nitrogen. High nitrate levels in drinking water can cause health problems in humans. Infants are particularly sensitive and can develop methemoglobinemia (blue-baby syndrome). Nitrogen is also considered a nutrient. Excessive amounts of nutrients can lead to other water quality impairments.

(a) **Algae**

Excessive growth of algae and/or other aquatic plants can degrade water quality. Algal blooms sometimes occur naturally, but they are often the result of excess nutrients (i.e., nitrogen, phosphorus) from waste discharges or nonpoint sources. These algal blooms can lead to problems with tastes, odors, color, and increased turbidity and can depress the dissolved oxygen content of the water, leading to fish kills. Floating algal scum and algal mats are also an aesthetically unpleasant nuisance.

The WQO for biostimulatory substances are based on Basin Plan narrative, "Waters shall not contain biostimulatory substances in concentrations that promote aquatic growth to the extent that such growth causes nuisance or adversely affects beneficial uses," and other relevant information to arrive at a mass based-limit intended to be protective of the beneficial uses, pursuant to 40 CFR section 122.44(d). Total inorganic nitrogen will be the indicator parameter intended to control algae, pursuant to 40 CFR section 122.44(d)(1)(vi)(C).

(b) **Concentration-based limit**

The Basin Plan WQO for Calleguas Creek Watershed above Potrero Road for total inorganic nitrogen ($\text{NO}_2\text{-N} + \text{NO}_3\text{-N}$) is 10 mg/L. However, the Nitrogen Compound TMDL for this Watershed has been in effect since July 16, 2003 and assigns a WLA of 9 mg/L to the Simi Valley WQCP. Therefore, a total inorganic nitrogen effluent limitation of 9 mg/L, which is based on the Nitrogen Compound TMDL, will apply in this permit.

(c) **Mass-based limit**

Since the Nitrogen Compound TMDL does not specify any mass-based WLA for nitrate plus nitrite as nitrogen, mass based limits are not included for $\text{NO}_2\text{-N} + \text{NO}_3\text{-N}$.

x. **Nitrite as Nitrogen and Nitrate as Nitrogen**

The effluent limit for nitrite as nitrogen ($\text{NO}_2\text{-N}$) of 0.9 mg/L is based on the Calleguas Creek Watershed Nutrient TMDL Waste Load Allocation which was assigned to the Simi Valley WQCP. The effluent limit for nitrate as nitrogen ($\text{NO}_3\text{-N}$) of 9 mg/L is based on the Calleguas Creek Watershed Nutrient TMDL Waste Load Allocation which was assigned to the Simi Valley WQCP. Since the TMDL does not specify any mass-

based WLA for nitrate as nitrogen or nitrite as nitrogen, mass based limits are not included for either of the two constituents.

xi. **Total Ammonia**

Ammonia is a pollutant routinely found in the wastewater effluent of POTWs, in landfill-leachate, as well as in run-off from agricultural fields where commercial fertilizers and animal manure are applied. Ammonia exists in two forms – un-ionized ammonia (NH_3) and the ammonium ion (NH_4^+). They are both toxic, but the neutral, un-ionized ammonia species (NH_3) is much more toxic, because it is able to diffuse across the epithelial membranes of aquatic organisms much more readily than the charged ammonium ion. The form of ammonia is primarily a function of pH, but it is also affected by temperature and other factors. Additional impacts can also occur as the oxidation of ammonia lowers the dissolved oxygen content of the water, further stressing aquatic organisms. Oxidation of ammonia to nitrate may lead to groundwater impacts in areas of recharge. Groundwater recharge may be impacted by the discharge. Ammonia also combines with chlorine (often both are present in POTW treated effluent discharges) to form chloramines – persistent toxic compounds that extend the effects of ammonia and chlorine downstream.

On October 24, 2002, the Regional Water Board adopted Resolution No. 02-017, *Amendment to the Water Quality Control Plant for the Los Angeles Region to Include a TMDL for Nitrogen Compounds and Related Effects in Calleguas Creek*. This Resolution was approved by the State Water Resources Control Board, Office of Administrative Law, and USEPA on March 19, 2003, June 5, 2003, and June 20, 2003, respectively.

On September 11, 2008, the Regional Water Board adopted Resolution No. R4-2008-009, *Amendment to the Water Quality Control Plan for the Los Angeles Region through revision of the Waste Load Allocation for the Calleguas Creek Watershed Nitrogen Compounds and Related Effects Total Maximum Daily Load (revised Nitrogen Compounds TMDL)*. This Basin Plan amendment corrects the mass based daily WLAs for ammonia to be used based upon MDEL, and updates the WLAs to be consistent with the current practice of recognizing that the flow is variable. The mass based WLAs for ammonia are corrected to be based on the maximum daily effluent limit, MDEL and the actual POTW effluent flow rate at the time the monitoring is conducted. This Order includes effluent limitations for nitrogen compounds established by the revised Nitrogen Compounds TMDL which became effective on October 15, 2009. The Calleguas Creek Nitrogen Compounds TMDL has ammonia nitrogen waste load allocations of 3.3 mg/L and 2.4 mg/L as maximum daily and average monthly effluent limitation, respectively. These waste load allocations will apply as end-of-pipe effluent limitations for Simi Valley WQCP.

xii. **Bacteria Indicator**

Total coliform bacteria are used to indicate the likelihood of pathogenic bacteria in surface waters. Given the nature of the facility, a wastewater treatment plant, pathogens are likely to be present in the effluent in cases where the disinfection process is not operating adequately. As such, the permit contains the following:

(a) Effluent Limitations:

- (1) The 7 day median number of coliform organisms at some point in the treatment process must not exceed a Most Probable Number (MPN) or Colony Forming Unit (CFU) of 2.2 per 100 milliliters,
- (2) The number of coliform organisms must not exceed an MPN or CFU of 23 per 100 milliliters in more than one sample within any 30-day period, and
- (3) No sample shall exceed an MPN of CFU of 240 total coliform bacteria per 100 milliliters.

The current effluent limitations in the NPDES permit are based on recycled water criteria (Cal. Code Reqs., tit. 22, Chapter 3), referred to as the Title 22 requirements. These disinfection-based effluent limitations for coliform are for human health protection and are consistent with requirements established by the California Department of Public Health. These limits for coliform must be met at the point of the treatment train immediately following disinfection, as a measure of the effectiveness of the disinfection process.

(b) Receiving Water Limitation

The *Escherichia coli* (*E. coli*) concentration in the receiving water shall not exceed the following, as a result of wastes discharged:

- (1) Geometric Mean (Six-week rolling) Limits
 - *E. coli* shall not exceed 100 cfu/100 mL.
- (2) Statistical Threshold Value (STV)
 - *E. coli* STV of 320 cfu/100mL shall not be exceeded by more than 10 percent of the samples collected in a calendar month, calculated in a static manner.

These receiving water limitations are based on *Part 3 of the Water Quality Control Plan for Inland Surface Waters, Enclosed Bays, and Estuaries (ISWEBE Plan) of California* - Bacteria Provisions and Water Quality Standards Variance Policy, which went into effect upon USEPA approval on March 22, 2019. The Bacteria Provisions establish *Escherichia coli* (*E. coli*) as the sole indicator of pathogens in freshwater. These *E. coli* water quality objectives supersede any numeric water quality objectives for bacteria for the protection of the REC-1 beneficial use in Regional Water Board Basin Plans prior to the effective date of

the Bacteria Provisions, except in certain circumstances, such as where there are site-specific numeric water quality objectives for bacteria.

xiii. **Temperature**

The Basin Plan contains the following water quality objective for temperature:

The natural receiving water temperature of all regional waters shall not be altered unless it can be demonstrated to the satisfaction of the Regional Board that such alteration in temperature does not adversely affect beneficial uses. Alterations that are allowed must meet the requirements below.

For waters designated WARM, water temperature shall not be altered by more than 5°F above the natural temperature. At no time shall these WARM-designated waters be raised above 80°F as a result of waste discharges.

Based on the review of the facility's temperature data for this permit cycle, the effluent temperature has never exceeded the effluent limitation of 86°F. The downstream receiving water station RSW-002 has never exceeded the 80°F maximum temperature. Likewise for the upstream receiving water station RSW-001, the temperature has never exceeded the 80°F maximum temperature. Therefore, it can be reasonably concluded that the receiving water temperature has never been raised above 80 °F as a result of the facility's effluent discharge.

Temperature can adversely affect beneficial uses. The USEPA document, *Quality Criteria for Water 1986* [EPA 440/5-86-001, May 1, 1986], also referred to as the *Gold Book*, discusses temperature and its effects on beneficial uses, such as recreation and aquatic life.

- (a) The Federal Water Pollution Control Administration in 1967 called temperature "a catalyst, a depressant, an activator, a restrictor, a stimulator, a controller, a killer, and one of the most important water quality characteristics to life in water." The suitability of water for total body immersion is greatly affected by temperature. Depending on the amount of activity by the swimmer, comfortable temperatures range from 20°C to 30°C (68 °F to 86 °F).
- (b) Temperature also affects the self-purification phenomenon in water bodies and therefore the aesthetic and sanitary qualities that exist. Increased temperatures accelerate the biodegradation of organic material both in the overlying water and in bottom deposits which makes increased demands on the dissolved oxygen resources of a given system. The typical situation is exacerbated by the fact that oxygen becomes less soluble as water temperature increases. Thus, greater demands are exerted on an increasingly scarce resource which may lead to total oxygen depletion and obnoxious septic conditions. Increased temperature may increase the odor of water because of the increased volatility of odor-causing

compounds. Odor problems associated with plankton may also be aggravated.

- (c) Temperature changes in water bodies can alter the existing aquatic community. Coutant (1972) has reviewed the effects of temperature on aquatic life reproduction and development. Reproductive elements are noted as perhaps the most thermally restricted of all life phases assuming other factors are at or near optimum levels. Natural short-term temperature fluctuations appear to cause reduced reproduction of fish and invertebrates.

xiv. **Turbidity**

Turbidity is an expression of the optical property that causes light to be scattered in water due to particulate matter such as clay, silt, organic matter, and microscopic organisms. Turbidity can result in a variety of water quality impairments. The effluent limitation for turbidity is based on the Basin Plan and section 60301.320 of Title 22, Chapter 3, "Filtered Wastewater" of the CCR. To protect water contact recreation, the Order limits turbidity of the wastewater such that it does not exceed: (a) a daily average of 2 Nephelometric turbidity units (NTU); (b) 5 NTU more than 5 percent of the time (72 minutes) during any 24 hour period; and (c) 10 NTU at any time."

xv. **Radioactivity**

Radioactive substances are generally present in natural waters in extremely low concentrations. Mining or industrial activities increase the amount of radioactive substances in waters to levels that are harmful to aquatic life, wildlife, or humans. Section 301(f) of the CWA contains the following statement with respect to effluent limitations for radioactive substances: "Notwithstanding any of other provisions of this Act it shall be unlawful to discharge any radiological, chemical, or biological warfare agent, any high-level radioactive waste, or any medical waste, into the navigable waters." Chapter 4.4 of the CWC contains a similar prohibition under section 13375, which reads as follows: "The discharge of any radiological, chemical, or biological warfare agent into the waters of the state is hereby prohibited." However, rather than an absolute prohibition on radioactive substances, this permit sets the following effluent limit for radioactivity: "Radioactivity of the wastes discharged shall not exceed the limits specified in Title 22, Chapter 15, Article 5, sections 64442 and 64443, of the CCR, or subsequent revisions." The limit is based on the Basin Plan incorporation of Title 22, CCR, *Drinking Water Standards*, by reference, to protect beneficial use. Therefore, the accompanying Order will retain the limit for radioactivity.

xvi. **CTR and SIP**

This Order implements the CTR and SIP, which specify numeric objectives for toxic substances and the procedures whereby these objectives are to be implemented. The procedures include those used to

conduct reasonable potential analysis (RPA) to determine the need for effluent limitations for priority pollutants. The Technical Support Document for Water Quality-based Toxics Control (TSD) specifies the procedures to conduct reasonable potential analyses for non-priority pollutants.

3. Determining the Need for WQBELs

The Regional Water Board developed WQBELs for ammonia-nitrogen, nitrite-nitrogen, nitrate-nitrogen, nitrite plus nitrite as nitrogen, TDS, sulfate, chloride, boron, copper, nickel, mercury, chlordane, 4,4'-DDD, 4,4'-DDE, 4,4'-DDT, dieldrin, PCBs, toxaphene, chlorpyrifos, diazinon, and chronic toxicity based upon Total Maximum Daily Loads (TMDLs). The effluent limitations for these pollutants were established on the basis of the wasteload allocations established in these TMDL. The Regional Water Board developed water quality-based effluent limitations for these pollutants pursuant to 40 CFR section 122.44(d)(1)(vii), which does not require or contemplate a separate reasonable potential analysis at the permitting stage. Similarly, the SIP at Section 1.3 recognizes that a separate reasonable potential analysis at the permitting stage is not necessary if a TMDL has been developed.

In accordance with Section 1.3 of the SIP, and noting the exceptions above, during the development of this Order, the Regional Water Board conducted a reasonable potential analysis for each priority pollutant with an applicable criterion or objective to determine if a WQBEL is required in the permit. The Regional Water Board analyzed effluent data to determine if a pollutant in a discharge has a reasonable potential to cause or contribute to an excursion above a state water quality standard. For all parameters that demonstrate reasonable potential, numeric WQBELs are required. The RPA considers water quality criteria from the CTR and NTR, and, when applicable, water quality objectives specified in the Basin Plan. To conduct the RPA, the Regional Water Board staff identified the maximum effluent concentration (MEC) and maximum background concentration in the receiving water for each constituent, based on data provided by the Permittee. The monitoring data cover the period from July 2014 to March 2019.

Section 1.3 of the SIP provides the procedures for determining reasonable potential to exceed applicable water quality criteria and objectives. The SIP specifies three triggers to complete an RPA:

Trigger 1 – If the MEC is greater than or equal to the CTR water quality criteria or applicable objective (C), a limitation is needed.

Trigger 2 – If background water quality (B) > C and the pollutant is detected in the effluent, a limitation is needed.

Trigger 3 – If other related information such as CWA 303(d) listing for a pollutant, discharge type, compliance history, then best professional judgment is used to determine that a limit is needed.

Sufficient effluent and ambient data are needed to conduct a complete RPA. If data are not sufficient, the Permittee will be required to gather the appropriate data for the Regional Water Board to conduct the RPA. Upon review of the data, and if the Regional Water Board determines that WQBELs are needed to protect the beneficial uses, the permit will be reopened for appropriate modification.

The RPA was performed for the priority pollutants regulated in the CTR for which data are available. Based on the RPA, pollutants that demonstrate reasonable potential are copper, mercury, nickel, chlordane, 4,4'-DDD, 4,4'-DDE, 4,4'-DDT, dieldrin, PCBs, and toxaphene because TMDLs are adopted for these constituents. Selenium also shows reasonable potential because receiving water concentration background (B) is greater than the criteria and detected at the effluent. Cyanide and bis (2-ethylhexyl) phthalate show reasonable potential because MEC is greater than the water quality criteria or applicable objective (C). The following Table summarizes results from RPA.

Table F-7. Summary of Reasonable Potential Analysis

CTR No.	Constituent	Applicable Water Quality Criteria (C) µg/L	Max Effluent Conc. (MEC) µg/L	Maximum Detected Receiving Water Conc. (B) µg/L	RPA Result - Need Limitation ?	Reason
1	Antimony	4300	0.98	0.87	No	C>B, C>MEC
2	Arsenic	150	2.6	3.7	No	C>B, C>MEC
3	Beryllium	Narrative	ND	ND	No	C>B, C>MEC
4	Cadmium	6.2	ND	0.43	No	C>B, C>MEC
5a	Chromium III	549.5	4.3	9.9	No	C>B, C>MEC
5b	Chromium VI	11	5	0.5	No	C>B, C>MEC
6	Copper	29	9.5	92	Yes	TMDL
7	Lead	10.8	ND	1.9	No	C>B, C>MEC
8	Mercury	0.012	0.1	0.5	Yes	TMDL
9	Nickel	166.6	4.5	27	Yes	TMDL
10	Selenium	5	2.4	29	Yes	Tier 2
11	Silver	36.8	ND	ND	No	C>B, C>MEC
12	Thallium	6.3	ND	ND	No	C>B, C>MEC
13	Zinc	376.1	65	31	No	TMDL; but no assigned WLA
14	Cyanide	5.2	8.1	2.5	Yes	Tier 1
15	Asbestos	7x10 ⁶ fibers/L	No sample	No sample	No	N/A

CTR No.	Constituent	Applicable Water Quality Criteria (C) µg/L	Max Effluent Conc. (MEC) µg/L	Maximum Detected Receiving Water Conc. (B) µg/L	RPA Result - Need Limitation ?	Reason
16	2,3,7,8-TCDD (Dioxin)	1.4x10 ⁻⁰⁸	ND	ND	No	C>B, C>MEC
17	Acrolein	780	ND	ND	No	C>B, C>MEC
18	Acrylonitrile	0.66	ND	ND	No	C>B, C>MEC
19	Benzene	71	ND	ND	No	C>B, C>MEC
20	Bromoform	360	ND	ND	No	C>B, C>MEC
21	Carbon Tetrachloride	4.4	ND	ND	No	C>B, C>MEC
22	Chlorobenzene	21,000	ND	ND	No	C>B, C>MEC
23	Dibromochloro-methane	34	1.6	ND	No	C>B, C>MEC
24	Chloroethane	No criteria	ND	ND	No	No criteria
25	2-chloroethyl vinyl ether	No criteria	ND	ND	No	No criteria
26	Chloroform	No criteria	6.6	ND	No	No criteria
27	Dichlorobromo-methane	46	3.7	ND	No	C>B, C>MEC
28	1,1-dichloroethane	No criteria	ND	ND	No	No criteria
29	1,2-dichloroethane	99	ND	ND	No	C>B, C>MEC
30	1,1-dichloroethylene	3.2	ND	ND	No	C>B, C>MEC
31	1,2-dichloropropane	39	ND	ND	No	C>B, C>MEC
32	1,3-dichloropropylene	1,700	ND	ND	No	C>B, C>MEC
33	Ethylbenzene	29,000	ND	ND	No	C>B, C>MEC
34	Methyl bromide	4,000	0.5	ND	No	C>B, C>MEC
35	Methyl chloride	No criteria	29	ND	No	No criteria
36	Methylene chloride	1,600	ND	ND	No	C>B, C>MEC
37	1,1,2,2-tetrachloroethane	11	ND	ND	No	C>B, C>MEC
38	Tetrachloroethylene	8.85	ND	ND	No	C>B, C>MEC
39	Toluene	200,000	ND	ND	No	C>B, C>MEC

CTR No.	Constituent	Applicable Water Quality Criteria (C) µg/L	Max Effluent Conc. (MEC) µg/L	Maximum Detected Receiving Water Conc. (B) µg/L	RPA Result - Need Limitation ?	Reason
40	Trans 1,2-Dichloroethylene	140,000	ND	ND	No	C>B, C>MEC
41	1,1,1-Trichloroethane	No criteria	ND	ND	No	C>B, C>MEC
42	1,1,2-Trichloroethane	42	ND	ND	No	C>B, C>MEC
43	Trichloroethylene	81	ND	2.7	No	C>B, C>MEC
44	Vinyl Chloride	525	ND	ND	No	C>B, C>MEC
45	2-chlorophenol	400	ND	ND	No	C>B, C>MEC
46	2,4-dichlorophenol	790	ND	ND	No	C>B, C>MEC
47	2,4-dimethylphenol	2,300	ND	ND	No	C>B, C>MEC
48	4,6-dinitro-o-cresol(aka 2-methyl-4,6-Dinitrophenol)	765	ND	ND	No	C>B, C>MEC
49	2,4-dinitrophenol	14,000	ND	ND	No	C>B, C>MEC
50	2-nitrophenol	No criteria	ND	ND	No	No criteria
51	4-nitrophenol	No criteria	ND	ND	No	No criteria
52	3-Methyl-4-Chlorophenol (aka P-chloro-m-cresol)	No criteria	ND	ND	No	No criteria
53	Pentachlorophenol	8.2	ND	ND	No	C>B, C>MEC
54	Phenol	4,600,000	2	ND	No	C>B, C>MEC
55	2,4,6-trichlorophenol	6.5	ND	ND	No	C>B, C>MEC
56	Acenaphthene	2,700	ND	ND	No	C>B, C>MEC
57	Acenaphthylene	No criteria	ND	ND	No	No criteria
58	Anthracene	110,000	ND	ND	No	C>B, C>MEC
59	Benzidine	0.00054	ND	ND	No	C>B, C>MEC
60	Benzo(a)Anthracene	0.049	ND	ND	No	C>B, C>MEC
61	Benzo(a)Pyrene	0.049	ND	ND	No	C>B, C>MEC
62	Benzo(b)Fluoranthene	0.049	ND	ND	No	C>B, C>MEC

CTR No.	Constituent	Applicable Water Quality Criteria (C) µg/L	Max Effluent Conc. (MEC) µg/L	Maximum Detected Receiving Water Conc. (B) µg/L	RPA Result - Need Limitation ?	Reason
63	Benzo(ghi)Perylene	No criteria	ND	ND	No	No criteria
64	Benzo(k)Fluoranthene	0.049	ND	ND	No	C>B, C>MEC
65	Bis(2-Chloroethoxy) methane	No criteria	ND	ND	No	No criteria
66	Bis(2-Chloroethyl)Ether	1.4	ND	ND	No	C>B, C>MEC
67	Bis(2-Chloroisopropyl) Ether	170,000	ND	ND	No	C>B, C>MEC
68	Bis(2-Ethylhexyl)Phthalate	4	5.3	7.7	Yes	Tier 1 and Tier 2
69	4-Bromophenyl Phenyl Ether	No criteria	ND	ND	No	C>B, C>MEC
70	Butylbenzyl Phthalate	5,200	ND	ND	No	C>B, C>MEC
71	2-Chloronaphthalene	4,300	ND	ND	No	C>B, C>MEC
72	4-Chlorophenyl Phenyl Ether	No criteria	ND	ND	No	No criteria
73	Chrysene	0.049	ND	ND	No	C>B, C>MEC
74	Dibenzo(a,h) Anthracene	0.049	ND	ND	No	C>B, C>MEC
75	1,2-Dichlorobenzene	17,000	ND	ND	No	C>B, C>MEC
76	1,3-Dichlorobenzene	2,600	ND	ND	No	C>B, C>MEC
77	1,4-Dichlorobenzene	2,600	ND	ND	No	C>B, C>MEC
78	3-3'-Dichlorobenzidine	0.077	ND	ND	No	C>B, C>MEC
79	Diethyl Phthalate	120,000	0.27	ND	No	C>B, C>MEC

CTR No.	Constituent	Applicable Water Quality Criteria (C) µg/L	Max Effluent Conc. (MEC) µg/L	Maximum Detected Receiving Water Conc. (B) µg/L	RPA Result - Need Limitation ?	Reason
80	Dimethyl Phthalate	2,900,000	ND	ND	No	C>B, C>MEC
81	Di-n-Butyl Phthalate	12,000	ND	ND	No	C>B, C>MEC
82	2-4-Dinitrotoluene	9.1	ND	ND	No	C>B, C>MEC
83	2-6-Dinitrotoluene	No criteria	ND	ND	No	No criteria
84	Di-n-Octyl Phthalate	No criteria	ND	ND	No	No criteria
85	1,2-Diphenylhydrazine	0.54	ND	ND	No	C>B, C>MEC
86	Fluoranthene	370	ND	ND	No	C>B, C>MEC
87	Fluorene	14,000	ND	ND	No	C>B, C>MEC
88	Hexachlorobenzene	50	ND	ND	No	C>B, C>MEC
89	Hexachlorobutadiene	50	ND	ND	No	C>B, C>MEC
90	Hexachlorocyclopentadiene	17,000	ND	ND	No	C>B, C>MEC
91	Hexachloroethane	8.9	ND	ND	No	C>B, C>MEC
92	Indeno(1,2,3-cd)Pyrene	0.049	ND	ND	No	C>B, C>MEC
93	Isophorone	600	ND	ND	No	C>B, C>MEC
94	Naphthalene	No criteria	ND	ND	No	No criteria
95	Nitrobenzene	1,900	ND	ND	No	C>B, C>MEC
96	N-Nitrosodimethylamine	8.1	ND	ND	No	C>B, C>MEC
97	N-Nitrosodi-n-Propylamine	1.4	ND	ND	No	C>B, C>MEC
98	N-Nitrosodiphenylamine	16	ND	ND	No	C>B, C>MEC
99	Phenanthrene	No criteria	ND	ND	No	No criteria
100	Pyrene	11,000	ND	ND	No	C>B, C>MEC
101	1,2,4-Trichlorobenzene	No criteria	ND	ND	No	No criteria
102	Aldrin	0.00014	ND	ND	No	C>B, C>MEC

CTR No.	Constituent	Applicable Water Quality Criteria (C) µg/L	Max Effluent Conc. (MEC) µg/L	Maximum Detected Receiving Water Conc. (B) µg/L	RPA Result - Need Limitation ?	Reason
103	Alpha-BHC	0.013	ND	0.016	No	C>MEC
104	Beta-BHC	0.046	0.0042	ND	No	C>B, C>MEC
105	Gamma-BHC (aka Lindane)	0.063	ND	ND	No	C>B, C>MEC
106	delta-BHC	No criteria	ND	ND	No	No criteria
107	Chlordane	0.00059	ND	ND	Yes	TMDL
108	4,4'-DDT	0.00059	ND	ND	Yes	TMDL
109	4,4'-DDE	0.00059	ND	ND	Yes	TMDL
110	4,4'-DDD	0.00084	ND	ND	Yes	TMDL
111	Dieldrin	0.00014	ND	ND	Yes	TMDL
112	Alpha-Endosulfan	0.056	0.04	ND	No	C>B, C>MEC
113	Beta-Endosulfan	0.056	ND	ND	No	C>B, C>MEC
114	Endosulfan Sulfate	240	ND	ND	No	C>B, C>MEC
115	Endrin	0.036	ND	ND	No	C>B, C>MEC
116	Endrin Aldehyde	0.81	ND	ND	No	C>B, C>MEC
117	Heptachlor	0.00021	ND	ND	No	C>B, C>MEC
118	Heptachlor Epoxide	0.00011	ND	ND	No	C>B, C>MEC
119	PCB 1016	0.00017	ND	ND	Yes	TMDL
120	PCB 1221	0.00017	ND	ND	Yes	TMDL
121	PCB 1232	0.00017	ND	ND	Yes	TMDL
122	PCB 1242	0.00017	ND	ND	Yes	TMDL
123	PCB 1248	0.00017	ND	ND	Yes	TMDL
124	PCB 1254	0.00017	ND	ND	Yes	TMDL
125	PCB 1260	0.00017	ND	ND	Yes	TMDL
126	Toxaphene	0.0002	ND	ND	Yes	TMDL

4. WQBEL Calculations

- a. **Calculation Options.** Once RPA has been conducted using either the TSD or the SIP methodologies, WQBELs are calculated. Alternative procedures for calculating WQBELs include:
 - i. Use WLA from applicable TMDL
 - ii. Use a steady-state model to derive MDELs and AMELs.

- iii. Where sufficient data exist, use a dynamic model which has been approved by the State Water Board.
- b. **Calleguas Creek Watershed Metals TMDL.** Concentration-based and mass-based WLAs are established for copper and nickel in total recoverable forms, and are applied to POTWs during both wet and dry weather. Mass-based WLAs are developed for mercury for POTWs. The TMDL does not include zinc WLAs because current information indicate that numeric targets for zinc are attained. The TMDL implementation plan includes a task to provide State Water Board data to support delisting of zinc. Waste load allocations for selenium are not set for POTWs because POTWs do not discharge to reaches listed for selenium. However, for Simi Valley WQCP's case, selenium effluent limitations are prescribed because the data show reasonable potential (Tier 2, B>C and was detected at the effluent).
- c. **SIP Calculation Procedure.** Section 1.4 of the SIP requires the step-by-step procedure to "adjust" or convert CTR numeric criteria into AMELs and MDELs, for toxics.

Step 3 of Section 1.4 of the SIP (page 8) lists the statistical equations that adjust CTR criteria for effluent variability.

Step 5 of Section 1.4 of the SIP (page 10) lists the statistical equations that adjust CTR criteria for averaging periods and exceedance frequencies of the criteria/objectives. This section also reads, "For this method only, maximum daily effluent limitations shall be used for POTWs in place of average weekly limitations.

Sample calculation for Cyanide:

Step 1: Identify applicable water quality criteria.

From California Toxics Rule (CTR), we can obtain the Criterion Maximum Concentration (CMC) and the Criterion Continuous Concentration (CCC).

Freshwater Aquatic Life Criteria:

CMC = 22 µg/L (CTR page 31712, column B1) and

CCC = 5.2 µg/L (CTR page 31712, column B2); and

Human Health Criteria for Organisms only = 220,000 µg/L (CTR page 31712, column D2).

Step 2: Calculate effluent concentration allowance (ECA)

ECA = Criteria in CTR, since no dilution is allowed.

Step 3: Determine long-term average (LTA) discharge condition

i. **Calculate CV:**

CV = Standard Deviation/Mean

= 0.7

Find the ECA Multipliers from SIP Table 1 (page 7), or by calculating them using equations on SIP page 6. When CV = 0.7, then:

ECA Multiplier acute = 0.281 and

ECA Multiplier chronic = 0.481

LTA acute = ECA acute x ECA Multiplier acute

$$= 22 \mu\text{g/L} \times 0.281 = 6.182 \mu\text{g/L}$$

LTA chronic = ECA chronic x ECA Multiplier chronic

$$= 5.2 \mu\text{g/L} \times 0.481 = 2.501 \mu\text{g/L}$$

Step 4: Select the lowest LTA

In this case, LTA chronic < LTA acute, therefore lowest LTA = 2.501 $\mu\text{g/L}$

Step 5: Calculate the Average Monthly Effluent Limitation (AMEL) & Maximum Daily Effluent Limitation (MDEL) for AQUATIC LIFE

- i. Find the multipliers. You need to know CV and n (frequency of sample collection per month). If effluent samples are collected 4 times a month or less, then $n = 4$. CV was determined to be 0.7 in a previous step.

AMEL Multiplier = 1.651

MDEL Multiplier = 3.559

- ii. AMEL aquatic life = lowest LTA (from Step 4) x AMEL Multiplier
 $= 2.501 \mu\text{g/L} \times 1.651 = 4.1 \mu\text{g/L}$
- iii. MDEL aquatic life = lowest LTA (from Step 4) x MDEL Multiplier
 $= 2.501 \mu\text{g/L} \times 3.559 = 8.9 \mu\text{g/L}$

Step 6: Find the Average Monthly Effluent Limitation (AMEL) & Maximum Daily Effluent Limitation (MDEL) for HUMAN HEALTH

- i. Find factors. Given CV = 0.7 and $n = 4$.

For AMEL human health limit, there is no factor.

The MDEL/AMEL human health factor = 2.16

- ii. AMEL human health = ECA = 220,000 $\mu\text{g/L}$
- iii. MDEL human health = ECA x MDEL/AMEL factor
 $= 220,000 \mu\text{g/L} \times 2.16 = 475,200 \mu\text{g/L}$

Step 7: Compare the AMELs for Aquatic life and Human health and select the lowest. Compare the MDELs for Aquatic life and Human health and select the lowest

- i. Lowest AMEL = 4.1 $\mu\text{g/L}$ (Based on Aquatic life protection)
- ii. Lowest MDEL = 8.9 $\mu\text{g/L}$ (Based on Aquatic life protection)

d. Impracticability Analysis

Federal NPDES regulations contained in 40 CFR section 122.45 (continuous discharges) states that, for POTWs, all permit limitations, standards, and

prohibitions, including those to achieve water quality standards, shall unless impracticable be stated as average weekly and average monthly.

As stated by USEPA in its long standing guidance for developing WQBELs average alone limitations are not practical for limiting acute, chronic, and human health toxic effects.

For example, a POTW sampling for a toxicant to evaluate compliance with a 7-day average limitation could fully comply with this average limit, but still be discharging toxic effluent on one, two, three, or up to four of these seven days and not be meeting 1-hour average acute criteria or 4-day average chronic criteria. Similarly, a 7-day average alone would not protect one, two, three, or four days of discharging pollutants in excess of the acute and chronic criteria. Fish exposed to these endocrine disrupting chemicals will be passed on to the human consumer. Endocrine disruptors alter hormonal functions by several means. These substances can:

- i. Mimic or partly mimic the sex steroid hormones estrogens and androgens (the male sex hormone) by binding to hormone receptors or influencing cell signaling pathways.
- ii. Block, prevent and alter hormonal binding to hormone receptors or influencing cell signaling pathways.
- iii. Alter production and breakdown of natural hormones.
- iv. Modify the making and function of hormone receptors.

For this reason, USEPA recommends daily maximum and 30-day average limits for regulating toxics in all NPDES discharges. For the purposes of protecting the acute effects of discharges containing toxicants (CTR human health for the ingestion of fish), daily maximum limitations have been established in this NPDES permit for certain priority pollutants because they are considered to be carcinogens, endocrine disruptors, and/or bioaccumulative.

- e. **Mass-based limits.** 40 CFR section 122.45(f)(1) requires that, except under certain conditions, or for certain pollutants, all permit limits, standards, or prohibitions be expressed in terms of mass units. 40 CFR section 122.45(f)(2) allows the permit writer, at its discretion, to express limits in additional units (e.g., concentration units). The regulations mandate that, where limits are expressed in more than one unit, the permittee must comply with both.

Generally, mass-based limits ensure that proper treatment, and not dilution, is employed to comply with the final effluent concentration limits. Concentration-based effluent limits, on the other hand, discourage the reduction in treatment efficiency during low-flow periods and require proper operation of the treatment units at all times. In the absence of concentration-based effluent limits, a permittee would be able to increase its effluent concentration (i.e., reduce its level of treatment) during low-flow periods and

still meet its mass-based limits. To account for this, this permit includes mass and concentration limits for some constituents.

Table F-8. Summary of WQBELs for Discharge Points 001 and 002

Parameter	Units	Average Monthly	Average Weekly	Maximum Daily	Remarks
Ammonia Nitrogen	mg/L	2.4	--	3.3	--
Ammonia Nitrogen	lbs/day	--	--	2.9 x Q ⁹	--
Nitrate + Nitrite (as N)	mg/L	9	--	--	--
Nitrate (as N)	mg/L	9	--	--	--
Nitrite (as N)	mg/L	0.9	--	--	--
Copper	µg/L	30.5	--	31	See note "a" below
Nickel	µg/L	169	--	960	See note "a" below
Mercury	lbs/month	0.031	--	--	See note "a" and "b" below
Selenium	µg/L	4.5	--	6.8	--
Selenium	lbs/day	0.47	--	0.71	See note "c" below
Cyanide	µg/L	4.1	--	8.9	--
Cyanide	lbs/day	0.43	--	0.93	See note "c" below
Bis (2-Ethylhexyl) Phthalate	µg/L	4	--	--	--
Bis (2-Ethylhexyl) Phthalate	lbs/day	0.42	--	--	See note "c" below
Chlorpyrifos	µg/L	0.014	--	0.025	See note "d" below
Diazinon	µg/L	0.1	--	0.1	See note "d" below

⁹ Q represents the POTW effluent flow at the time the water quality measurement is collected (not to exceed 12.5 MGD) and a conversion factor to lbs/day based on the units of measure for the flow.

Parameter	Units	Average Monthly	Average Weekly	Maximum Daily	Remarks
Chronic Toxicity ^{10,11}	Pass or Fail, % Effect (TST)	Pass ¹²	--	Pass or % Effect < 50	--
Chlordane	µg/L	0.00059	--	0.0012	--
4,4'-DDD	µg/L	0.00084	--	0.0017	--
4,4'-DDE	µg/L	0.00059	--	0.0012	--
4,4'-DDT	µg/L	0.00059	--	0.0012	--
Dieldrin	µg/L	0.00014	--	0.00028	--
PCBs ¹³	µg/L	0.00017	--	0.00034	--
Toxaphene	µg/L	0.00016	--	0.00033	--

- This limitation is derived from the final waste load allocation, as set forth in the Calleguas Creek Watershed Metals TMDL, established by the Regional Water Board on June 8, 2006. The TMDL became effective on March 26, 2007.
- According to Metals TMDL, the mercury (in suspended sediment) effluent limitation, in lbs/month, is assumed that the total load in effluent water is equal to the suspended sediment load.
- The mass based limitations are based on the plant design flow rate of 12.5 MGD, and are calculated as follows: Flow (MGD) x Concentration (mg/L) x 8.34 (conversion factor) = lbs/day. During wet-weather storm events in which the flow exceeds the design capacity, the mass discharge rate limitations

¹⁰ The *Calleguas Creek Watershed Toxicity TMDL* includes a WLA of 1.0 TUC for toxicity, which is required to be implemented in accordance with USEPA, State Water Board, and Regional Water Board resolutions, guidance and policy at the time of permit issuance or renewal. In addition, a numeric WQBEL is established because effluent data showed that there is reasonable potential for the effluent to cause or contribute to an exceedance of the chronic toxicity water quality objective. The numeric WLA is protective of both the numeric acute toxicity and the narrative toxicity Basin Plan water quality objective. Consistent with the *Toxicity TMDL*, these chronic toxicity WLA-based final effluent limitations will be implemented using the *Short Term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Freshwater Organisms* (USEPA 2002, EPA-821-R-02-013), and current USEPA guidance in *National Pollutant Discharge Elimination System Test of Significant Toxicity Implementation Document* (EPA 833-R-10-003, June /2010) and *EPA Regions 8, 9 and 10 Toxicity Training Tool* (January 2010), <http://www2.epa.gov/region8/epa-regions-8-9-and-10-toxicity-training-tool-january-2010>.

¹¹ The Median Monthly Effluent Limitation (MMEL) shall be reported as "Pass" or "Fail." The Maximum Daily Effluent Limitation (MDEL) shall be reported as "Pass" or "Fail" and "% Effect." The MMEL for chronic toxicity shall only apply when there is a discharge on more than one day in a calendar month period. During such calendar months, up to three independent toxicity tests may be conducted when one toxicity test results in "Fail."

¹² This is a Median Monthly Effluent Limitation.

¹³ Applies to sum of all congener or isomer or homolog or Aroclor analyses.

shall not apply, and concentration limitations shall be the only applicable effluent limitations.

- d. This limitation is derived from the final waste load allocation as set forth in the Calleguas Creek Watershed Toxicity TMDL, established by the Regional Water Board on July 7, 2005. The TMDL became effective on March 24, 2006. Consistent with the TMDL, the final WLA-based limit became operative on March 23, 2008. The interim limits specified in this TMDL lapsed prior to the date this permit was renewed. Therefore, only final WLA-based limits are incorporated into this permit.

5. Whole Effluent Toxicity (WET)

Whole effluent toxicity (WET) testing protects the receiving water quality from the aggregate toxic effect of a mixture of pollutants in the effluent. An acute toxicity test is conducted over a short time period and measures mortality. A chronic toxicity test is conducted over a longer period of time and may measure mortality, reproduction, and growth. Chronic toxicity is a more stringent requirement than acute toxicity. A chemical at a low concentration can have chronic effects but no acute effects until it gets to the higher level.

The effluent limitations for chronic toxicity were established regardless of whether or not there is reasonable potential for the pollutants to be present in the discharge at levels that would cause or contribute to a violation of water quality standards, since the *Toxicity TMDL* establishes a chronic toxicity WLA for the Simi Valley WQCP. The Regional Water Board developed water quality-based effluent limitations for these pollutants pursuant to section 122.44(d)(1)(vii), which does not require or contemplate a reasonable potential analysis.

In the past, the State Water Board reviewed circumstances warranting a numeric chronic toxicity effluent limitation for POTWs when there is reasonable potential. (See SWRCB/OCC Files A-1496 & A-1496(a) [Los Coyotes/Long Beach Petitions].) On September 16, 2003, at a public hearing, the State Water Board adopted Order No. 2003-0012 deferring the issue of numeric chronic toxicity effluent limitation for POTWs until a subsequent Phase of the SIP is adopted. In the meantime, the State Water Board replaced the numeric chronic toxicity limit with a narrative effluent limitation and a 1 TUc trigger, in the Long Beach and Los Coyotes WRP NPDES permits. Simi Valley WQCP's NPDES permit contained a similar narrative chronic toxicity effluent limitation, with a numeric trigger for accelerated monitoring, consistent with the State Water Board's precedential Order.

However, many facts have changed since the State Water Board adopted the Los Coyotes Order in 2003. Namely, the Regional Water Board adopted the *Calleguas Creek Toxicity TMDL* containing a numeric WLA for chronic toxicity for the five POTWs located in the watershed; USEPA published two new guidance documents with respect to chronic toxicity; the Los Angeles Regional Water Board adopted NPDES permits for industrial facilities incorporating TST-based effluent limitations for chronic toxicity and has adopted numeric chronic toxicity effluent limitations for industrial facilities with TMDL WLAs of 1.0 TUc; the Santa

Ana Regional Water Board adopted an NPDES permit for a POTW incorporating TST-based effluent limitations for chronic toxicity. In addition to these factual developments, the State Water Board has not adopted a revised policy that addresses chronic toxicity effluent limitations in NPDES permits for inland discharges, as anticipated by the Los Coyotes Order. Because the Los Coyotes Order explicitly “declined to make a determination ... regarding the propriety of the final numeric effluent limitations for chronic toxicity...,” (Los Coyotes Order, p. 9) and because of the differing facts before the Regional Water Board in 2014 as compared to the facts that were the basis for the Los Coyotes Order in 2003, the Regional Water Board concludes that the Los Coyotes Order does not require inclusion of narrative rather than numeric effluent limitations for chronic toxicity. Further, the Regional Water Board finds that numeric effluent limitations for chronic toxicity are necessary, feasible, and appropriate.

On October 19, 2018, the State Water Board released a revised version of the Chronic Toxicity Plan for public comment. The item is tentatively scheduled for State Water Board consideration in 2019.

Because effluent data exhibited reasonable potential to cause or contribute to an exceedance of the water quality objective, this Order contains numeric chronic toxicity effluent limitations. Compliance with the chronic toxicity requirement contained in this Order shall be determined in accordance to sections VII.J of this Order. This Order contains a reopener to allow the Regional Water Board to modify the permit, if necessary, to make it consistent with any new policy, law, or regulation.

For this permit, chronic toxicity in the discharge is evaluated using USEPA’s 2010 Test of Significant Toxicity (TST) hypothesis testing approach. Chronic toxicity limitations are expressed as “Pass” for the median monthly summary results and “Pass” and “<50% Effect” for each maximum daily individual results. The chronic toxicity effluent limitations are as stringent as necessary to protect the narrative Basin Plan Water Quality Objective for chronic toxicity. Those limitations are also consistent with the chronic toxicity WLA of 1.0 TUC and the assumptions of the *Calleguas Creek Toxicity TMDL* which went into effect on March 24, 2006, and the implementation language which reads as follows: “The toxicity WLAs will be implemented in accordance with USEPA, State Board and Regional Board resolutions, **guidance** (emphasis added) and policy at the time of permit issuance or renewal.”

In January 2010, USEPA published a guidance document titled, “EPA Regions 8, 9 and 10 Toxicity Training Tool,” which among other things discusses permit limit expression for chronic toxicity. The document acknowledges that NPDES regulations at 40 CFR section 122.45(d) requires that all permit limitations be expressed, unless impracticable, as an Average Weekly Effluent Limitation (AWEL) and an Average Monthly Effluent Limitation (AMEL) for POTWs. Following Section 5.2.3 of the Technical Support Document (TSD), the use of an AWEL is not appropriate for WET. In lieu of an AWEL for POTWs, USEPA recommends establishing an MDEL for toxic pollutants and pollutants in water quality permitting, including WET. This is appropriate for two reasons. The basis for the average weekly requirement for POTWs derives from secondary treatment

regulations and is not related to the requirement to assure achievement of water quality standards (WQS). Moreover, an average weekly requirement comprising of up to seven daily samples could average out daily peak toxic concentrations for WET and therefore, the discharge's potential for causing acute and chronic effects would be missed. It is impracticable to use an AWEL, because short-term spikes of toxicity levels that would be permissible under the 7-day average scheme would not be adequately protective of all beneficial uses. The MDEL is the highest allowable value for the discharge measured during a calendar day or 24-hour period representing a calendar day. The AMEL is the highest allowable value for the average of daily discharges obtained over a calendar month. For WET, this is the average of individual WET test results for that calendar month. However, in cases where a chronic mixing zone is not authorized, USEPA Regions 9 and 10 continue to recommend that the AMEL for chronic WET should be expressed as a median monthly limit (MMEL).

Later in June 2010, USEPA published another guidance document titled, *Test of Significant Toxicity Implementation Document* (EPA 833-R-10-003, June 2010), in which they recommend the following: "Permitting authorities should consider adding the TST approach to their implementation procedures for analyzing valid WET data for their current NPDES WET Program." The TST approach is another statistical option for analyzing valid WET test data. Use of the TST approach does not result in any changes to EPA's WET test methods. Section 9.4.1.2 of the USEPA's *Short-term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Freshwater Organisms* (EPA/821/R-02/013, 2002), recognizes that, "the statistical methods in this manual are not the only possible methods of statistical analysis." The TST approach can be applied to acute (survival) and chronic (sublethal) endpoints and is appropriate to use for both freshwater and marine EPA WET test methods.

USEPA's WET testing program and acute and chronic WET methods rely on the measurement result for a specific test endpoint, not upon achievement of specified concentration-response patterns to determine toxicity. USEPA's WET methods do not require achievement of specified effluent or ambient concentration-response patterns prior to determining that toxicity is present¹⁴. Nevertheless, USEPA's acute and chronic WET methods require that effluent and ambient concentration-response patterns generated for multi-concentration acute and chronic toxicity tests be reviewed - as a component of test review following statistical analysis - to ensure that the calculated measurement result for the toxicity test is interpreted appropriately. (EPA-821-R-02-012, section 12.2.6.2; EPA-821-R-02-013, section 10.2.6.2.). In 2000, USEPA provided guidance for such reviews to ensure that test endpoints for determining toxicity based on the statistical approaches utilized at the time the guidance was written (NOEC, LC50's, IC25's) were calculated appropriately (EPA 821-B-00-004).

USEPA designed its 2000 guidance as a standardized step-by step review process that investigates the causes for 10 commonly observed concentration-

¹⁴ See, Supplementary Information in support of the Final Rule establishing WET test methods at 67 Fed.Reg. 69952, 69963, November 19, 2002.

response patterns and provides for the proper interpretation of the test endpoints derived from these patterns for NOECs, LC50s, and IC25s, thereby reducing the number of misclassified test results. The guidance provides one of three determinations based on the review steps: (1) that calculated effect concentrations are reliable and should be reported, (2) that calculated effect concentrations are anomalous and should be explained, or (3) that the test was inconclusive and should be repeated with a newly collected sample. The standardized review of the effluent and receiving water concentration-response patterns provided by USEPA's 2000 guidance decreased discrepancies in data interpretation for NOEC, LC50, and IC25 test results, thereby lowering the chance that a truly nontoxic sample would be misclassified and reported as toxic.

Appropriate interpretation of the measurement result from USEPA's TST statistical approach (pass/fail) for effluent and receiving water samples is, by design, independent from the concentration-response patterns of the toxicity tests for those samples. Therefore, when using the TST statistical approach, application of USEPA's 2000 guidance on effluent and receiving waters concentration-response patterns will not improve the appropriate interpretation of TST results as long as all Test Acceptability Criteria and other test review procedures - including those related to Quality Assurance for effluent and receiving water toxicity tests, reference toxicity tests, and control performance (mean, standard deviation, and coefficient of variation) - described by the WET test methods manual and TST guidance, are followed. The 2000 guidance may be used to identify reliable, anomalous, or inconclusive concentration-response patterns and associated statistical results to the extent that the guidance recommends review of test procedures and laboratory performance already recommended in the WET test methods manual. The guidance does not apply to single-concentration (IWC) and control statistical t-tests and does not apply to the statistical assumptions on which the TST is based. The Regional Water Board will not consider a concentration-response pattern as sufficient basis to determine that a TST t- test result for a toxicity test is anything other than valid, absent other evidence. In a toxicity laboratory, unexpected concentration-response patterns should not occur with any regular frequency and consistent reports of anomalous or inconclusive concentration-response patterns or test results that are not valid will require an investigation of laboratory practices.

Any Data Quality Objectives or Standard Operating Procedure used by the toxicity testing laboratory to identify and report valid, invalid, anomalous, or inconclusive effluent or receiving water toxicity test measurement results from the TST statistical approach which include a consideration of concentration-response patterns and/or PMSDs must be submitted for review by the Regional Water Board, in consultation with USEPA and the State Water Board's Quality Assurance Officer and Environmental Laboratory Accreditation Program (40 CFR section 122.41(h)). As described in the bioassay laboratory audit directives to the San Jose Creek Water Quality Laboratory from the State Water Resources Control Board dated August 7, 2014, and from the USEPA dated December 24, 2013, the PMSD criteria only apply to compliance for NOEC and the sublethal endpoints of the NOEC, and therefore are not used to interpret TST results.

D. Final Effluent Limitation Considerations

1. Anti-Backsliding Requirements

Sections 402(o) and 303(d)(4) of the CWA and federal regulations at 40 CFR section 122.44(l) prohibit backsliding in NPDES permits. These anti-backsliding provisions require effluent limitations in a reissued permit to be as stringent as those in the previous permit, with some exceptions where limitations may be relaxed. The effluent limitations in this Order are at least as stringent as the effluent limitations in the previous Order No. R4-2014-0066-A01 and there were no effluent limitations removed from this previous Order.

2. Antidegradation Policies

40 CFR section 131.12 requires that state water quality standards include an antidegradation policy consistent with the federal antidegradation policy. On October 28, 1968, the State Water Board established California's antidegradation policy when it adopted Resolution No. 68-16, *Statement of Policy with Respect to Maintaining the Quality of the Waters of the State*. Resolution No. 68-16 requires that existing water quality be maintained unless degradation is justified based on specific findings. The State Water Board has, in State Water Board Order No. 86-17 and an October 7, 1987 guidance memorandum, interpreted Resolution No. 68-16 to be fully consistent with the federal antidegradation policy contained in 40 CFR section 131.12. Similarly, CWA section 303(d)(4)(B) and 40 CFR section 131.12 require that all permitting actions be consistent with the federal antidegradation policy. Together, the state and federal antidegradation policies are designed to ensure that a water body will not be degraded resulting from the permitted discharge. The Regional Water Board's Basin Plan implements, and incorporates by reference, both the state and federal antidegradation policies.

The renewal of the NPDES permit will not lower the surface water quality because the conditions in this Order are at least as stringent as the previous permit. There are no changes to the plant's treatment facilities or processes are planned that would impact the concentrations of those constituents that have limitations in the discharged effluent. Monitoring for these constituents in the effluent and receiving waters continue to be required under this Order. The Regional Water Board may modify the terms of this Order to prevent degradation of high quality waters based on any change in the concentration of these constituents in the effluent or receiving water that indicates that a degradation of receiving water quality may occur. The treatment required by this Order is the best practicable treatment or control of the discharge necessary to assure that a pollution or nuisance will not occur and the highest water quality consistent with maximum benefit to the people of the State will be maintained. Therefore, discharges permitted in this Order are consistent with the antidegradation provisions of 40 CFR section 131.12 and State Water Board Resolution No. 68-16.

3. Stringency of Requirements for Individual Pollutants

This Order contains both TBELs and WQBELs for individual pollutants. The technology-based effluent limitations consist of restrictions on BOD, TSS, pH, and percent removal of BOD and TSS. Restrictions on BOD, TSS and pH are discussed in section IV.B. of the Fact Sheet. This Order's technology-based pollutant restrictions implement the minimum, applicable federal technology-based requirements. In addition, this Order contains effluent limitations more stringent than the minimum, federal technology-based requirements that are necessary to meet water quality standards.

Water quality-based effluent limitations have been scientifically derived to implement WQOs that protect beneficial uses. Both the beneficial uses and the WQOs have been approved pursuant to federal law and are the applicable federal water quality standards. To the extent that toxic pollutant WQBELs were derived from the CTR, the CTR is the applicable standard pursuant to 40 CFR section 131.38. The scientific procedures for calculating the individual water quality-based effluent limitations for priority pollutants are based on the CTR-SIP, which was approved by USEPA on May 18, 2000. All beneficial uses and WQOs contained in the Basin Plan were approved under state law and submitted to and approved by USEPA prior to May 30, 2000. Any WQOs and beneficial uses submitted to USEPA prior to May 30, 2000, but not approved by USEPA before that date, are nonetheless "applicable water quality standards for purposes of the CWA" pursuant to 40 CFR section 131.21(c)(1). Collectively, this Order's restrictions on individual pollutants are no more stringent than required to implement the requirements of the CWA and the applicable water quality standards for purposes of the CWA.

Table F-9. Summary of Final Effluent Limitations for Discharge Points 001 and 002

Parameter	Units	Average Monthly	Average Weekly	Maximum Daily	Basis	Remarks
BOD ₅ 20°C	mg/L	20	30	45	Existing/ Secondary treatment	--
BOD ₅ 20°C	lbs/day	2,080	3,130	4,690	Existing/ Secondary treatment	See note "2" below
TSS	mg/L	15	40	45	Existing/ Secondary treatment	--
TSS	lbs/day	1,560	4,170	4,690	Existing/ Secondary treatment	See note "2" below
Temperature	°F	--	--	86	Existing	--

Parameter	Units	Average Monthly	Average Weekly	Maximum Daily	Basis	Remarks
Removal Efficiency for BOD and TSS	%	≥85	--	--	Existing/ Technology base	--
Oil and Grease	mg/L	10	--	15	Existing/ Technology base	--
Oil and Grease	lbs/day	1,040	--	1,560	Existing/ Technology base	See note "2" below
Settleable Solids	ml/L	0.1	--	0.3	Existing/ BPJ	--
Total Residual Chlorine	mg/L	--	--	0.1	Existing/ Basin Plan	--
Radioactivity ¹⁵	--	--	--	--	--	--
Combined Radium-226 and Radium 228	pCi/L	5	--	--	Existing/ Title 22	--
Gross Alpha particle activity (excluding radon and uranium)	pCi/L	15	--	--	Existing/ Title 22	--
Uranium	pCi/L	20	--	--	Existing/ Title 22	--
Gross Beta/photon emitters	millirem/ year	4	--	--	Existing/ Title 22	--
Strontium-90	pCi/L	8	--	--	Existing/ Title 22	--
Tritium	pCi/L	20,000	--	--	Existing/ Title 22	--
Total Dissolved Solids (dry-weather)	lbs/day	88,610	--	--	Existing, TMDL, Basin Plan	See note "4" and "5" below
Total Dissolved Solids (wet-weather)	mg/L	850	--	--	Existing, TMDL, Basin Plan	See note "6" below

¹⁵ These effluent limitations are derived from Title 22, chapter 15, article 5, sections 64442 and 64443, of the California Code of Regulations (CCR). The incorporation by reference is prospective including future changes to the incorporated provisions as the changes take effect

Parameter	Units	Average Monthly	Average Weekly	Maximum Daily	Basis	Remarks
Sulfate (dry-weather)	lbs/day	26,060	--	--	Existing, TMDL, Basin Plan	See note "4" and "5" below
Sulfate (wet-weather)	mg/L	250	--	--	Existing, TMDL, Basin Plan	See note "6" below
Chloride (dry-weather)	lbs/day	15,640	--	--	Existing, TMDL, Basin Plan	See note "4" and "5" below
Chloride (wet-weather)	mg/L	150	--	--	Existing, TMDL, Basin Plan	See note "6" below
Boron(dry-weather)	lbs/day	104	--	--	Existing, TMDL, Basin Plan	See note "4" and "5" below
Boron (wet-weather)	mg/l	1.0	--	--	Existing, TMDL, Basin Plan	See note "6" below
MBAS	mg/L	0.5	--	--	Existing, Basin Plan	--
MBAS	lbs/day	50	--	--	Existing, TMDL, Basin Plan	See note "2" below
Ammonia Nitrogen	mg/L	2.4	--	3.3	TMDL	See note "7" below
Ammonia Nitrogen	lbs/day	--	--	$2.9 \times Q^{16}$	TMDL	See note "7" below
Nitrate + Nitrite (as N)	mg/L	9	--	--	TMDL	See note "7" below
Nitrate (as N)	mg/L	9	--	--	TMDL	See note "7" below
Nitrite (as N)	mg/L	0.9	--	--	TMDL	See note "7" below
Copper	µg/L	30.5	--	31	TMDL	See note "8" below
Nickel	µg/L	169	--	960	TMDL	See note "8" below
Mercury	lbs/month	0.031	--	--	TMDL	See note "9" below

¹⁶ Q represents the POTW effluent flow at the time the water quality measurement is collected (not to exceed 12.5 mgd) and a conversion factor to lbs/day based on the units of measure for the flow.

Parameter	Units	Average Monthly	Average Weekly	Maximum Daily	Basis	Remarks
Selenium	µg/L	4.5	--	6.8	SIP/CTR	-
Selenium	lbs/day	0.47	--	0.71	SIP/CTR	See note "2" below
Cyanide	µg/L	4.1	--	8.9	SIP/CTR	--
Cyanide	lbs/day	0.43	--	0.93	SIP/CTR	See note "2" below
Bis (2-Ethylhexyl) Phthalate	µg/L	4	--	--	SIP/CTR	--
Bis (2-Ethylhexyl) Phthalate	lbs/day	0.42	--	--	SIP/CTR	See note "2" below
Chlorpyrifos	µg/L	0.014	--	0.025	TMDL	See note "10" below
Diazinon	µg/L	0.1	--	0.1	TMDL	See note "10" below
Chronic Toxicity ^{17, 18}	Pass or Fail, % Effect (TST)	Pass ¹⁹	--	Pass or % Effect < 50	TMDL, TST and USEPA Guidance	--
Chlordane	µg/L	0.00059	--	0.0012	TMDL	--
4,4'-DDD	µg/L	0.00084	--	0.0017	TMDL	--
4,4'-DDE	µg/L	0.00059	--	0.0012	TMDL	--
4,4'-DDT	µg/L	0.00059	--	0.0012	TMDL	--
Dieldrin	µg/L	0.00014	--	0.00028	TMDL	--
PCBs	µg/L	0.00017	--	0.00034	TMDL	--
Toxaphene	µg/L	0.00016	--	0.00033	TMDL	--

¹⁷ The *Calleguas Creek Watershed Toxicity TMDL* includes a WLA of 1.0 TUC for toxicity, which is required to be implemented in accordance with USEPA, State Water Board, and Regional Water Board resolutions, guidance and policy at the time of permit issuance or renewal. In addition, a numeric WQBEL is established because effluent data showed that there is reasonable potential for the effluent to cause or contribute to an exceedance of the chronic toxicity water quality objective. The numeric WLA is protective of both the numeric acute toxicity and the narrative toxicity Basin Plan water quality objective. Consistent with the *Toxicity TMDL*, these chronic toxicity WLA-based final effluent limitations shall be implemented using the *Short Term Methods for Estimating the Chronic Toxicity of Effluents and Receiving Waters to Freshwater Organisms* (USEPA 2002, EPA-821-R-02-013), and current USEPA guidance in *National Pollutant Discharge Elimination System Test of Significant Toxicity Implementation Document* (EPA 833-R-10-003, June /2010) and *EPA Regions 8, 9 and 10 Toxicity Training Tool* (January 2010), <http://www2.epa.gov/region8/epa-regions-8-9-and-10-toxicity-training-tool-january-2010>.

¹⁸ The Median Monthly Effluent Limitation (MMEL) shall be reported as "Pass" or "Fail." The Maximum Daily Effluent Limitation (MDEL) shall be reported as "Pass" or "Fail" and "% Effect." The MMEL for chronic toxicity shall only apply when there is a discharge on more than one day in a calendar month period. During such calendar months, up to three independent toxicity tests may be conducted when one toxicity test results in "Fail."

¹⁹ This is a Median Monthly Effluent Limitation.

1. The effluent values for pH shall be maintained within the limits of 6.5 standard units and 8.5 standard units.
2. The mass based limitations are based on the plant design flow rate of 12.5 MGD, and are calculated as follows: $\text{Flow (MGD)} \times \text{Concentration (mg/L)} \times 8.34$ (conversion factor) = lbs/day. During wet-weather storm events in which the flow exceeds the design capacity, the mass discharge rate limitations shall not apply, and concentration limitations shall be provide the only applicable effluent limitations.
3. The wastes discharged to water courses shall at all times be adequately disinfected. For the purpose of this requirement, the wastes shall be considered adequately disinfected if (1) the median number of total coliform bacteria at some point in the treatment process does not exceed a 7-day median of 2.2 Most Probable Number (MPN) or Colony Forming Unit (CFU) per 100 milliliters utilizing the bacteriological results of the last seven (7) days for which an analysis has been completed, (2) the number of total coliform bacteria does not exceed 23 MPN or CFU per 100 milliliters in more than one sample within any 30-day period, and (3) no sample shall exceed 240 MPN or CFU of total coliform bacteria per 100 milliliters. Samples shall be collected at a time when wastewater flow and characteristics are most demanding on treatment facilities and disinfection processes.
4. Dry-weather is defined in the *Calleguas Creek Watershed Salts Total Maximum Daily Load (Salts TMDL)* as the condition when the flows in the receiving water are below the 86th percentile flow, as explained in section VII.O of this Order.
5. This final effluent limitation is derived from the final Waste Load Allocations (WLAs) in the *Salts TMDL*, established by the Regional Water Board on October 4, 2007. The *Salts TMDL*, which became effective on December 2, 2008, following USEPA's approval, specifies interim WLAs for total dissolved solids (TDS), sulfate, and chloride. However, interim effluent limits based on the interim WLAs in the *Salts TMDL* have not been incorporated into this Order because the effluent data demonstrates that the Facility's discharge is currently able to comply with the final WLA-based limitations for TDS and sulfate and chloride. Consistent with the *Salts TMDL*, these limits apply only during dry- weather (as defined in the *Salts TMDL*, and as explained in section VII.O of this Order.
6. Wet-weather is defined in the *Salts TMDL* as the condition when the flows in the receiving water are greater than or equal to the 86th percentile flow, as explained in section VII.O of this Order.
7. This final effluent limitation is derived from the final waste load allocation for ammonia nitrogen, as set forth in Resolution No. R4-2008-009, *Amendment to the Water Quality Control Plan for the Los Angeles Region through revision of the Waste Load Allocations for the Calleguas Creek Watershed Nitrogen Compounds and Related Effects Total Maximum Daily Load*, adopted by the Regional Water Board on September 11, 2008, and became effective on October 15, 2009.

8. This limitation is derived from the final WLA, as set forth in the *Calleguas Creek Watershed Metals TMDL*, established by the Regional Water Board on June 8, 2006. The TMDL became effective on March 26, 2007.
9. According to the *Metals TMDL*, the mercury (in suspended sediment) effluent limitation, in lbs/month, is assumed that the total load in effluent water is equal to the suspended sediment load.
10. This limitation is derived from the final waste load allocation as set forth in the *Calleguas Creek Watershed Toxicity TMDL*, established by the Regional Water Board on July 7, 2005. The TMDL became effective on March 24, 2006. Consistent with the TMDL, the implementation of final WLA-based limit became operative on March 25, 2008. The interim limits specified in this TMDL lapsed prior to the date this permit was renewed. Therefore, only final WLA-based limits are incorporated into this permit.

E. Interim Effluent Limitations

No interim limits are included in this Order.

F. Land Discharge Specifications – Not Applicable

G. Recycling Specifications

The Permittee shall continue to investigate the feasibility of recycling, conservation, and/or alternative disposal methods for wastewater (such as groundwater injection), and/or beneficial use of storm water and dry-weather urban runoff. The Permittee shall submit an update to this feasibility study as part of the submittal of the Report of Waste Discharge (ROWD) for the next permit renewal.

V. RATIONALE FOR RECEIVING WATER LIMITATIONS

A. Surface Water

Receiving water limitations are based on WQOs contained in the Basin Plan and are a required part of this Order.

B. Groundwater

Limitations in this Order must protect not only surface receiving water beneficial uses, but also, the beneficial uses of underlying groundwater where there is a groundwater recharge (GWR) beneficial use of the surface water. In addition to a discharge to surface water, the discharge may impact groundwater. Sections of the Arroyo Simi and Calleguas Creek, near Simi Valley WQCP discharge points, are designated with the GWR beneficial use. Surface water from the Arroyo Simi percolates into the Simi Valley and Ventura Central Groundwater Basins with MUN beneficial use specified in the Basin Plan. Since groundwater from the Basin is used to provide drinking water to the community, the groundwater aquifers must be protected.

VI. RATIONALE FOR PROVISIONS

A. Standard Provisions

Standard Provisions, which apply to all NPDES permits in accordance with 40 CFR section 122.41, and additional conditions applicable to specified categories of permits in accordance with 40 CFR section 122.42, are provided in Attachment D. The Permittee must comply with all standard provisions and with those additional conditions that are applicable under section 122.42.

40 CFR sections 122.41(a)(1) and (b) through (n) establish conditions that apply to all state-issued NPDES permits. These conditions must be incorporated into the permits either expressly or by reference. If incorporated by reference, a specific citation to the regulations must be included in the Order. Section 123.25(a)(12) of 40 CFR allows the state to omit or modify conditions to impose more stringent requirements. In accordance with 40 CFR section 123.25, this Order omits federal conditions that address enforcement authority specified in 40 CFR section 122.41(j)(5) and (k)(2) because the enforcement authority under the CWC is more stringent. In lieu of these conditions, this Order incorporates by reference CWC section 13387(e).

B. Special Provisions

1. Reopener Provisions

- a. This provision is based on 40 CFR part 123. The Regional Water Board may reopen the permit to modify permit conditions and requirements. Causes for modifications include the promulgation of new regulations, modification in sludge use or disposal practices, or adoption of new regulations by the State Water Board or Regional Water Board, including revisions to the Basin Plan.

2. Special Studies and Additional Monitoring Requirements

- a. **Antidegradation Analysis and Engineering Report for Any Proposed Plant Expansion.** This provision is based on the State Water Board Resolution No. 68-16, which requires the Regional Water Board in regulating the discharge of waste to maintain high quality waters of the state. The Permittee must demonstrate that it has implemented adequate controls (e.g., adequate treatment capacity) to ensure that high quality waters will be maintained. This provision requires the Permittee to clarify that it has increased plant capacity through the addition of new treatment system(s) to obtain alternative effluent limitations for the discharge from the treatment system(s). This provision requires the Permittee to report specific time schedules for the plant's projects. This provision requires the Permittee to submit a report to the Regional Water Board for approval.
- b. **Operations Plan for Proposed Expansion.** This provision is based on section 13385(j)(1)(D) of the CWC and allows a time period not to exceed 90 days in which the Permittee may adjust and test the treatment system(s). This provision requires the Permittee to submit an Operations Plan describing the actions the Permittee will take during the period of adjusting and testing to prevent violations.

- c. **Treatment Plant Capacity.** The treatment plant capacity study required by this Order shall serve as an indicator for the Regional Water Board regarding Facility's increasing hydraulic capacity and growth in the service area.

3. **Best Management Practices and Pollution Prevention**

- a. **Pollutant Minimization Program (PMP).** This provision is based on the requirements of section 2.4.5 of the SIP.

4. **Construction, Operation, and Maintenance Specifications**

- a. This provision is based on the requirements of 40 CFR section 122.41(e) and the previous Order.

5. **Special Provisions for Municipal Facilities (POTWs Only)**

- a. **Biosolids Requirements.** To implement CWA section 405(d), on February 19, 1993, USEPA promulgated 40 CFR part 503 to regulate the use and disposal of municipal sewage sludge. This regulation was amended on September 3, 1999. The regulation requires that producers of sewage sludge meet certain reporting, handling, and disposal requirements. It is the responsibility of the Permittee to comply with said regulations that are enforceable by USEPA, because California has not been delegated the authority to implement this program. The Permittee is also responsible for compliance with WDRs and NPDES permits for the generation, transport and application of biosolids issued by the State Water Board, other Regional Water Boards, Arizona Department of Environmental Quality or USEPA, to whose jurisdiction the Facility's biosolids will be transported and applied.
- b. **Pretreatment Requirements.** This permit contains pretreatment requirements consistent with applicable effluent limitations, national standards of performance, and toxic and performance effluent standards established pursuant to sections 208(b), 301, 302, 303(d), 304, 306, 307, 403, 404, 405, and 501 of the CWA, and amendments thereto. This permit contains requirements for the implementation of an effective pretreatment program pursuant to section 307 of the CWA; 40 CFR parts 35 and 403; and/or Title 23, CCR section 2233.
- c. **Filter Bypass Requirements.** Conditions pertaining to bypass are contained in Attachment D, Section I. Standard Provisions – Permit Compliance, subsection G. The bypass or overflow of untreated or partially treated wastewater to waters of the State is prohibited, except as allowed under conditions stated in 40 CFR section 122.41(m) and (n). During periods of elevated, wet weather flows, a portion of the secondary treated wastewater is diverted around the tertiary filters. These anticipated discharges are approved under the bypass conditions when the resulting combined discharge of fully treated (tertiary) and partially treated (secondary) wastewater complies with the effluent and receiving water limitations in this Order.

- d. **Spill Reporting Requirements.** This Order established a reporting protocol for how different types of spills, overflow or bypasses of raw or partially treated sewage from its collection system or treatment plant covered by this Order shall be reported to regulatory agencies.

The State Water Board issued General Waste Discharge Requirements for Sanitary Sewer Systems, Water Quality Order 2006-0003-DWQ (SSO WDR) on May 2, 2006. The Monitoring and Reporting Requirements for the SSO WDR were amended by Water Quality Order WQ 2008-0002-EXEC on February 20, 2008. The SSO WDR requires public agencies that own or operate sanitary sewer systems with greater than one mile of pipes or sewer lines to enroll for coverage under the SSO WDR. The SSO WDR requires agencies to develop sanitary sewer management plans (SSMPs) and report all sanitary sewer overflows (SSOs), among other requirements and prohibitions.

Furthermore, the SSO WDR contains requirements for operation and maintenance of collection systems and for reporting and mitigating sanitary sewer overflows. Inasmuch that the Permittee's collection system is part of the system that is subject to this Order, certain standard provisions are applicable as specified in Provisions, section VI.C.5. For instance, the 24-hour reporting requirements in this Order are not included in the SSO WDR. The Permittee must comply with both the SSO WDR and this Order. The Permittee and public agencies that are discharging wastewater into the Facility were required to obtain enrollment for regulation under the SSO WDR by December 1, 2006.

In the past, the region has experienced loss of recreational use at coastal beaches and in Arroyo Conejo as a result of major sewage spills. The SSO requirements are intended to prevent or minimize impacts to receiving waters as a result of spills.

6. Other Special Provisions (Not Applicable)

7. Compliance Schedules (Not Applicable)

VII. RATIONALE FOR MONITORING AND REPORTING REQUIREMENTS

CWA section 308 and 40 C.F.R. sections 122.41(h), (j)-(l), 122.44(i), and 122.48 require that all NPDES permits specify monitoring and reporting requirements. CWC sections 13267 and 13383 also authorizes the Regional Water Board to establish monitoring, inspection, entry, reporting, and recordkeeping requirements. The MRP of this Order establishes monitoring, reporting, and recordkeeping requirements that implement federal and state requirements. The following provides the rationale for the monitoring and reporting requirements contained in the MRP for this Facility.

A. Influent Monitoring

Influent monitoring is required:

1. To determine compliance with the permit conditions for BOD₅ 20°C and suspended solids removal rates.
2. To assess treatment plant performance.
3. To assess the effectiveness of the Pretreatment Program.
4. As a requirement of the PMP.

B. Effluent Monitoring

The Permittee is required to conduct monitoring of the permitted discharges in order to evaluate compliance with permit conditions. Monitoring requirements are given in the MRP Attachment E. This provision requires compliance with the MRP, and is based on 40 CFR sections 122.44(i), 122.62, 122.63, and 124.5. The MRP is a standard requirement in almost all NPDES permits (including this Order) issued by the Regional Water Board. In addition to containing definition of terms, it specifies general sampling/analytical protocols and the requirements of reporting spills, violation, and routine monitoring data in accordance with NPDES regulations, the CWC, and Regional Water Board policies. The MRP also contains sampling program specific for the Permittee's wastewater treatment plant. It defines the sampling stations and frequency, pollutants to be monitored, and additional reporting requirements. Pollutants to be monitored include all pollutants for which effluent limitations are specified. Further, in accordance with section 1.3 of the SIP, a periodic monitoring is required for all priority pollutants defined by the CTR, for which criteria apply and for which no effluent limitations have been established, to evaluate reasonable potential to cause or contribute to an excursion above a water quality standard.

Monitoring for those pollutants expected to be present in the discharge from the Facility, will be required as shown on the MRP and as required in the SIP. Semi-annual monitoring for priority pollutants in the effluent is required in accordance with the Pretreatment requirements.

Table F-10. Monitoring Frequency Comparison

Parameter	Monitoring Frequency (2014 Permit)	Monitoring Frequency (2019 Permit)
Total waste flow	continuous	continuous
Total residual chlorine	daily	daily
Turbidity	continuous	continuous
Temperature	weekly	weekly
pH	weekly	weekly
Settleable solids	weekly	weekly
Total suspended solids	weekly	weekly
Oil and grease	quarterly	quarterly
BOD	weekly	weekly
Total coliform	daily	daily
Fecal Coliform	daily	deleted

Parameter	Monitoring Frequency (2014 Permit)	Monitoring Frequency (2019 Permit)
E.coli	daily (as necessary)	daily (as necessary)
Total Dissolved Solids	monthly	monthly
Sulfate	monthly	monthly
Chloride	monthly	monthly
Boron	monthly	monthly
MBAS	monthly	quarterly
Ammonia nitrogen	monthly	monthly
Nitrate + nitrite (as nitrogen)	monthly	monthly
Nitrite nitrogen	monthly	monthly
Chronic toxicity	monthly	monthly
Bis(2-ethylhexyl)phthalate	quarterly	monthly
Total Nitrogen	monthly	monthly
Total Phosphorus	monthly	monthly
Iron	semiannually	semiannually
Fluoride	semiannually	semiannually
Antimony	semiannually	semiannually
Arsenic	semiannually	semiannually
Beryllium	semiannually	semiannually
Cadmium	semiannually	semiannually
Chromium III	semiannually	semiannually
Chromium VI	semiannually	semiannually
Copper	monthly	monthly
Lead	semiannually	semiannually
Mercury	monthly	monthly
Nickel	monthly	monthly
Selenium	monthly	monthly
Silver	semiannually	semiannually
Thallium	semiannually	semiannually
Zinc	quarterly	quarterly
Cyanide	monthly	monthly
2,3,7,8-TCDD (Dioxin)	semiannually	semiannually
Bromoform	quarterly	quarterly
Dibromochloromethane	quarterly	quarterly
Chloroform	quarterly	quarterly
Bromodichloromethane	quarterly	quarterly
Tetrachloroethylene	semiannually	semiannually
N-Nitrosodi-n-propylamine	semiannually	semiannually

Parameter	Monitoring Frequency (2014 Permit)	Monitoring Frequency (2019 Permit)
Gamma-BHC (Lindane)	semiannually	semiannually
Chlordane	quarterly	quarterly
4,4'-DDT	quarterly	quarterly
4,4'- DDE	quarterly	quarterly
4,4'-DDD	quarterly	quarterly
Dieldrin	quarterly	quarterly
Endrin	semiannually	semiannually
Heptachlor epoxide	semiannually	semiannually
PCBs	quarterly	quarterly
Aroclor 1016	quarterly	quarterly
Aroclor 1221	quarterly	quarterly
Aroclor 1232	quarterly	quarterly
Aroclor 1242	quarterly	quarterly
Aroclor 1248	quarterly	quarterly
Aroclor 1254	quarterly	quarterly
Aroclor 1260	quarterly	quarterly
Toxaphene	quarterly	quarterly
Chlorpyrifos	quarterly	quarterly
Diazinon	quarterly	quarterly
Methoxychlor	semiannually	semiannually
Barium	semiannually	semiannually
2,4-D	semiannually	semiannually
2,4,5-TP (Silvex)	semiannually	semiannually
Halomethanes ²⁰	quarterly	quarterly
Ammonium perchlorate	semiannually	semiannually
1,4-Dioxane	semiannually	semiannually
1,2,3-Trichloropropane	semiannually	semiannually
Methyl-tert-butyl-ether (MTBE)	semiannually	semiannually
Remaining USEPA priority pollutant not listed on this Table	semiannually	semiannually

This permit has maintained the same monitoring frequency in the previous permit except for fecal coliform which was removed as a result of Basin Plan Resolution No. R10-005 that eliminates the limitation for fecal coliform. In addition, the monitoring for bis(2-ethylhexyl) phthalate was increased from quarterly to monthly because it shows reasonable potential to exceed the criteria.

²⁰ Halomethanes shall mean the sum of bromoform, bromodichloromethane, chloroform, and dibromochloromethane.

C. Whole Effluent Toxicity Requirements

WET testing protects the receiving water quality from the aggregate toxic effect of a mixture of pollutants in the effluent. An acute toxicity test is conducted over a short time period and measures mortality. A chronic toxicity test is conducted over a short or longer period of time and may measure mortality, reproduction, and growth. Chronic toxicity is a more stringent requirement than acute toxicity. A chemical at a low concentration can have chronic effects but no acute effects until it gets to the higher level. For this permit, chronic toxicity in the discharge is evaluated using USEPA's 2010 TST hypothesis testing approach. The chronic toxicity effluent limitations are as stringent as necessary to protect the Basin Plan Water Quality Objective for chronic toxicity. Those limitations are also consistent with the assumptions of the *Calleguas Creek Toxicity TMDL* which went into effect on March 24, 2006, and the implementation language which reads as follows: "The toxicity WLAs will be implemented in accordance with USEPA, State Board and Regional Board resolutions, **guidance** (emphasis added) and policy at the time of permit issuance or renewal." The rationale for WET has been discussed extensively in section IV.C.5 of this Fact Sheet.

D. Receiving Water Monitoring

1. Surface Water

Receiving water monitoring is required to determine compliance with receiving water limitations and to characterize the water quality of the receiving water.

2. Groundwater – (Not Applicable)

E. Other Monitoring Requirements

1. Watershed Monitoring and Bioassessment Monitoring

The goals of the Watershed-wide Monitoring Program including the bioassessment monitoring for the Ventura River Watershed are to:

- a. Determine compliance with receiving water limits.
- b. Monitor trends in surface water quality.
- c. Ensure protection of beneficial uses.
- d. Provide data for modeling contaminants of concern.
- e. Characterize water quality including seasonal variation of surface waters within the watershed.
- f. Assess the health of the biological community.
- g. Determine mixing dynamics of effluent and receiving waters in the estuary.

2. Discharge Monitoring Report-Quality Assurance (DMR-QA) Study Program

Under the authority of section 308 of the CWA (33 U.S.C. § 1318), USEPA requires major and selected minor dischargers under the NPDES Program to

participate in the annual DMR-QA Study Program. The DMR-QA Study evaluates the analytical ability of laboratories that routinely perform or support self-monitoring analyses required by NPDES permits. There are two options to satisfy the requirements of the DMR-QA Study Program: (1) The Discharger can obtain and analyze a DMR-QA sample as part of the DMR-QA Study; or (2) Per the waiver issued by USEPA to the State Water Board, the Discharger can submit the results of the most recent Water Pollution Performance Evaluation Study from its own laboratories or its contract laboratories. A Water Pollution Performance Evaluation Study is similar to the DMR-QA Study. Thus, it also evaluates a laboratory's ability to analyze wastewater samples to produce quality data that ensure the integrity of the NPDES Program. The Discharger shall ensure that the results of the DMR-QA Study or the results of the most recent Water Pollution Performance Evaluation Study are submitted annually to the State Water Board. The State Water Board's Quality Assurance Program Officer will send the DMR-QA Study results or the results of the most recent Water Pollution Performance Evaluation Study to USEPA's DMR-QA Coordinator and Quality Assurance Manager.

VIII. CONSIDERATION OF NEED TO PREVENT NUISANCE AND CWC SECTION 13241 FACTORS.

- A. Need to prevent nuisance: The state law requirements in this Order are required to prevent pollution or nuisance as defined in section 13050, subdivisions (l) and (m), of the CWC. Many are also required in accordance with narrative water quality objectives in the Basin Plan. These state requirements include, but are not limited to, groundwater limitations, spill prevention plans, operator certification, sanitary sewer overflow reporting, and requirements for standby or emergency power.
- B. Past, present, and probable future beneficial uses of water: Chapter 2 of the Basin Plan identifies designated beneficial uses for water bodies in the Los Angeles Region. Beneficial uses of water relevant to this Order are also identified above in Section III.C.1.
- C. Environmental characteristics of the hydrographic unit under consideration, including the quality of water available thereto: The environmental characteristics are discussed in the Region's Watershed Management Initiative Chapter, as well as available in State of the Watershed reports and the State's CWA Section 303(d) List of impaired waters. The environmental characteristics of the hydrographic unit, including the quality of available water, will be improved by compliance with the requirements of this Order. Additional information on the CCW is available at http://www.waterboards.ca.gov/losangeles/water_issues/programs/regional_program/Water_Quality_and_Watersheds/ws_calleguas.shtml.
- D. Water quality conditions that could reasonably be achieved through the coordinated control of all factors which affect water quality in the area: The beneficial uses of the waterbodies in the Calleguas Creek Watershed can reasonably be achieved through the coordinate control of all factors that affect water quality in the area. TMDLs have been developed (as required by the CWA) for many of the impairments in the watershed. A number of Regional Water Board programs and actions are in place to address the water quality impairments in the watershed, including regulation of point

source municipal and industrial discharges with appropriate NPDES permits and non-point source discharges such as irrigated agriculture. All of these regulatory programs control the discharge of pollutants to surface and ground waters to prevent nuisance and protect beneficial uses. These regulatory programs have resulted in watershed solutions and have improved water quality. Generally, improvements in the quality of the receiving waters impacted by the permittee's discharges can be achieved by reducing the volume of discharges to receiving waters (e.g., through increased recycling), reducing pollutant loads through source control/pollution prevention, including operational source control such as public education (e.g., disposal of pesticides, pharmaceuticals, and personal care products into the sewer) and product or materials elimination or substitution, and removing pollutants through treatment.

- E. Economic considerations: The Permittee did not present any evidence regarding economic considerations related to this Order. However, the Regional Water Board has considered the economic impact of requiring certain provisions pursuant to state law. The additional costs associated with complying with state law requirements are reasonably necessary to prevent nuisance and protect beneficial uses identified in the Basin Plan. Further, the loss of, or impacts to, beneficial uses would have a detrimental economic impact. Economic considerations related to costs of compliance are therefore not sufficient, in the Regional Water Board's determination, to justify failing to prevent nuisance and protect beneficial uses.
- F. Need for developing housing within the region: The Regional Water Board has no evidence regarding the need for developing housing within the region or how the Permittee's discharge will affect that need. The Regional Water Board, however, does not anticipate that these state law requirements will adversely impact the need for housing in the area. The region generally relies on imported water to meet many of its water resource needs. Imported water makes up a vast majority of the region's water supply, with local groundwater, local surface water, and reclaimed water making up the remaining amount. This Order helps address the need for housing by controlling pollutants in discharges, which will improve the quality of local surface and ground water, as well as water available for recycling and re-use. This in turn may reduce the demand for imported water thereby increasing the region's capacity to support continued housing development. A reliable water supply for future housing development is required by law, and with less imported water available to guarantee this reliability, an increase in local supply is necessary. Therefore, the potential for developing housing in the area will be facilitated by improved water quality.
- G. Need to develop and use recycled water: The State Water Board's Recycled Water Policy requires the Regional Water Boards to encourage the use of recycled water. In addition, as discussed immediately above, a need to develop and use recycled water exists within the region, especially during times of drought. To encourage recycling, the Permittee is required by this Order to continue to explore the feasibility of recycling to maximize the beneficial reuse of tertiary treated effluent.

IX. PUBLIC PARTICIPATION

The Regional Water Board has considered the issuance of WDRs that will serve as an NPDES permit for Simi Valley WQCP. As a step in the WDR adoption process, the

Regional Water Board staff has developed tentative WDRs and has encouraged public participation in the WDR adoption process.

A. Notification of Interested Parties

The Regional Water Board notified the Permittee and interested agencies and persons of its intent to prescribe WDRs for the discharge and provided an opportunity to submit written comments and recommendations. Notification was provided through the following: (1) one copy at the front window of the main entrance of 600 West Los Angeles Avenue, Simi Valley and (2) at the front window of the Public Services Center located at 500 W. Los Angeles Avenue, Simi Valley, 93065.

The public had access to the agenda and any changes in dates and locations through the Regional Water Board's website at: <http://www.waterboards.ca.gov/losangeles/>.

B. Written Comments

Interested persons were invited to submit written comments concerning tentative amended WDRs as provided through the notification process. The Board will accept comments only with respect to the proposed changes to the tentative amended requirements marked in underline and strikeout format. Comments were due either in person or by mail to the Executive Office at the Regional Water Board at the address above on the cover page of this Order, or by email submitted to losangeles@waterboards.ca.gov.

To be fully responded to by staff and considered by the Regional Water Board, the written comments were due at the Regional Water Board office by **5:00 p.m. on October 18, 2019**.

C. Public Hearing

The Regional Water Board held a public hearing on the tentative amended WDRs during its regular Board meeting on the following date and time and at the following location:

Date: **November 14, 2019**
Time: 9:00 a.m.
Location: City of Monrovia
415 S. Ivy Avenue
Monrovia, California 91016

Interested persons were invited to attend. At the public hearing, the Regional Water Board heard testimony pertinent to the discharge, WDRs, and permit. For accuracy of the record, important testimony was requested in writing.

D. Reconsideration of Waste Discharge Requirements

Any aggrieved person may petition the State Water Board to review the decision of the Regional Water Board regarding the final WDRs. The petition must be received by the State Water Board at the following address within 30 calendar days of the Regional Water Board's action:

State Water Resources Control Board
Office of Chief Counsel
P.O. Box 100, 1001 I Street
Sacramento, CA 95812-0100

Or by email at waterqualitypetitions@waterboards.ca.gov

For instructions on how to file a petition for review, see
http://www.waterboards.ca.gov/public_notices/petitions/water_quality/wqpetition_instr.shtml

E. Information and Copying

The ROWD, other supporting documents, and comments received are on file and may be inspected at the address above at any time between 8:30 a.m. and 4:45 p.m., Monday through Friday. Copying of documents may be arranged through the Regional Water Board at the address below or by calling (213) 576-6600.

Los Angeles Regional Water Quality Control Board
320 W. 4th Street, Suite 200
Los Angeles, CA 90013-2343

F. Register of Interested Persons

Any person interested in being placed on the mailing list for information regarding the WDRs and NPDES permit should contact the Regional Water Board, reference this facility, and provide a name, address, and phone number.

G. Additional Information

Requests for additional information or questions regarding this order should be directed to Raul Medina at (213) 620-2160 or via email at raul.medina@waterboards.ca.gov.

ATTACHMENT G – TOXICITY REDUCTION EVALUATION (TRE) WORK PLAN OUTLINE

- A.** Gather and Review Information and Data
 - 1. POTW Operations and Performance
 - 2. POTW Influent and Pretreatment Program
 - 3. Effluent Data, including Toxicity Results
 - 4. Sludge (Biosolids) Data
- B.** Evaluate Facility Performance
- C.** Conduct Toxicity Identification Evaluation (TIE)
- D.** Evaluate Sources and In-Plant Controls
- E.** Implement Toxicity Control Measures
- F.** Conduct Confirmatory Toxicity Testing

ATTACHMENT H – BIOSOLIDS AND SLUDGE MANAGEMENT

BIOSOLIDS USE AND DISPOSAL REQUIREMENTS

(Note: “Biosolids” refers to non-hazardous sewage sludge defined in 40 CFR part 503.9. Sewage sludge that is hazardous, as defined in 40 CFR part 261, must be disposed of in accordance with the Resource Conservation and Recovery Act (RCRA).) 40 CFR part 503 requirements identified below are for information only and are not regulated by this Order.

- A.** All biosolids generated by the Discharger shall be reused or disposed of in compliance with the applicable portions of:
 - 1. 40 CFR part 503: for biosolids that are land applied, placed in surface disposal sites (dedicated land disposal sites or monofills), or incinerated; 40 CFR part 503 Subpart B (land application) applies to biosolids placed on the land for the purpose of providing nutrients or conditioning the soil for crops or vegetation. 40 CFR part 503 Subpart C (surface disposal) applies to biosolids placed on the land for the purpose of disposal.
 - 2. 40 CFR part 258: for biosolids disposed of in Municipal Solid Waste landfills.
 - 3. 40 CFR part 257: for all biosolids disposal practices not covered under 40 CFR part 258 or 503.
- B.** The Discharger is responsible for assuring that all biosolids from its facility are used or disposed of in accordance with 40 CFR part 503, whether the Discharger reuses or disposes of the biosolids itself or transfers them to another party for further treatment, reuse, or disposal. The Discharger is responsible for informing subsequent preparers, applicers, or disposers of the requirements they must meet under 40 CFR part 503.
- C.** Duty to mitigate: The Discharger shall take all reasonable steps to prevent or minimize any biosolids use or disposal which may adversely impact human health or the environment.
- D.** No biosolids shall be allowed to enter wetland or other waters of the United States.
- E.** Biosolids treatment, storage, and use or disposal shall not contaminate groundwater.
- F.** Biosolids treatment, storage, and use or disposal shall not create a nuisance such as objectionable odors or flies.
- G.** The Discharger shall assure that haulers who transport biosolids off site for further treatment, storage, reuse, or disposal take all necessary measures to keep the biosolids contained.
- H.** If biosolids are stored for over two years from the time they are generated, the Discharger must ensure compliance with all the requirements for surface disposal under 40 CFR part 503 Subpart C, or must submit a written request to USEPA with the information in section 503.20 (b), requesting permission for longer temporary storage.
- I.** Sewage sludge containing more than 50 mg/kg PCB's shall be disposed of in accordance with 40 CFR part 761.
- J.** Any off-site biosolids treatment, storage, use or disposal site operated by the Discharger within Region 4 (Los Angeles Region of RWQCB) that is not subject to its own Waste Discharge Requirements shall have facilities adequate to divert surface runoff from the adjacent area, to protect the site boundaries from erosion, and to prevent any conditions that would cause drainage from the materials in the disposal site to escape from the site. Adequate protection is defined as protected from at least a 100-year storm and from the highest tidal stage that may occur.

K. Inspection and Entry: The USEPA or an authorized representative thereof, upon the presentation of credentials, shall be allowed by the Discharger, directly or through contractual arrangements with their biosolids management contractors, to:

1. Enter upon all premises where biosolids are produced by the Discharger and all premises where Discharger biosolids are further treated, stored, used, or disposed, either by the Discharger or by another party to whom the Discharger transfers the biosolids for further treatment, storage, use, or disposal.
2. Have access to and copy any records that must be kept under the conditions of this permit or of 40 CFR part 503, by the Discharger or by another party to whom the Discharger transfers the biosolids for further treatment, storage, use, or disposal.
3. Inspect any facilities, equipment (including monitoring and control equipment), practices, or operations used in the production of biosolids and further treatment, storage, use, or disposal by the Discharger or by another party to whom the Discharger transfers the biosolids for further treatment, storage, use, or disposal.

L. Monitoring shall be conducted as follows:

1. Biosolids shall be tested for the metals required in section 503.16 (for land application) or section 503.26 (for surface disposal), using the methods in "Test Methods for Evaluating Solids Waste, Physical/Chemical Methods" (SW-846), as required in 503.8(b)(4), at the following minimum frequencies:

<u>Volume (dry metric tons/year)</u>	<u>Frequency</u>
0 – 290	once per year
290 – 1500	once per quarter
1500 – 15000	once per 60 days
> 15000	once per month

For accumulated, previously untested biosolids, the Discharge shall develop a representative sampling plan, which addresses the number and location of sampling points, and collect representative samples.

Test results shall be expressed in mg pollutant per kg biosolids on a 100% dry weight basis.

Biosolids to be land applied shall be tested for Organic-N, ammonium-N, and nitrate-N at the frequencies required above.

2. Prior to land application, the Discharger shall demonstrate that the biosolids meet Class A or Class B pathogen reduction levels by one of the methods listed in 40 CFR section 503.32. Prior to disposal in a surface disposal site, the Discharger shall demonstrate that the biosolids meet Class B levels or shall ensure that the site is covered at the end of each operating day.
3. For biosolids that are land applied or placed in a surface disposal site, the Discharger shall track and keep records of the operational parameters used to achieve Vector Attraction Reduction requirements in 40 CFR section 503.33 (b).
4. Class 1 facilities (facilities with pretreatment programs or others designated as Class 1 by the Regional Administrator) and Federal facilities with > 5 MGD influent flow shall sample biosolids for pollutants listed under section 307 (a) of the Act (as required in the pretreatment section of the permit for POTWs with pretreatment programs.) Class 1 facilities and Federal Facilities with > 5 MGD influent flow shall test dioxins/dibenzofurans

using a detection limit of < 1 pg/g during their next sampling period if they have not done so within the past 5 years and once per 5 years thereafter.

5. The biosolids shall be tested annually or more frequently if necessary to determine hazardousness in accordance with California Law.
6. If biosolids are placed in a surface disposal site (dedicated land disposal site or monofill), a qualified groundwater scientist shall develop a groundwater monitoring program for the site, or shall certify that the placement of biosolids on the site will not contaminate an aquifer.
7. Biosolids placed in a municipal landfill shall be tested semi-annually by the Paint Filter Test (SW-846, Method 9095) to demonstrate that there are no free liquids.

M. The Discharger either directly or through contractual arrangements with their biosolids management contractors shall comply with the following 40 CFR part 503 notification requirements:

1. A reuse/disposal plan shall be submitted to USEPA Region IX Coordinator and, in the absence of other state or regional reporting requirements, to the state permitting agency, prior to the use or disposal of any biosolids from this facility to a new or previously unreported site. The plan shall be submitted by the land applier of the biosolids and shall include, a description and a topographic map of the proposed site(s) for reuse or disposal, names and addresses of the applier(s) and site owner(s), and a list of any state or local permits which must be obtained. For land application sites, the plan shall include a description of the crops or vegetation to be grown, proposed nitrogen loadings to be used for the crops, and a groundwater monitoring plan if one exists.
2. If the Discharger biosolids do not meet 40 CFR section 503.13 Table 3 metals concentration limits, the Discharger must require their land applier to contact the state permitting authority to determine whether bulk biosolids subject to the cumulative pollutant loading rates in 40 CFR section 503.12(b)(2) have been applied to the site since July 20, 1993, and, if so, the cumulative amount of pollutants applied to date, and background concentration, if known. The Discharger shall then notify USEPA Region IX Coordinator of this information.
3. For biosolids that are land applied, the Discharger shall notify the applier in writing of the nitrogen content of the biosolids, and the applier's requirements under 40 CFR part 503, including the requirements that the applier certify that the requirement to obtain information in Subpart A, and that the management practices, site restrictions, and any applicable vector attraction reduction requirements Subpart D have been met. The Discharger shall require the applier to certify at the end of 38 months following application of Class B biosolids that those harvesting restrictions in effect for up to 38 months have been met.
4. If bulk biosolids are shipped to another State or to Indian Lands, the Discharger must send written notice prior to the initial application of bulk biosolids to the permitting authorities in the receiving State or Indian Land (the USEPA Regional Office for the area and the State/Indian authorities).
5. Notification of 40 CFR part 503 non-compliance: The Discharger shall require appliers of their biosolids to notify USEPA Region 9 and their state permitting agency of any noncompliance within 24 hours if the non-compliance may seriously endanger health or the environment. For other instances of non-compliance, the Discharger shall require appliers of their biosolids to notify USEPA Region 9 and their state permitting agency of

the non-compliance in writing within 10 working days of becoming aware of the non-compliance.

- N.** The Discharger shall submit an annual biosolids report to USEPA Region IX Biosolids Coordinator and the Los Angeles Regional Water Quality Control Board by February 19 of each year for the period covering the previous calendar year. The report shall include:
1. The amount of biosolids generated that year, in dry metric tons, and the amount accumulated from previous years.
 2. Results of all pollutant monitoring required in the Monitoring Section above.
 3. Descriptions of pathogen reduction methods, and vector attraction reduction methods, as required in 40 CFR sections 503.17 and 503.27.
 4. Results of any groundwater monitoring or certification by groundwater scientist that the placement of biosolids in a surface disposal site will not contaminate an aquifer.
 5. Names and addresses of land appliers and surface disposal site operators, and volumes applied (dry metric tons).
 6. Names and addresses of persons who received biosolids for storage, further treatment, disposal in a municipal waste landfill, or for other reuse/disposal methods not covered in N.3, above, and volumes delivered to each.
- O.** The Discharger shall require all parties contracted to manage their biosolids to submit an annual biosolids report to USEPA Region IX Biosolids Coordinator by February 19 of each year for the period covering the previous calendar year. The report shall include:
1. Names and addresses of land appliers and surface disposal site operators, name, location (latitude/longitude), and size (hectares) of site(s), volumes applied/disposed (dry metric tons) and for land application, biosolids loading rates (metric tons per hectare), nitrogen loading rates (kg/ha), dates of applications, crops grown, dates of seeding and harvesting and certifications that the requirement to obtain information in 40 CFR section 503.12(e)(2), management practices in section 503.14 and site restrictions in section 503.32(b)(5) have been met.

ATTACHMENT I – PRETREATMENT REPORTING REQUIREMENTS

The City of Simi Valley (Permittee) is required to submit annual Pretreatment Program Compliance Report (Report) to the Regional Water Board and United States Environmental Protection Agency, Region 9 (USEPA). This Attachment outlines the minimum reporting requirements of the Report. If there is any conflict between requirements stated in this attachment and provisions stated in the Waste Discharge Requirements (WDR), those contained in the WDR will prevail.

A. Pretreatment Requirements

1. The Permittee shall be responsible and liable for the performance of all Control Authority pretreatment requirements contained in 40 CFR part 403, including any subsequent regulatory revisions to part 403. Where part 403 or subsequent revision places mandatory actions upon the Permittee as Control Authority but does not specify a timetable for completion of the actions, the Permittee shall complete the required actions within six months from the issuance date of this permit or the effective date of the part 403 revisions, whichever comes later. For violations of pretreatment requirements, the Permittee shall be subject to enforcement actions, penalties, fines and other remedies by the USEPA or other appropriate parties, as provided in the Act. USEPA may initiate enforcement action against a nondomestic user for noncompliance with applicable standards and requirements as provided in the act.
2. The Permittee shall enforce the requirements promulgated under sections 307(b), 307(c), 307(d) and 402(b) of the Act with timely, appropriate and effective enforcement actions. The Permittee shall cause all nondomestic users subject to federal categorical standards to achieve compliance no later than the date specified in those requirements or, in the case of a new nondomestic user, upon commencement of the discharge.
3. The Permittee shall perform the pretreatment functions as required in 40 CFR part 403 including, but not limited to:
 - a. Implement the necessary legal authorities as provided in 40 CFR section 403.8(f)(1);
 - b. Enforce the pretreatment requirements under 40 CFR sections 403.5 and 403.6;
 - c. Implement the programmatic functions as provided in 40 CFR section 403.8(f)(2); and
 - d. Provide the requisite funding and personnel to implement the pretreatment program as provided in 40 CFR section 403.8(f)(3).
4. The Permittee shall submit annually a report to USEPA Pacific Southwest Region, and the State describing its pretreatment activities over the previous year. In the event the City is not in compliance with any conditions or requirements of this permit, then the City shall also include the reasons for noncompliance and state how and when the City shall comply with such conditions and requirements. This annual report shall cover operations from January 1 through December 31 and is due on April 15 of each year. The report shall contain, but not be limited to, the following information:
 - a. A summary of analytical results from representative, flow proportioned, 24-hour composite sampling of the publicly-owned treatment works (POTW) influent and effluent for those pollutants USEPA has identified under section 307(a) of the Act which are known or suspected to be discharged by nondomestic users. This will consist of an annual full priority pollutant scan, with quarterly samples analyzed only

for those pollutants detected in the full scan. The City is not required to sample and analyze for asbestos. Sludge sampling and analysis are covered in the sludge section of this permit. The City shall also provide any influent or effluent monitoring data for nonpriority pollutants which the City believes may be causing or contributing to interference or pass through. Sampling and analysis shall be performed with the techniques prescribed in 40 CFR part 136;

- b. A discussion of Upset, Interference or Pass Through incidents, if any, at the treatment plant which the City knows or suspects were caused by nondomestic users of the POTW system. The discussion shall include the reasons why the incidents occurred, the corrective actions taken and, if known, the name and address of the nondomestic user(s) responsible. The discussion shall also include a review of the applicable pollutant limitations to determine whether any additional limitations, or changes to existing requirements, may be necessary to prevent pass through or interference;
- c. An updated list of the City's significant industrial users (SIUs) including their names and addresses, and a list of deletions, additions and SIU name changes keyed to the previously submitted list. The City shall provide a brief explanation for each change. The list shall identify the SIUs subject to federal categorical standards by specifying which set(s) of standards are applicable to each SIU. The list shall also indicate which SIUs are subject to local limitations;
- d. The City shall characterize the compliance status of each SIU by providing a list or table which includes the following information:
 - i. Name of the SIU;
 - ii. Category, if subject to federal categorical standards;
 - iii. The type of wastewater treatment or control processes in place;
 - iv. The number of samples taken by the POTW during the year;
 - v. The number of samples taken by the SIU during the year;
 - vi. For an SIU subject to discharge requirements for total toxic organics, whether all required certifications were provided;
 - vii. A list of the standards violated during the year. Identify whether the violations were for categorical standards or local limits;
 - viii. Whether the facility is in significant noncompliance (SNC) as defined at 40 CFR section 403.8(f)(2)(viii) at any time during the year; and
 - ix. A summary of enforcement or other actions taken during the year to return the SIU to compliance. Describe the type of action, final compliance date, and the amount of fines and penalties collected, if any. Describe any proposed actions for bringing the SIU into compliance.
- e. A brief description of any programs the POTW implements to reduce pollutants from nondomestic users that are not classified as SIUs;
- f. A brief description of any significant changes in operating the pretreatment program which differ from the previous year including, but not limited to, changes concerning the program's administrative structure, local limits, monitoring program or monitoring frequencies, legal authority, enforcement policy, funding levels, or staffing levels;
- g. A summary of the annual pretreatment budget, including the cost of pretreatment program functions and equipment purchases; and

- h. A summary of activities to involve and inform the public of the program including a copy of the newspaper notice, if any, required under 40 CFR section 403.8(f)(2)(viii).

B. LOCAL LIMITS EVALUATION

1. In accordance with 40 CFR section 122.44(j)(2)(ii), the POTW shall provide a written technical evaluation of the need to revise local limits under 40 CFR section 403.5(c)(1) within 180 days of issuance or reissuance of the Facility's NPDES permit.

C. SIGNATORY REQUIREMENTS AND REPORT SUBMITTAL

1. Signatory Requirements.

The annual report must be signed by a principal executive officer, ranking elected official or other duly authorized employee if such employee is responsible for the overall operation of the POTW. Any person signing these reports must make the following certification [40 CFR section 403.6(a)(2)(ii)]:

I certify under penalty of law that this document and all attachments were prepared under my direction or supervision in accordance with a system designed to assure that qualified personnel properly gather and evaluate the information submitted. Based on my inquiry of the person or persons who manage the system, or those persons directly responsible for gathering the information, the information submitted is, to the best of my knowledge and belief, true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment for knowing violations.

2. Report Submittal.

The Annual Pretreatment Report shall be submitted electronically using the State Water Board's California Integrated Water Quality System (CIWQS) Program website <http://www.waterboards.ca.gov/ciwqs/index.html>. The CIWQS website will provide additional information for SMR submittal in the event there will be a planned service interruption for electronic submittal.

A copy of the Annual Pretreatment Report must be sent to USEPA electronically to the following address:

R9Pretreatment@epa.gov